



ИНОСТРАННЫЙ ЯЗЫК В СФЕРЕ ПРОФЕССИОНАЛЬНЫХ КОММУНИКАЦИЙ

Учебно-методическое пособие



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Учебно-методическое пособие соответствует содержанию рабочей программы по дисциплине «Иностранный язык в сфере профессиональных коммуникаций» и предназначено для учащихся магистратуры инженерно-технологических направлений.

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Предисловие

Настоящее пособие предназначено для работы на практических занятиях по учебной дисциплине «Иностранный язык в сфере профессиональных коммуникаций» с учащимися 1 курса магистратуры инженерно-технологических направлений. Работа с пособием призвана способствовать овладению компетенцией УК-4 «Способен применять современные коммуникативные технологии, в том числе на иностранном(ых) языке(ах), для академического и профессионального взаимодействия».

Первая часть пособия «Academic Writing» имеет целью научить студентов-магистрантов методике работы с такими текстами академического дискурса, как аннотация, научная статья, обзор, а также соответствующим способам репрезентации результатов академической и профессиональной деятельности в устной и письменной формах. Данный блок знакомит магистрантов с особенностями научного стиля английского языка, структурой и современными требованиями к научной статье на английском языке, а также способствует выработке практических навыков написания отдельных частей научной статьи.

Разделы «Введение», «Методология», «Результаты», «Обсуждение» и «Заключение», а также разделы, посвященные аннотации, названию и ключевым словам нацелены на то, чтобы помочь студентам понять, как выглядит традиционная модель соответствующего раздела исследовательской статьи. Молодой исследователь, начинающий писать в соответствии с простой и общепринятой моделью, вскоре развивает навыки самостоятельного и профессионального письма на более высоком уровне. Раздел «How to Write a Conference Paper for 'Science and Technology: 21st Century'» готовит студентов к выступлению с докладом на ежегодной студенческой конференции, организованной кафедрой иностранных языков, а также к эффективному написанию тезисов к конференции.

Теме академической конференции посвящена вторая часть пособия. Цель этой части – формирование основ профессионального взаимодействия на иностранном языке, исходя из цели общения. Данный раздел учит использованию ключевой лексики, связанной с конференцией, развивает навыки публичной речи для выступления на международных конференциях на английском языке. Комплекс заданий отражает основные этапы деятельности, направленной на представление результатов научных исследований на международных конференциях. В двух модулях раздела отражены следующие этапы: поиск конференции, изучение информационного письма, написание и отправка тезисов, выступление с устным докладом, нетворкинг, руководство секцией на конференции. Студенты получают информацию о лучших методах оформления презентации, привлечения аудитории и эффективного управления вниманием, учатся применять приемы, позволяющие четко и уверенно понимать вопросы и отвечать на них.

Третья часть пособия посвящена изучению грамматики английского языка. Данный раздел систематизирует грамматический материал, отражающий особенности стиля научного изложения. Теоретическая информация четко систематизирована в таблицах и диаграммах и подкреплена практическими упражнениями. В рамках дисциплины студенты учатся активно владеть основными грамматическими явлениями, характерными для профессиональной речи: времена глагола активного и страдательного залога, типы вопросов; относительные придаточные предложения; номинализация; исчисляемые и неисчисляемые существительные; предлоги; модальные глаголы, выражающие предположение; условные предложения; причастие и герундий; инфинитив; инфинитивная конструкция «Сложное подлежащее»; инфинитивная конструкция «Сложное дополнение». Грамматический материал представлен в соответствии с контекстным подходом: используются учебные профессионально-ориентированные предложения и ситуации, развивающие профессиональное мышление и дающие представление о том, как

грамматические правила применяются на практике. Упражнения повышенной трудности даны со звездочкой.

Приложение в систематизированном виде содержит необходимые для изучения курса английского языка вспомогательные материалы: «Числа и основы математики»; «Единицы СИ»; «Как читать формулы по механике и инженерному делу»; «Как читать химические формулы и реакции»; «Сокращения и акронимы»; «Транскрипция личных имен»; «Английские названия факультетов и кафедр Санкт-Петербургского государственного технологического института»; «Перевод названий компаний, научно-исследовательских и образовательных учреждений»; «Заглавные буквы в академическом письме».

1. The IMRAD format. General Information

Objectives

By the end of this module, students will be able to:

1. Understand the IMRAD structure.
2. Label IMRAD sections in field-specific articles and state its section's role.
3. Choose a journal, apply its author guidelines, and draft an IMRAD-style outline.

1.1 Warming-up. Discuss the following questions:

1. Have you ever read research papers in English? What types of research papers can you name?
2. What are the challenges in understanding research articles in English?
3. Have you heard of IMRAD? If so, in what context?
4. What are the key components you typically expect to find in a research paper?
5. Why is it important to have a standardized format for a research article?

1.2 Read the text to find out more about the IMRAD format:

Most scientific papers are prepared according to a format called IMRAD. The term represents the first letters of the words Introduction, Materials and Methods, Results, And, Discussion. It indicates a pattern or format rather than a complete list of headings or components of research papers; the missing parts of a paper are: Title, Authors, Keywords, Abstract, Conclusions, and References. Additionally, some papers include Acknowledgments and Appendices. The Introduction explains the scope and objective of the study in the light of current knowledge on the subject; the Materials and Methods describes how the study was conducted; the Results section reports what was found in the study; and the Discussion section explains meaning and significance of the results and provides suggestions for future directions of research.

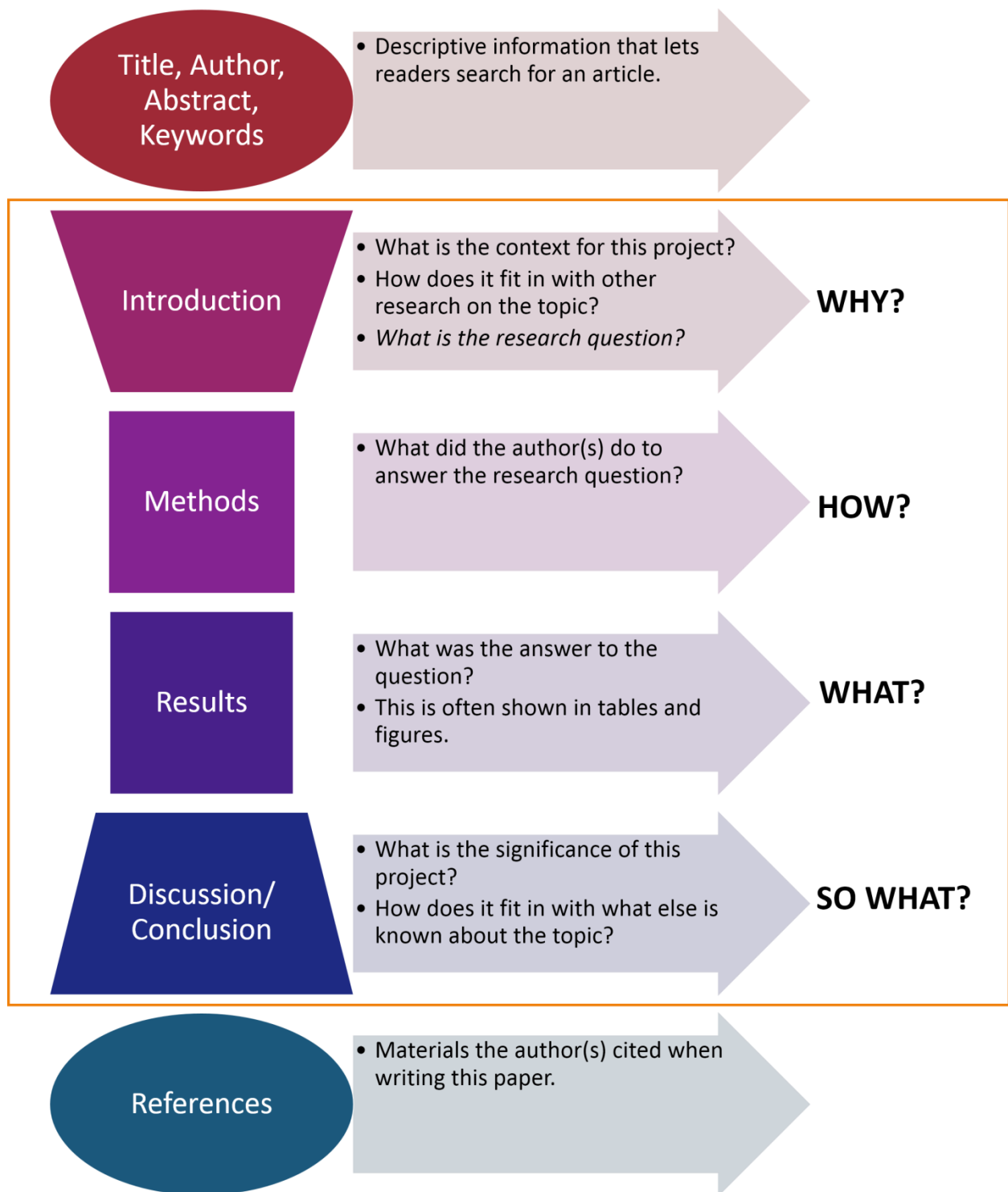


Fig.1. The shape of a research article or thesis.

(<https://i.pinimg.com/originals/2d/b2/9c/2db29c6e7c3aa5619be3d7740949cfea.png>)

Something you should notice about the shape of the diagram is that it narrows towards the central report section, and widens after it. This represents the way information is ordered in the Introduction and the Discussion/Conclusion: in the Introduction you start out by being fairly general and gradually narrow your focus, whereas the opposite is true in the Discussion/Conclusion.

An important point to keep in mind is that there is no standard or uniform style that is followed by all journals. Each journal has its own style; but they all have their own Instructions to Authors (or other word combinations to mean the same thing). Once you select a journal to which you wish to submit your manuscript, please FOLLOW THE JOURNAL'S

INSTRUCTIONS TO AUTHORS, which can usually be found in each volume of the journal (note that a volume may contain several numbers, and there could be multiple volumes in a year), or easily accessed from the journal's webpage. Some authors may not be fully convinced about the logic of some of these instructions, but it is a futile effort to argue with the journal or complain about its instructions. Remember that authors are free to choose from a number of journals in which to publish their papers.

1.3 Find a research article in your field. Identify the parts of the IMRAD format. Some of them might have different subheadings.

1.4. Match the sections of a research article with their purposes (1-8).

1. It summarises the research study and results of the study; it is included in article databases and is usually free to a large audience.
2. It describes both specific techniques and the overall experimental strategy used by researchers.
3. It clearly indicates the content of the article and helps interested readers to find the article quickly in a database.
4. It explains research findings and often deals with their applications.
5. It summarises the data collected and calculates totals or trends, statistically significant findings, etc.
6. It describes the theoretical background, indicates why the work is important, states a specific research question, and poses a specific hypothesis to be tested.
7. It states implications and recommends further research.
8. It lists the sources cited by the author(s) of the article.

1.5. Discuss the following questions.

1. What is a standard structure of a scientific paper?
2. Which part of the article contains the gist of the research?
3. Which part is a literature review included into?
4. In which part should you be very specific and why?
5. Where would you put most of your tables and graphs?
6. Where do you discuss the implications?
7. Where would you talk about the novelty and significance of your work?
8. In what situation is it really important to make sure that the paper reads with one voice?
9. What submission systems do most papers have? What should you do to submit your paper to a journal?
10. What demands should your paper meet to be accepted?
11. Does it often happen that the article gets accepted first time?

1.6 Choose a journal in your field where you could publish an article. Study Instructions to Authors on its website. Share them with your class.

2. Reading and Understanding a Research Paper

Objectives

By the end of this module, students will be able to:

1. Identify the purpose of each paper section.
2. Link section content to the main research aim.
3. Check that methods match the aims and results support the conclusions.
4. Spot strengths and weaknesses in the argument flow and methodology.
5. Use examples to answer guided questions about each section.
6. Apply these insights to outline their own IMRAD draft.

Warming-up. Discuss the following questions:

1. What parts can you identify in the research paper you are reading?
2. How do they correspond to the IMRAD format?
3. In which part of the paper should you look for literature review?
4. Which part presents your findings?
5. What is the last part of the paper?
6. Beyond the main IMRAD sections, what other parts of the research paper can you identify?

2.1. Introduction

The Introduction to a research paper is where the authors set up their topic and approach for the reader. It has several key goals:

- Present the topic and get the reader interested
- Provide background or summarize existing research
- Position the author's approach
- Detail specific research problem and problem statement
- Give an overview of the paper's structure

Basically, authors should organize the Introduction around five key questions:

1. What were you investigating?
2. Why was it important?
3. What was already known about the subject of your research?
4. What did you expect to know after doing the research?
5. How did you approach the problem? What research methods did you use?

2.1.1 Watch the video at <https://rutube.ru/video/f10149751810d611eb625dfc4ed22e2e/> and answer the following questions:

1. Why is the Introduction to a research paper important?
2. What are the two ways of starting the Introduction to a research paper?
3. What is the second element of the Introduction?
4. How should a literature review be organized?
5. What is the third element of the Introduction? Why is it the most important one? What phrases are used to state the research gap?
6. What part usually follows the research gap?
7. What are the optional elements of the Introduction?

2.1.2 Read the Introduction below. What standard components of a typical Introduction do the sections 1-8 correspond to? Build a Model of the Introduction by writing a short description of what the writer is doing in each section. Do they correspond to the structure discussed in the video above?

The Synthesis of flexible polymer blends from polylactide and rubber

Introduction

1 Polylactide (PLA) has received much attention in recent years due to its biodegradable properties, which offer important economic benefits. **2** PLA is a polymer obtained from corn and is produced by the polymerization of lactide. It has many possible uses in the biomedical field and has also been investigated as a potential

In this sentence the authors:

- 1.** _____
- 2.** _____

engineering material. **3** However, it has been found to be too weak under impact to be used commercially.

4 One way to toughen polymers is to incorporate a layer of rubber particles and there has been extensive research regarding the rubber modification of PLA. For example, Penny et al. showed that PLA composites could be prepared using blending techniques and more recently, Hiller established the toughness of such composites.

5 However, although the effect of the rubber particles on the mechanical properties of copolymer systems was demonstrated over two years ago, little attention has been paid to the selection of an appropriate rubber component.

6 The present paper presents a set of criteria for selecting such a component. **7** On the basis of these criteria it then describes the preparation of a set of polymer blends using PLA and a hydrocarbon rubber (PI). **8** This combination of two mechanistically distinct polymerizations formed a novel copolymer in which the incorporation of PI significantly increased flexibility.

(From Glasman-Deal H. *Science Research Writing for Non-Native Speakers of English*. Imperial College Press. London, UK. 2010, p. 16-17)

2.1.3 Read the Introduction to the research paper you are reading with your group. Answer the questions:

1. Do the authors provide general *background information* on the research?
2. Do the authors establish the *importance of their research topic*?
3. How do they describe the *general problem* or the current research focus of the field?
What is the *research gap*?
4. Is there any brief *overview of key research projects (literature review)* in this area?
5. What *hypothesis* is offered by the authors?
6. Do the authors give any *details about the methodology*?
7. Do the authors announce their *findings*?

2.2. Materials and Methods

The materials and methods section follows the Introduction section. This is generally the easiest section to write because the authors are simply reproducing what they did in their experiments. It is always a good idea to start writing your research paper with the materials and methods section. Materials and methods should describe how you did your research and detail the experimental procedure. One of the most important things to bear in mind while writing the materials and methods section is that it should have enough detail so that other researchers in your field can replicate your experiments and reproduce your results. You should provide all the steps in a logical order so that your readers can follow your description easily.

Grammar Tip: The Materials and Methods section is written in the past tense. Passive voice is often used.

Using Acronyms and Abbreviations. Acronyms and abbreviations are often used for long method names. Abbreviations and acronyms are a great way to make writing concise and save time. Acronyms and abbreviations are defined during their first occurrence, then the short form is used in the rest of the text. The common practice is to put the acronym and abbreviations in parentheses after the full term. Commonly known abbreviations like IR, SEM, UV are not defined. Please refer to *Appendix E* for the list of most commonly used abbreviations.

2.2.1 Read the following Methodology section and answer the questions below.

Changes in the chemistry of groundwater in the chalk of the London Basin

The current investigation involved sampling and analysing six sites to measure changes in groundwater chemistry. The sites were selected from the London Basin area, which is located in the south-east of England and has been frequently used to interpret groundwater evolution. A total of 18 samples was collected and then analysed for the isotopes mentioned earlier. Samples 1–9 were collected in thoroughly-rinsed 25 ml brown glass bottles which were filled to the top and then sealed tightly to prevent contamination. The filled bottles were shipped directly to two separate laboratories at Reading University, where they were analysed using standard methods suitably miniaturised to handle small quantities of water. Samples 10–18 were prepared in our laboratory using a revised version of the precipitation method established by the ISF Institute in Germany. This method obtains a precipitate through the addition of $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$; the resulting precipitate can be washed and stored easily. The samples were subsequently shipped to ISF for analysis by accelerator mass spectrometry (AMS). All tubing used was stainless steel, and although two samples were at risk of CFC contamination as a result of brief contact with plastic, variation among samples was negligible.

(From Glasman-Deal H. *Science Research Writing for Non-Native Speakers of English*. Imperial College Press, London, UK. 2010, p. 58)

Answer the questions and summarise the basic components of the Methodology section.

1. Where do the authors offer a general overview of the subsection and include the *purpose* of the investigation?
2. How do the authors *justify the choice* of location?
3. What details about what was done and used do the authors provide? What specific *details about materials or methods* do they provide? (quantities, temperatures, duration, sequence, conditions, locations, sizes)?
4. How do the authors *give justifications* for their steps? What language do they use to show that *care was taken*?
5. Do the authors relate their method to other studies? Where do they describe what was done by *referring to existing methods in the literature*?
6. In what way do the authors underline that *their method has been a good choice*?
7. What *possible difficulty* in the methodology do the authors mention?

2.2.2 Analyze the Materials and Methods section of the paper you are reading and answer the following questions:

1. Did the authors propose a new method or did they use a standard method from the literature?
2. What were the reasons for choosing this method?
3. How was the method implemented?
4. What equipment was used to perform the measurements?
5. What parameters were measured during the experiment?
6. What was the measurement procedure?
7. How were the samples prepared?
8. Were the measurements performed under special conditions?
9. How were the calculations performed and how was the data produced?
10. What visuals were used in the section? What made them effective?
11. What abbreviations and acronyms did the authors use? Did they define them? Why?

2.3. Results

The results section summarizes and presents the findings of the study to put them in context with the research question(s). The study's data should be presented in a logical sequence without bias or interpretation. Findings may be reported in written text, tables, graphs, and other illustrations. It is important to include a contextual analysis of the data by tying it back to the research question(s). Only relevant data and findings that connect with the goal of the study should be shared, because too much data may overwhelm a reader. In addition, your results may be problematic; perhaps some experiments were not fully successful and you want to suggest possible reasons for this.

The title of this section varies in different disciplines, and also in different journals. Instead of Results, it is sometimes called 'Analysis' or 'Data Analysis' or 'Findings' or 'Results and Discussion'.

Using Visuals in the Results

In the Results section, visual data can be effectively presented using various graphic techniques, including figures, tables, flow diagrams, and infographics, to illustrate experimental procedures, equipment, and data collection methods. Choose the appropriate visualization type based on the data and research question, keeping the presentation clear, concise, and understandable. The general rule is: present the data in the text, or in a table, or in a figure. Almost never present the same data in more than one way. Graphs are very similar to tables as a means of presenting data in an organized way. In fact, the data can be presented either as tables or as graphs. How do we decide which is preferable? This is often a difficult question. A good rule might be this: If the data show pronounced trends, making an interesting picture, use a graph. If the numbers just sit there, with no exciting trend in evidence, a table should be satisfactory. Tables are also preferred for presenting exact numbers.

2.3.1 Study the following vocabulary related to visuals:

bar chart – столбчатая диаграмма

cell - ячейка

column - столбец

curve – кривая

figure - рисунок

flow chart/flow diagram – технологическая схема

graph - график

histogram - гистограмма

line graph – линейная диаграмма, график

pie chart – круговая диаграмма

row - ряд

scatter plot – диаграмма рассеивания, точечная диаграмма

table – таблица

X-axis – ось абсцисс

Y-axis – ось ординат

How Does a Results Section Differ from a Discussion Section?

The results section of a research paper tells the reader what you found, while the discussion section tells the reader what your findings mean. The results section should present the facts in an academic and unbiased manner, avoiding any attempt at analyzing or interpreting the data. Think of the results section as setting the stage for the discussion section by making all the necessary information known to the reader. It is not uncommon for these sections to be combined, but researchers will often use sub-headings to distinguish between the two.

2.4. Discussion

The discussion reviews the findings and puts them into the context of the overall research. It brings together all the sections that came before it and allows a reader to see the connections between each part of the research paper. In a discussion section, the author engages in three necessary steps: interpretation, analysis, and explanation. An effective discussion section will tell a reader why the research results are important and where they fit in the current literature, while also being self-critical and candid about the shortcomings of the study.

How Does a Discussion Section Differ from a Conclusion?

A conclusion summarizes parts of the paper. A discussion, however, is a much more thorough and rigorous examination of the results. It requires the author to interpret those results by looking at how or why they are the way they are. Additionally, the discussion section is the space where the author acknowledges the limits of the research and identifies gaps for future research. Finally, this section investigates the implications of the research based on the findings and results, and it draws meaningful conclusions from those implications. So, where a conclusion is brief and touches on the main points of the paper, the discussion is much longer and more detailed.

2.4.1 Read the Results and Discussion section of your paper and answer the following questions:

1. What are the *key findings* of the research? How are they linked to the initial research question?
2. Were there any *unexpected* or *secondary results*? Do the authors explain why they have appeared?
3. Do the authors address any *limitations* or *weaknesses* in the research?
4. Do the authors mention any *gap in the research*? Do they recommend *areas for further investigation*?

2.5. Conclusions

Conclusions are not simply an overview of a paper. Instead, they should reiterate why the research is important. Effective conclusions help readers reflect on what they just read, draw connections to existing knowledge, and spark their desire to further explore the subject. Ultimately, the conclusion should help readers answer the following question: Why should I care about this topic?

Conclusions are usually short. While there is no hard and fast rule on length, conclusions are typically one paragraph long; however, you may find some that are two or three paragraphs long.

2.5.1 Study the Conclusions of the paper you are reading and answer the following questions:

1. How do the authors *summarize the main achievement* or contribution of their study, presenting the key innovation or method developed?
2. What proof do the authors give that their new technique works well? *What methods did they use to check this?*
3. How do the authors discuss *the broader applicability of the developed method*, explaining how it can be used beyond the specific example presented in the study?
4. How do the authors show that their method is special or different from old ways of doing things? *What makes it unique?*
5. How do the authors speculate on the *future impact and potential applications* of the developed method?

3. How to Write an Abstract

Objectives

By the end of this module, students will be able to:

1. Identify the structure of an abstract.
2. Understand grammar and style in abstracts.
3. Critically assess abstract content (strengths, weaknesses, common mistakes, and areas for improvement).
4. Write a concise abstract that effectively communicates the research, adhering to academic standards.

3.1 Warming-up. Discuss the following questions:

1. What is the purpose of an abstract? Why is it important?
2. Where do you typically find abstracts?
3. What information does the abstract usually include?
4. What common mistakes do you think people make when writing an abstract?
5. What types of abstracts do you know?

3.2 Read the text and answer the questions after it:

An abstract is a mini-version of the paper. It enables readers to identify the basic content of a document quickly and accurately, to determine its relevance to their interests, and thus to decide whether they need to read the entire document. Therefore, it is extremely important that the abstract be written clearly. The abstract should be definitive rather than descriptive; i.e., it should give facts rather than say the paper is ‘about’ something. Since an abstract will usually be read by an average of 100 times more people than will read the full paper, it should convey the information itself, not just promise it. When writing an abstract, avoid phrases such as ‘... are described’ or ‘... will be presented’; instead, describe them, present them. In other words, reading a scientific article shouldn’t feel like reading a detective story. Journals have strict limitations on the length of abstracts, usually in the range of 150–250 words, and written in one paragraph. The abstract should stand on its own, i.e., be complete in itself. It starts with a statement of rationale (обоснование важности темы исследования) and objectives and reports the methods used, the main results including any newly observed facts, and the principal conclusions and their significance. If keywords are not listed separately, the abstract should contain the keywords by which the paper should be indexed. Because the abstract is a short version of the full paper, it contains a mixture of tenses representing the tense used in reporting the respective sections of the paper. Thus, in the abstract, statements referring to the rationale and introduction, interpretation of results, and conclusions are in present tenses, whereas materials and methods and results are in past tense. The abstract should not contain:

- abbreviations or acronyms unless they are standard or explained;
- references to tables or figures in the paper;
- literature citations;
- any information or conclusion not in the paper itself;
- general statements;
- complex, winding, verbose sentences.

Furthermore, in order to facilitate smooth reading, excessive quantitative data with statistical details should be avoided in the abstract. Experienced writers prepare or fine-tune their title and abstract after the rest of the paper is written.

Very often journals specify their abstract structure, but even if they do not, you should keep close to IMRAD structure. You are recommended to answer the following questions in your abstract:

- What do we know about the topic?
- What question do we aim to answer (hypothesis and reason for the research)

- What have we done?
- What have we found?
- What are our conclusions?
- Why is our research important?

Novice researchers are advised to use the copy-and-paste method: select the most important sentences from each section of their article and put them in order.

Questions:

1. What is the purpose of an abstract?
2. Should an abstract be written as a detective story?
3. How long should an abstract be?
4. How many paragraphs does an abstract usually have?
5. What parts are included in the contents of an abstract?
6. Does an abstract contain keywords?
7. Which grammar tenses are used in an abstract?
8. What type of information should not be included in an abstract?
9. Do you write an abstract before or after your article is completed?
10. What questions should you answer when writing an abstract?
11. What recommendation can novice researchers use for writing an abstract?

3.3 Read the information from MDPI Layout Style Guide about writing an abstract. Summarize the main points.

Abstracts

The abstract contains a summary of the entire paper and can be up to 200 words long with only one paragraph. It must not contain any images or tables (although a graphical abstract may also be submitted). Do not include running title (a shortened version of the paper's main title), website links, equations, figures (or other graphical elements), tables, or structures that require display on a line separate from the text.

Authors should follow the style of a structured abstract, which is based on the IMRAD structure of a paper but without using headings. In other words, give a background and motivation to the paper, a brief description of the methods, the principal results, and then conclusions or interpretations. Some journals in the medical field may require subheadings within the abstract; you may refer to the instructions for authors to see if this is required. Abstracts without headings should consist of a single paragraph.

Abstracts must be self-contained: they are often displayed and read independently of the rest of the paper. This means that any abbreviations used must be defined in the abstract, and no reference can be made to the bibliography or any figures. Citations to previously published papers are not required in abstracts.

The abstract, along with the main title, is the first part of your paper that a reader will see. It should give them a good overview of all the major aspects of the work carried out. It should not be thought of as a sales pitch to encourage readers to download and read the full article, although including some motivation is a good idea. Instead, you should focus on making it informative and comprehensive. A well-written abstract will mean that someone who goes on to read the full article will already have a good idea of the content and will be able to focus on the parts they are most interested in.

From https://www.mdpi.com/authors/layout#_bookmark5

3.4. Read the following abstract of a research paper 'Feeling How Old I am: Subjective Age is Associated with Estimated Brain Age'. Match its parts to the parts of the paper (Introduction, Method, Results, Discussion). Give reasons for your answer.

'While the aging process is a universal phenomenon, people perceive and experience one's aging considerably differently. Subjective age (SA), referring to how individuals experience themselves as younger or older than their actual age, has been highlighted as an

important predictor of late-life health outcomes. However, it is unclear whether and how SA is associated with the neurobiological process of aging. In this study, 68 healthy older adults underwent a SA survey and magnetic resonance imaging (MRI) scans. T1-weighted brain images of open-access datasets were utilized to construct a model for age prediction. We utilized both voxel-based morphometry (VBM) and age-prediction modeling techniques to explore whether the three groups of SA (i.e., feels younger, same, or older than actual age) differed in their regional gray matter (GM) volumes, and predicted brain age. The results showed that elderly individuals who perceived themselves as younger than their real age showed not only larger GM volume in the inferior frontal gyrus and the superior temporal gyrus, but also younger predicted brain age. Our findings suggest that subjective experience of aging is closely related to the process of brain aging and underscores the neurobiological mechanisms of SA as an important marker of late-life neurocognitive health.’

From Kwak S, Kim H, Chey J, Youm Y. *Feeling How Old I Am: Subjective Age Is Associated With Estimated Brain Age*. *Front Aging Neurosci*. 2018 Jun 7;10:168. doi: 10.3389/fnagi.2018.00168. PMID: 29930506; PMCID: PMC5999722.

Grammar tip:

The use of VERB TENSE is very important in the abstract because the strict word limit means that you may need to omit phrases that tell the reader whose work you are referring to, or what you think about your results. In this case, careful and accurate use of a verb tense is very important.

The **problem** of your research is normally expressed in the Present Simple Tense:

The main problem, however, is...

We examine why these models have difficulty with...

However, this assessment cannot be based solely on...

Although it is known theoretically that...

When you are referring to **what the paper itself does** or **what is actually in the paper itself**, use the Present Simple Tense, for example:

This paper presents a new methodology for...

In this paper we apply...

We consider a novel system for...

When you are referring to your **methodology**, or **what you did during the research period**, it is common to use the Past Simple Tense, for example:

The data obtained from my field study were analyzed...

The results were evaluated using...

The findings were synthesized into ten categories...

Results can be expressed in either the Present Simple or the Past Simple Tense, for example:

The research suggests that one of the future directions for...

This study reveals that...

This trend was found to produce...

The research showed a marked increase in...

Achievements can be expressed in the Present Perfect Tense, as in the Discussion/Conclusions:

This investigation has revealed that...

We have devised a strategy which allows...

We have demonstrated that this approach is...

and also in the Present Simple Tense:

The framework described here is both simple and universal...

The value of our approach lies in...

This provides a powerful tool for...

Applications are normally stated in the Present Simple Tense:

This framework can be used to evaluate...
This approach can be applied to...
This demonstrates potential for general applicability...

3.5 Write a sentence about each of the following: 1) what your research actually does, 2) the methodology you used, 3) your results, 4) achievements, and 5) applications. Use the verb tenses described above.

3.6 The text of an abstract must be concise. Replace the underlined words in sentences 1-5 from abstracts with that or those.

1. The hormone increased the power output of healthy volunteers by 16 per cent after four weeks of taking the drug. Healthy volunteers who took the drug could also exercise 50 per cent longer than control subjects.
2. We compare photographic exposure from scattered light with light from direct light.
3. The target yield is the yield which can be produced in 'perfect' conditions.
4. Structures like the structures described in this paper are not known in glyptodonts recorded before the Great American Biotic Interchange.
5. The lithology of failed carbonate strata differs from the lithology if their basal shear surfaces.

3.7 Homework assignment. Think about some research you have done recently. Write:

- one or two sentences which provide the key background to the research
- a sentence which states your research question
- a sentence which presents your hypothesis
- two or three sentences outlining the main methods used
- one or two sentences presenting the key results
- a sentence which states the key implication of your findings.

Connect these sentences to form the draft of an abstract. Check your work for repetition of words or ideas.

4. Giving a Title to Your Paper

Objectives

By the end of this module, students will be able to:

1. Understand the importance of a research paper title.
2. Identify key characteristics of a good title.
3. Avoid common mistakes like vague phrases, informal language, or low-impact words.
4. Evaluate and improve existing titles.
5. Apply best practices to create their own title.

4.1 Warming-up. Discuss the following questions:

1. Why is the title of a research paper so important?
2. What are some common mistakes students make when creating titles for their research papers?
3. What are the key characteristics of a good title for a research paper?

4.2 Read the text and decide if the statements following it are true or false:

The title of the paper will be read more than any other part. The way in which a paper is 'browsed' by readers is in the order: Title—Abstract—Results (Tables and Figures)—Full paper. The prevailing trend is said to be that, on average, the number of readers from one section to the next in the above sequence decreases by a factor of 10. That means for every 10 readers who look at the title, one reads the abstract; for every 10 who read the abstract, one goes to the

Results section, especially Tables and Figures; for every 10 who read the Results, one reads the full paper. Thus, for every person who reads the full paper, 1,000 read the title. Titles are read both by scientists scanning the contents of a journal and by those depending on searches through secondary sources, which always carry the title and author but may or may not carry abstracts. The title may be reprinted in bibliographies and subject indexes, stored in bibliographic databases and cited in other articles. Therefore, the title is an extremely important component of the paper. A good title will attract readers who might not otherwise read the paper and may help future researchers find important information. A good title of a research paper should:

- contain as few words as possible (preferably not more than 20). In fact, many journals limit titles to 12 words;
- be easy to understand;
- describe the contents of the paper accurately and specifically;
- not make your article seem more important than it really is;
- avoid abbreviations, formulas, and jargon;
- not include any finite forms of verbs;
- not contain low-impact words such as ‘Some notes on ...,’ ‘Observations on ...,’ ‘Investigations on ...,’ ‘Study of ...,’ and ‘Effect of ...’;
- not be flashy as in newspapers (e.g., avoid statements like ‘Agroforestry can stop deforestation’);
- report the subject of the research rather than the results;
- follow the style preference of the target journal.

True or False:

1. The title is usually read after the abstract.
2. For every 20 readers who look at the title, two read the abstract.
3. The longer the title is, the better.
4. A good title should not make your article look more important than it really is.
5. A title to a research paper should not include any finite verb forms.
6. It is a good idea to begin your title with a phrase like ‘Study of...’ or ‘Observations on...’

4.3 Read the titles of six research papers. Three of them are fine, while the other three need improvement. Decide, which titles are most helpful for the reader and which are not very successful. Explain why.

Tip: you may use the following vocabulary:

- The title is vague and lacks specificity; it needs clarity regarding the study's focus.
 - The title contains informal language and includes words which should be avoided.
 - The title is quite effective, but could be more specific regarding the context of the study.
 - The title is too lengthy and complex; it needs focus.
 - The title is not direct and clear.
 - The title is good, concise, and informative.
 - The title avoids low-impact words.
 - The title clearly defines the subject without being overly sensational or vague.
 - The title poses a question, which is better suited for an exploratory article.
1. Advancements in Renewable Energy Storage: Enhancing Battery Efficiency through Nanotechnology
 2. Some Thoughts on Energy and Stuff
 3. A Study About Things in Cities
 4. The Impact of Urban Green Spaces on Mental Health: A Longitudinal Study
 5. Machine Learning Algorithms for Early Detection of Alzheimer's Disease Using MRI Data
 6. How You Can Use Computers to Do Medical Things

4.4 Read seven suggestions for writing the title of a research paper. Which suggestions should you use to write a good title? Which suggestions do not give good advice?

1. Make it about fifty words long
2. Write it as a question
3. Begin with a phrase like 'A study of...' or 'An investigation into...'
4. Include a joke or play on words
5. Include important key words for internet search tools
6. Include information such as the species studied, the treatment used, etc.
7. Present the key result

4.5 Read the four alternatives for the title of a research paper. Decide which one is the best. Explain, why. You may use the vocabulary from exercise 4.3.

- "Climate Change and Farming: What Happens?"
- "An Analysis of Weather and Plants"
- "How Climate Affects Agriculture"
- "Assessing the Effects of Climate Change on Crop Yield and Soil Health in Semi-Arid Regions"

4.6 Discuss the titles of the papers you found earlier. What are their strong and weak points? How can they be improved?

4.7 Homework assignment. Think of a title you could give your paper. Discuss it with the class.

5. How to Create Keywords

Objectives

By the end of this module, students will be able to:

1. Understand why keywords are important in a research paper and where they go.
2. Explain how keywords help researchers find papers.
3. Recognize common mistakes when choosing keywords.
4. Identify poor and better keywords using provided tips.

5.1 Warming-up. Discuss the following questions:

1. Why are keywords important in a research paper?
2. Where do you typically find keywords in a published research paper?
3. What are some common mistakes people make when choosing keywords?
4. How do keywords help researchers find relevant papers?
5. How many keywords should you typically include in a research paper?

5.2 Read the text and summarize its content:

Keywords are words that capture the essence of your paper. Keywords make your paper searchable and ensure that you get more citations. Therefore, it is important to include the most relevant keywords that will help other authors find your paper. Here are a few tips that will help you create relevant and effective keywords for your paper:

1. Think from the point of view of the reader. What keywords would the reader search for that would help retrieve your article?
2. Keywords should ideally be phrases of 2-4 words; single word keywords are acceptable, but they may lead to many false matches.

- Keywords should contain words and phrases that suggest what the topic is about. Also include words and phrases that are closely related to your topic. (For example, if the paper is about heart diseases, use words like stroke, circulatory system, blood, etc.)
- Also use variant terms or phrases that readers are likely to use (For example, if the paper is about spine disorders, use words like spinal cord, vertebral column, backbone, etc.)
- The full forms of shortened words or acronyms and abbreviations should be included as well.

Journals ask for anywhere between 3-8 keywords. Therefore, the general recommendation is to have 4-5 keywords ready, and add more depending on the journal requirement.

5.3 Look at the following examples of manuscript titles and keywords. Identify at least two specific problems with the 'Poor keywords' based on the tips provided in the text. Identify at least two specific strengths of the 'Better keywords'. What tips do they demonstrate?

- Manuscript title:** Direct observation of nonlinear optics in an isolated carbon nanotube
Poor keywords: molecule, optics, lasers, energy lifetime
Better keywords: single-molecule interaction, Kerr effect, carbon nanotubes, energy level structure
- Manuscript title:** Region-specific neuronal degeneration after okadaic acid administration
Poor keywords: neuron, brain, OA (an abbreviation), regional-specific neuronal degeneration, signaling
Better keywords: neurodegenerative diseases; CA1 region, hippocampal; okadaic acid; neurotoxins; MAP kinase signaling system; cell death
- Manuscript title:** Increases in levels of sediment transport at former glacial-interglacial transitions
Poor keywords: climate change, erosion, plant effects
Better keywords: quaternary climate change, soil erosion, bioturbation

6. Common Mistakes and Vocabulary Practice

Objectives

By the end of this module, students will be able to:

- Write clear English abstracts, avoiding Russian-style phrases and direct translations.
- Use varied and precise language, employing synonyms and academic vocabulary correctly.
- Construct grammatically correct sentences, applying proper word order, punctuation, and article usage.

Instead of writing their Abstract in English, students often directly translate their Russian «аннотация» using translation tools such as Google or Yandex translator. Why is it better to avoid this? Let us look at the potential stumbling blocks:

- Russian fixed phrases.** Academic papers and abstracts written in English typically do not include certain fixed phrases that are customary in the Russian academic tradition (e.g. «Актуальность/Новизна исследования состоит в том, что...», «Объектом/предметом исследования является», «Цели и задачи исследования», etc.). Therefore, it is advisable to avoid using such phrases in your abstract.

- **The use of synonyms.** It is important to avoid using the same word repeatedly and instead employ synonyms. They make your text more interesting and less repetitive and help you express ideas more precisely and clearly.
- **Conjunctions.** Conjunctions are important because they create logical connections between ideas, helping to organize information clearly and coherently. Very often, students either do not use conjunctions or use them incorrectly.
- **Word order.** When translating from Russian into English, students often use the Russian word order in their sentences, neglecting the fact that English requires a fixed word order.
- **Russian punctuation.** Students often resort to Russian punctuation in their texts, for example, use a dash (–) in place of a predicate or a conjunction: ‘*X can be a source of enzymes – it is easy to cultivate and research.*’
- **Ignoring articles.** Since the Russian language does not have articles, students often omit them entirely. When reviewing your text, consider whether an article is required whenever a noun appears.
- **Using single and plural forms of nouns.** It is important to remember that certain nouns, such as ‘knowledge’ and ‘research’ are uncountable in English and therefore are only used in the singular form. To express plural concepts, it is preferable to use terms such as ‘studies’, ‘research studies’, or ‘pieces of research’.

6.1 Read the following examples from abstracts written by Russian students. What is wrong with them? Improve them by replacing the underlined phrases with more precise, concise, or formal language from the box.

... a widely discussed topic...;
 The objective of this study was ...;
 Hence, the issue of ...;
 The results demonstrate ...;
 As part of our research ...;
 It is widely accepted that ...

1. Due to this, the problem of finding new producers remains a hot topic in biotechnology.
2. The aim and objectives of this work was to obtain a milk clotting enzyme from X.
3. Based on the results obtained in the course of this work, we came to the conclusion that the obtained enzyme could be used in the cheese industry.
4. It is known that many X bacteriophages use Y as a primary receptor.
5. In the course of scientific work, several recombinant bacteriophage depolymerases were obtained.

6.2 In the following extract, replace the word ‘obtain’ and its forms with more suitable synonyms from the provided list. You may need to make other changes to improve the text.

Synonym Options:

*Synthesize, produce, generate, yield, determine,
 identify, acquire, prepare, isolate*

Having analyzed a number of substances obtained at different parameters and calculated on the basis of the molecular structure of the interacting substances and the structure of the products obtained, the dependence of the composition of the substance obtained on the reaction conditions was obtained. After determining the parameters necessary to obtain X, an experiment was carried out, as a result of which Y was obtained. Further work will be aimed at obtaining the liquid glass necessary for this technology.

6.3 Replace the underlined conjunctions with the correct ones:

1. An experiment was carried out, because of which X was obtained.
2. An experiment was carried out, thanks to what X was synthesized.
3. In parallel to determine the number of cells, X culture was sown on a selective medium.

6.4 Rewrite the sentences using the correct word order:

1. As the object of research was used X and its modified strains.
2. The supernatant is removed, and to the cells is added 100 µg of carrier DNA along with the transforming DNA in minimum volume.
3. In this paper as a nanostructured filler were used graphene and nanodiamonds.

6.5 Match the neutral words with their more formal equivalents. Then use them to write five sentences about your research.

Neutral	More formal
1. in short, briefly, basically	b) attempt
2. only	c) characteristic of
3. almost/more or less	d) in sum, to sum up
4. try	e) primarily
5. mainly/mostly	f) virtually
6. typical of	g) sole(ly)

6.6 Complete the following sentences with the words: generate, turn, solid, confirm, identify, underline, character, pose, nature, focus. Make the necessary grammar changes. Translate the sentences into Russian.

1. This microorganism _____ a threat to biodiversity in the region.
2. We should _____ our attention on the most important issues.
3. The data _____ my hypothesis.
4. This issue always _____ debate among academics.
5. Progress is closely _____ with technology.
6. Let us now _____ to the subject of the research.
7. The study _____ the fact that very little research exists on the subject.
8. The study lacks _____ evidence and therefore its conclusions are doubtful.

6.7 Match the beginning of the sentence with the most appropriate ending. Then use the sentences to write about your research.

1 The study revealed a regular	a scope of your research.
2 The research focused on one particular	b awareness of the problem.
3 The writer makes a powerful	c issues facing the world today.
4 The writers take an original	d into the environmental effects of nanoparticles.
5 Until recently there was little	e approach to their theme.
6 I think you should broaden	f aspect of modern society.
7 To date, there has been little research	g pattern of changes in temperature.
8 There are many important	h case for restructuring parliament.

6.8 Prepositions. Correct the preposition errors in these sentences:

1. A lengthy discussion of the advantages of solar power is not relevant with an essay that required you to focus on wind turbines.
2. It is typical to the disease to start with an itchy rash.
3. This methodology is not appropriate about the kind of research you are planning.
4. The use of original metaphors is characteristic from the writer's style.
5. Relative with the previous attempts to explain the phenomenon, this interpretation is quite persuasive.
6. The dark hair and eyes are common for all people from the region.

6.9 Antonyms. Rewrite each sentence using the opposite of the adjective in italics:

1. Karlsson checked the figures and agreed with me that they were *accurate*.
2. The solution to the problem is a *simple* one.
3. Make *rough* calculations before you begin to write up your results.
4. The army played a *significant* role in events.
5. Hernandez prefers to discuss ideas in abstract terms.

6.10 Use the following word combinations to complete the sentences: potential problem, principal cause, qualitative approach, specific conditions, rigorous methodology, apparent discrepancy

1. There is an _____ in your figures.
2. Leakage of the condenser heat exchanger tubes can be the _____ of gross impurity ingress to the secondary circuit.
3. Lamaque devised a _____ which has since been used successfully by many other researchers in the field.
4. We spotted a _____ with our procedure and so we changed it in two areas.
5. The plant is difficult to grow and needs very _____ to survive.
6. The rates of chemical reactions can be explained by a _____.

6.11 Change the sentences using adverbs which mean the opposite of the ones in italics:

1. There were *roughly* equal numbers of male and female participants using this simulation.
2. Researchers in materials sciences, physics and interfacial science *seldom* seek to actively collaborate with those in microbiological research.
3. We investigated the problem and *initially* found some small errors in the calculations.
4. The temperature was *exactly* half a degree lower than the average.
5. Singh (1998) is *explicitly* critical of the existing theories.
6. Soil erosion is *specifically* caused by water or wind.
7. Nucleophiles are *almost always* negatively charged, and form a dative covalent (coordinate) bond with the carbon.
8. The disease is *directly* linked to environmental factors.

7. How to Write a Conference Paper (тезисы) for 'Science and Technology: 21st Century Advances'

A conference paper (тезисы) is a brief yet concise summary of your research, highlighting the main points of your scientific work, article or other research projects. Your conference paper serves as a 'business card' for your research, clearly presenting its key advantages and enabling reviewers to decide whether to publish your work and accept it for the conference. The paper for the conference 'Science and Technology: 21st Century Advances' should be one page long, including the information about the authors and references.

Structure and content. The paper's content should correspond to that of the abstract. The difference is that the abstract is written as a single paragraph, whereas the paper consists of several paragraphs. Additionally, the paper may include tables, figures, formulas, diagrams, and references. The references should include only the works cited in your paper.

What you should pay attention to when checking your conference paper:

- Your paper should comply with all submission requirements.
- Use English-language websites to verify the correct English terms for your methods and equipment. For example, “метод ИК-Фурье спектроскопии” in English is ‘Fourier-transform infrared spectroscopy (FT-IR)’, not ‘the method of IR-Fourier spectroscopy’. The same applies to equipment names, chemical laws, and similar terms.
- Check the official English names of institutions on their websites. You can find the names of some research and educational institutions of St. Petersburg in Appendix H.
- All abbreviations, except for commonly known ones, should be spelled out in full when they first occur in the text. Please check Appendix E for the list of the most popular abbreviations used in chemistry.
- If your paper includes tables or figures, refer to them in the text. For example: ‘The results are shown in Table 1’, ‘Figure 2 presents...’. If there is only one table or figure, numbers after ‘Table’ or ‘Figure’ are unnecessary.
- Write numbers correctly. The decimal symbol is a point, not a comma. For instance, 48,453 in English means forty-eight thousand four hundred fifty-three, not forty-eight point four hundred fifty-three thousandths, as in Russian. Separate numerals are often written as words rather than digits in the text, e.g., ‘seven days’ instead of ‘7 days’, and ‘three times’ instead of ‘3 times’.
- Quotation marks are ‘...’ in British English and “...” in American English.

8. A Critical Review

Objectives

By the end of this module, students will be able to:

1. Understand the purpose of a critical review.
2. Explain why critical reading of research is important.
3. Identify the main parts of a critical review.
4. Use helpful phrases to summarize a paper, evaluate its strengths and weaknesses, conclude, and connect it to their own research.
5. Analyze if a critical review's introduction has all needed information.
6. Compare evaluation sections of different reviews.
7. Determine if critical reviews offer constructive suggestions for improving papers.
8. Identify if the reviews state why they chose the article and how it connects to their own research in the conclusion.

8.1 Warming-up. Discuss the following questions:

- 1) Have you ever written a critical review?
- 2) What do you think is the difference between writing an abstract and a critical review of a research paper?
- 3) In your opinion, what makes a research paper "good" or "bad"? What criteria do you use?
- 4) Why are critical reviews important in the academic world? Who benefits from them?
- 5) Think about a research paper you've read recently. What were its strengths and weaknesses?

8.2 Now read the text about a critical review and answer the questions after it:

A critical review of a research paper evaluates the strengths and weaknesses of its ideas and content (particularly the method used and the conclusions drawn from the results). It provides description, analysis and interpretation that allow readers to assess the article's value.

You should always read research critically because it allows you to judge how reliable the results obtained are and how credible the conclusions drawn are.

If research is **credible** we can believe the results. If it is **original**, the research has not been done before. If it is **reliable**, the research could be repeated and the same results would be found. **Significant** research produces findings which are important. If research is **valid**, it tests what it claims to test. A good piece of research should be all of these things.

A critical review usually consists of **Introductory part, Summary, Evaluation and Conclusion**. In the **Introductory part** you state the title of the article, the year of publication, the title of the journal, the authors and their affiliation. Then you give a brief **summary** of the paper you have read. After that you give your opinion and **evaluate** the paper (both the positive and negative points of the research). You can include some recommendations on how the article can be improved. In **Conclusion** you should also mention why you have chosen this article for reading and how it corresponds to your research.

Questions:

- 1) What is the purpose of a critical review?
- 2) Why should you always read research critically?
- 3) What are the qualities of a good piece of research?
- 4) What are the components of a critical review?

It might be a good idea to think of the questions you want answered while reading the paper. Another recommendation is to complete the table below before you write your critical review:

Paper title, authors, journal where it was published		
Section of paper	Summary	Opinion
Introduction		
Materials and Methods		
Results		
Discussion		

A critical review should not copy the text of the article. You should write it in your own words. Learn the phrases below and use them in your critical review:

Useful phrases:

To give a brief summary of the paper:

- The text comes from... *(if you're analyzing only a part of an article)*
- The paper's title is...
- The authors are...
- It was published in... (journal, year)
- The study provides an overview of...
- The authors address the issues of...
- The text/paper describes/examines/reveals/exposes/explains/addresses/discusses/presents/covers/outlines/states/offers/considers/looks into/treats...
- The authors' attention is focused on...
- The study highlights important implications for...
- The text/paper is divided into ... parts
- The text/paper consists of...

- The first part covers the issue of...
- The second part touches on...
- The research found that...
- The research findings are...
- In the final part, the authors come to the conclusion that...
- The authors conclude that...

To evaluate strengths:

- The findings are clearly presented and supported by [data/analysis].
- The methodology employed is robust and appropriate for...
- The discussion part effectively relates the key findings to prior studies.
- This research makes a significant contribution to [field/topic].
- The theoretical framework is innovative and well-grounded.

To highlight weaknesses:

- In summary, while the study offers [specific strengths], it also has notable weaknesses such as...
- The sample size seems insufficient to draw conclusions.
- It might have been helpful to provide more details about...
- There is no explanation for the absence of [specific data/timeframe].
- The study fails to account for ...
- The analysis lacks depth in addressing (the issue of...).
- However, one problem with the research is...
- There was also no control group in the study...
- It is impossible to conclude whether...
- Furthermore, only ... were investigated.
- The results cannot be applied to...
- The study would have been more relevant if it had included...
- These findings could be strengthened by considering...
- Future research should focus on exploring...
- It would be beneficial to expand the scope of the analysis to include...

To conclude and refer to your research:

- Overall, this article presents...
- To conclude, the article provides a clear example of...
- The study is valuable from the point of view of its practical application.
- Theoretical ideas and conclusions are supported by the empirical data.
- Figures and schematic representations help readers better understand the topic.
- This article correlates with my research as...
- This paper is relevant to my research as...

8.3 Carefully read the two examples of critical reviews provided below and answer the following questions:

- 1) Does the **Introductory** part include all the required information (title, year, journal, authors, affiliation)?
- 2) Find two or three phrases used in either of the examples from:
- 3) the "To give a brief summary of the paper" list;
- 4) the "To evaluate strengths" list;
- 5) the "To highlight weaknesses" list;

- 6) the "To conclude and refer to your research" list.
- 7) Compare the **Evaluation** sections of Example 1 and Example 2. Which example offers more specific and technical critiques of the research methodology or results? Provide a specific example from each review to support your point.
- 8) Do the authors of the critical reviews offer constructive suggestions for improvement? Provide one example of such a suggestion from each review.
- 9) In the **Conclusion**, do the authors clearly state why they chose the article and how it relates to their own research?

Example 1:

The title of the paper is "A Facile Approach for Preparing MnO₂-Graphene Composite as Anode Material for Lithium-Ion Batteries". The article was published in 2015 in journal "International journal of electrochemical science". The authors are Kang Wen and a group of scientists from Central South university in China.

This research presents a new method of synthesis of an anode material for lithium-ion batteries (LIB). Lithium-ion batteries now are taking leading positions in power sources market. Their versatility allows them to be applied across a wide range of fields, including consumer electronics like smartphones and laptops, medical devices, aerospace and military equipment, and numerous other applications, underscoring their key role in modern technology. That is why nowadays many researchers try to create new electrode materials which can improve parameters of LIB. The novelty of this research is the facile method of synthesizing composite material that can provide high specific capacity and cycle stability.

The article is divided into 4 parts. In the introduction the authors emphasize the importance of lithium-ion battery technology at present and the necessity of improving its characteristics to satisfy the demands of modern electronic devices. The scientists describe advantages and disadvantages of MnO₂ and graphene as anodes to clarify the justification for choosing these types of materials and the imperative to enhance their properties through the synthesis of a composite material. Furthermore, this section provides a description of the material's structure and outlines the methods employed to characterize its electrochemical performance.

The second section, titled 'Experimental', provides all details about the materials and equipment utilized in the study. It offers a detailed description of the synthesis process, outlining each step. Additionally, it specifies the methods employed to characterize the material's structure and chemical composition. The preparation of the electrode mixture is thoroughly presented, including the precise proportions of materials and the drying conditions. Furthermore, the assembly of the testing coin cell is described, along with the methods used to evaluate the electrochemical parameters of the material.

In the section 'Results and Discussion' the authors refer back to their methodology and add more information about it, inviting to view results in graphs and images, which helps the readers to understand the findings. The XRD pattern, raman spectra and TGA curve collectively proves that the material obtained consists of graphene and MnO₂. SEM and TEM images show that the surface of graphene was decorated randomly with MnO₂ and this structure provides the ability to reduce the volume expansion of MnO₂ during the charge/discharge process. The cyclic voltammetry results reveal that while discharging there is a two-step electrochemical oxidation of Mn even though the reduction process goes only in one step. The scientists also demonstrate the results of cycling showing that material presents high specific capacity of 752 mA h g⁻¹ at a current density of 100 mA g⁻¹ after 65 cycles.

The conclusion summarizes that the MnO₂ nanoparticles are well-anchored on the graphene surface and sandwiched between layers, providing improved electrochemical performance. This makes the nanocomposite a promising electrode material for lithium-ion batteries.

Although this research is original and its findings are credible enough, I would like to give some comments that may improve the current study. Firstly, the proportion of materials used is noteworthy. Typically, no more than 5% of acetylene black is employed to enhance electrode capacity. Secondly, the drying process commences at 120 °C. However, initial drying at 80 °C is generally preferred as it facilitates the slow evaporation of the solvent, thereby preventing rapid changes that could lead to defects in the electrode structure. Lastly, the scanning rate for cyclic voltammetry was set at 1 mV/s. It is advisable to use slower scan rates, as they provide more time for electrochemical reactions to occur, resulting in sharper and more defined peaks. Thus, this will help in identifying specific redox processes and understanding the electrochemical behavior of the battery materials.

Overall, this article is a practical study that shows an original method of synthesizing a new promising anode material for lithium ion batteries. The study is valuable from the point of view of its practical application. Theoretical ideas and conclusions are supported by the empirical data. Figures and schematic representations help readers better understand the topic.

This article correlates with my research work and will be beneficial for analyzing the synthesis parameters of this material type, as well as understanding the properties that result from such synthesis method.

Example 2:

The paper's title is "Chemical Composition and Particle Size Analysis of Kaolin". The authors are Shehu Yahaya, Suzi Salwah Jikan, Nur Azam Badarulzaman and Ajiya Dahiru Adamu from universities of Malaysia and Nigeria. The article was published in Path of Science in 2017. The study discusses the physicochemical properties of kaolin, focusing on its elemental composition and particle size distribution. Kaolin is a crucial industrial material used in ceramics, paper, cosmetics, and pharmaceuticals, making its characterization essential for quality control and application suitability. The authors' attention is focused on the examination of chemical constituents and particle size distribution of kaolin from Malaysia. The paper is divided into 4 parts – introduction, materials and methods, results and discussion, and conclusion. The first part covers the issue of appliance of kaolin and explains the difference in its properties. The second part touches on kaolin's investigation via different methods for its particle size distribution. The research findings are represented by tables with characteristics of kaolin from XRF and XRD analyses. The authors conclude that kaolin is a prospective material to be used in manufacturing industry.

The employed X-ray fluorescence (XRF) is robust and appropriate for chemical composition, and laser diffraction (XRD) for particle size analysis is appropriate and well-documented. These techniques are standard for such studies, ensuring their reliability. The article provides detailed quantitative results, including SiO₂, Al₂O₃, and trace element concentrations, as well as particle size distribution (PSD) curves, which are useful for comparative studies.

However, one problem with the research is that the paper does not compare its findings with other kaolin sources. The kaolin sample's source is mentioned, but its geological context is not discussed, while it could influence its composition and particle size distribution. While trace elements are reported, their potential impact (e.g., Fe₂O₃ on whiteness, TiO₂ on refractoriness) is not analyzed, which is crucial for industrial applications. The analysis lacks depth in addressing measurement reproducibility (no multiple tests), which raises questions about data robustness. Future research should focus on application in a specific manufacturing, as there is lack of understanding of its necessity for now.

Overall, the article provides a clear example of particle size characterization of kaolin. However, while the analytical method is solid, the lack of comparative data and thorough interpretation of results reduce its impact.

This paper is relevant to my research as it represents an example of a method for measuring the particle size distribution of granular bulk materials. The comparison of the

method presented in the paper with the method we are developing will make a significant contribution to my thesis.

8.4 Write a critical review of a field-specific research paper, using the examples provided as a guide. Ensure your review includes:

1. A comprehensive introduction with the paper's title, year, journal, authors, and affiliations.
2. A clear summary of the paper's main points.
3. A detailed evaluation of the paper's strengths and weaknesses, with specific and technical critiques of its methodology and results.
4. Constructive suggestions for improvement.
5. A conclusion that explains why you chose the article and how it relates to your own research interests.

Part 2. Academic Conferences

1. Taking Part in a Conference. General Vocabulary

Objectives

By the end of this lesson, students will be able to:

1. Define the purpose and format of academic conferences.
2. Understand and use key conference-related vocabulary.
3. Read and comprehend a text about conference processes.
4. Apply vocabulary in matching, gap-fill, and summary exercises.
5. Reflect on benefits of attending/presenting at conferences.

Warm-Up Discussion

1.1 In pairs or small groups, discuss:

1. Have you ever attended a conference? How would you feel giving a talk to a large audience?
2. What benefits do conferences offer researchers and their institutions?
3. Where do you usually find information about upcoming conferences?

Vocabulary Introduction

1.2 Check your understanding of the vocabulary related to academic conferences.

annual conference	ежегодная конференция
application deadline	последний срок подачи заявки
call for papers	информационное письмо (о конференции)
chairperson and conferees	председатель и участники конференции
closing remarks	заключительная речь
coffee break	короткий перерыв / перерыв на кофе
conference proceedings	сборник материалов конференции
conference venue	место проведения конференции
keynote speaker	пленарный докладчик
multiple track conference	конференция, в которой секции идут параллельно
networking	установление контактов (нетворкинг)
opening (plenary) session	открытие конференции, пленарное заседание
panel discussion	обсуждение экспертной группы
parallel sessions	параллельные заседания
peer review	экспертная оценка (статьи)
questions from the floor	вопросы из зала
registration fee	регистрационный взнос
single track conference	конференция, в которой секции идут друг за другом
to be stressed	чувствовать напряжение
to chair a session	вести заседание
to establish contacts	устанавливать контакты
to give a paper at a conference	выступить с докладом на конференции
to give a talk to a large audience	выступить перед большой аудиторией
to present a lecture	выступить с сообщением, представлять доклад
to stay calm	не волноваться
to submit an abstract	отправить аннотацию
workshop	тренинг / практический семинар

Reading Comprehension Practice.

1.3 Read the text about annual academic conferences and answer the questions after it.

Annual Academic Conferences

Academic conferences play a crucial role in the sharing of research results and the establishment of professional contacts. Every year, many scholars take part in **annual conferences** in their field, which offer a unique opportunity to present research, learn from peers, and engage in meaningful discussions.

The process usually begins months in advance with a **call for papers**, inviting researchers to **submit an abstract** by the **application deadline**. Submitted papers undergo a **peer review** to ensure quality and relevance. Successful applicants are then invited **to give a paper at a conference** or **to present a lecture** during the event.

Conferences can be organized as a **single track conference**, where all participants attend the same sessions, or a **multiple track conference**, featuring **parallel sessions** that cover different topics simultaneously. This format allows attendees to choose sessions most relevant to their interests. The **conference venue** is often a university or another center equipped to host various activities, including the **opening (plenary) session**, **panel discussions** and **workshops**.

The **keynote speaker** usually delivers a high-profile talk to a large audience, setting the tone for the event. Throughout the conference, participants enjoy **coffee** and lunch **breaks**, which are perfect moments for **networking** and **establishing contacts** with fellow **conferees**. The **chairperson** plays a vital role in **chairing sessions** and ensuring discussions stay on track and speakers **stay calm** under pressure.

During sessions, **questions from the floor** encourage interactive dialogue, enriching the learning experience. At the end of the conference, the **closing remarks** summarize key insights and thank participants. The **conference proceedings** are often published afterward, providing a valuable resource for the wider academic community.

Attending academic conferences requires paying a **registration fee**, but the benefits—knowledge exchange, professional growth, and collaboration—are invaluable for any researcher's career.

Answer the questions:

1. What is the purpose of an annual conference in academia?
2. What is the first step researchers usually take when preparing to participate in a conference?
3. How are submitted papers evaluated before being accepted for presentation?
4. What is the difference between a single track conference and a multiple track conference?
5. Where are academic conferences typically held?
6. What role does the keynote speaker play during a conference?
7. Why are coffee and lunch breaks important during academic conferences?
8. What responsibilities does a chairperson have during a conference session?
9. How do questions from the floor contribute to the conference experience?
10. What happens during the opening (plenary) session of a conference?
11. What is the significance of the closing remarks at the end of a conference?
12. What are conference proceedings, and why are they important?
13. Why is networking considered a valuable part of attending conferences?
14. How can presenting a paper or lecture at a conference benefit a researcher's career?
15. What does it mean to "stay calm" in the context of presenting at a conference?

Vocabulary exercises

1.4 Match the vocabulary words/phrases with their correct definitions:

Part I

1. Title
2. Field
3. To take place
4. Plenary session
5. Registration fee

6. *To host a conference*

Definitions:

- A. To provide the facilities for the event to take place.
- B. The specific academic or professional category someone works in.
- C. A session where all conference participants come together to hear key presentations.
- D. To be scheduled to happen (about the event).
- E. The price required to register for a conference.
- F. The name given to something, such as a conference or a research paper.

Part II

- 7. *To cover the topics*
- 8. *To submit a paper (proposal)*
- 9. *Area of research*
- 10. *Keynote speakers*
- 11. *To fit the theme of the conference*
- 12. *The scope of the conference*

Definitions:

- G. A specific subject that a researcher focuses on.
- H. Individuals giving main presentations at a conference.
- I. To align with the topics presented at the conference (about a research).
- J. To present a written work for consideration at a conference.
- K. The range of topics that the conference will address.
- L. To deal with specific subjects by discussing or analyzing their aspects at the conference.

Part III

- 13. *Notifications of acceptance or rejection*
- 14. *The deadline for*
- 15. *Networking*
- 16. *Panel discussion*
- 17. *A workshop*
- 18. *To be affiliated with*

Definitions:

- M. A meeting where experts discuss various subjects.
- N. Information about whether your submission was accepted or not.
- O. A seminar where participants engage in interactive learning.
- P. The process of making professional connections.
- Q. The point in time by which a task must be completed.
- R. To be connected with an organization or institution.

1.5 Complete the sentences using the vocabulary from the box.

cover, title, scope, keynote, field, take place, affiliated, fit, networking, penal, workshop, submit, area of research, hosts, notifications, plenary session, deadline

- 1. The conference on climate change will _____ next month in Saint Petersburg.
- 2. The SPSIT(TU) usually _____ the annual science conference in April.
- 3. The conference will _____ various topics related to renewable energy.
- 4. Participants need to _____ their registration forms and fees by April 15th.
- 5. The _____ of the conference includes discussions on technology and innovation.
- 6. Renowned scientists will serve as the _____ speakers to inspire attendees.
- 7. My research _____ is environmental policy and sustainability.
- 8. The conference is _____ with several leading journals in the field.

9. After the review process, attendees will receive _____ about their proposal status.
10. It is important for my research to _____ the theme of the conference on sustainable development.
11. The _____ for submitting papers is May 1st.
12. Attending the conference is a great opportunity for _____ with other researchers.
13. A _____ discussion is scheduled to explore various approaches in environmental chemistry.
14. There will also be a _____ focusing on hands-on tools for researchers.
15. The conference _____ is "Innovations in Science and Technology."
16. My _____ is in chemical engineering, and I am excited to reach out to people in my field.
17. The conference will begin with a _____ featuring a keynote address by a leading expert in the field.

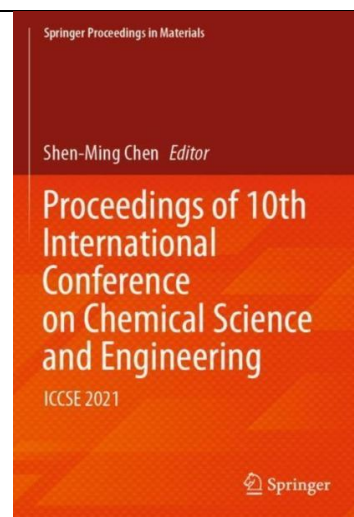
1.6 Use the words provided below to complete the sentences about submitting a conference proposal:

Findings, submission process, proposal requirements, to be peer-reviewed, referees, proofread, to be concise, descriptive title, research question, overall methodology, conclusions, implications

1. Prospective presenters are usually asked to follow the conference's specific _____ and guidelines.
2. All submissions must go through the _____, where they are evaluated by experts in the field.
3. The proposals often need _____ by members of the program committee or qualified _____.
4. It is essential to _____ the proposal carefully to eliminate typos and grammatical errors.
5. A successful proposal should include a _____ that grabs the reader's attention and clearly states the topic.
6. It is important _____ in the proposal but at the same time provide the readers with information necessary to evaluate your findings.
7. A clear _____ helps guide the research and focus the study.
8. The proposal should outline the _____ you will use to conduct your research.
9. It's essential to briefly summarize your expected _____ in the proposal's _____.
10. Make sure to state the broader _____ of your research and its contributions to the field.

1.7 Explain the meanings of the following phrases in your own words.

1. "Call for Papers" _____
2. "keynote speaker" _____
3. "peer review" _____
4. "single-track conference" _____
5. "multiple-track conference" _____
6. "to submit a proposal" _____
7. "plenary session" _____
8. "research question" _____
9. "conference proceedings" _____
10. "a referee" _____
11. "networking" _____
12. "abstract" _____
13. "affiliation" _____



<https://link.springer.com/book/10.1007/978-981-19-4290-7>

1.8 Look at the agenda of the WHO Conference on Health and Climate provided below. Was it a single track or a multiple track conference? How many plenary sessions and panel discussions did it include?

 World Health Organization	
WHO Conference on Health and Climate	
Agenda	
Day 1, 27 August 2014: Defining the agenda	
07:45 – 09:00	Registration
09:00 – 09:15	Welcome
09:30 – 10:00	Opening Plenary Session
10:00 – 10:10	Video statements by high-level supporters
10:10 – 10:20	The science of climate and climate change.
10:20 – 10:40	Current state of knowledge on climate and health
10:40 – 12:30	Panel discussion: Linking international climate, sustainable development and health policy
12:30 – 14:00	Lunch break
14:00 – 14:20	Introduction to the public health response to climate change
14:30 – 15:45	Plenary Session: A multi-stakeholder dialogue on Climate Change and Health Round Table 1 – Health Resilience
15:45 – 16:15	Coffee break

<https://storage.googleapis.com/dkiwevmjyozcfe/roundtable-meeting-format.html>

1.9 Examine the CUSME 2020 Call for Papers poster and answer the questions.

1. When and where did the conference *take place*?
2. What was the official *title* of the conference?
3. What specific *field* or *area* did the conference *focus on*? What was its *scope*?
4. Who were the *keynote speakers*?
5. What was the *registration fee* for participants?
6. Which journals was the conference *affiliated with*?
7. Do these journals have a good reputation? Why might this have been a strong selling point for the conference?
8. By when did *full papers* need to be submitted?
9. When was *the deadline for fee payment*?
10. When did participants *receive notifications regarding the acceptance* of their papers?

CUSME 2020



2020 International Conference on Urban Sustainability, Management, and Engineering

Urban Life
and Technology

20-21 January 2020,
Jakarta - Indonesia

Call for Papers

Scope

- Urban technology
- Urban planing, development, building and infrastructure
- Urban economics
- Urban policy government
- Urban lifestyle and life quality

Publication

- IOP: Earth and Environmental Science, (SCOPUS, ISI WoS and EI Compendex)
- Journal of Geographia Technica (Q3), (SCOPUS with SCIE and under collection of WoS)

Keynote Speakers



Prof. Ir. Frederik Josep Putuhena M.Sc., Ph.D.
Center for Urban Studies,
Universitas Pembangunan Jaya, Indonesia

Dr. Md Firoz Khan
Senior lecturer in the Department of Chemistry,
University of Malaya, Malaysia



Important Dates

Full Paper Submission Deadline
Notification of acceptance
Conference fee payment deadline
Camera ready paper
Conference date

18th October 2019
1st November 2019
10th November 2019
15th November 2019
20th - 21st January 2020

Conference Fee

Payment for	Early Bird Payment 10th November 2019	Regular Payment 10th December 2019	On Site Payment 20th January 2020
Presentation and Publication (Full Paper)	USD 300	USD 350	USD 400
Presentation Only (Abstract)	USD 200	USD 225	USD 250
Non Presenter (Attendees)	USD 100	USD 125	USD 150

The conference fee includes conference seminar kit, lunch, and coffee break. Valid for one person with one paper only.

Contact

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Lembaga Penelitian dan Pengabdian Masyarakat
Universitas Pembangunan Jaya
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<https://grad.rmutt.ac.th/EN/cusme-2020-urban-life-and-technology-2020-international-conference-on-urban-sustainability-management-and-engineering-20-21-january-2020-jakarta-indonesia/>

1.10 Skim the text and list the factors one should consider when choosing a conference. Pay special attention to the words in bold.

Conferences involve costs like travel, accommodation, and registration. Identify your funding sources and think if the potential benefits of a conference justify the expense.

Ensure your research **aligns with** the conference's focus. A **good fit** increases the likelihood of acceptance and relevant feedback. Consider if the conference **scope** aligns with your work.

Determine who is organizing the conference. Opt for conferences **hosted** by **reputable** universities or government agencies. Avoid **predatory conferences** focused solely on profit. **Well-established annual** conferences usually have a strong reputation and attract many attendees in your field, which provide valuable **networking** and learning opportunities.

Check if the conference has partnerships with journals that align with your publication goals. Publication opportunities are a **significant benefit**. Ensure any **affiliated journals** fit your overall **publication strategy**.

Chairing a Session

1.11 Study the key vocabulary related to the role of a chairperson at a conference. Use this vocabulary to complete the sentences below.

To keep an eye on the time

To manage questions from the floor

To ensure the event flows smoothly

To introduce speakers

1. The chairperson's main job is _____ so that everyone can enjoy the conference.
2. It's important for the chairperson _____ at the beginning of each session to help the audience know who is speaking.
3. The chairperson needs _____ to allow attendees to engage with the speakers.
4. Lastly, the chairperson must _____ to make sure the conference finishes on schedule.

1.12 Study the picture and explain the qualities a chair must have. Do you agree with them? What can you add to the list?

Qualities of a Conference Chair



1.13 Study the vocabulary that can be used by a chairperson or a moderator at a conference.

Getting attention:

- Right, is everybody here?
- Ladies and gentlemen, could I have your attention for a moment?
- Good, I think we can start.

Welcoming the audience:

- Good morning, everyone.
- I'd like to welcome you all on behalf of the department/faculty/the organizers.

Looking at the session programme:

- You can see from the programme that we have a lot to get through today.
- I'd like to remind participants that they have a maximum of 10 minutes for their presentation with 10 minutes allocated for questions at the end.
- We have made some changes to the programme.
- The presentation on ... will now take place at ...
- I would also like to bring your attention to a change in the lunchtime arrangements.

Making apologies:

- Professor ... has sent his apologies that he will not be able to make his presentation today, but instead, Professor ... has kindly agreed to bring his presentation forward from tomorrow afternoon.

Presenting the speaker:

- OK, now let's move on to the first speaker. I'm delighted to introduce Professor ... from the University of ... , who will tell us something about ...
- Ladies and gentlemen, I'd like to welcome Professor/Doctor Brown/ Mr/Ms Brown, who will speak on ...
- It is my pleasure to introduce our speaker for today, ...
- Now, we're going to hear from (name) about (topic).

Inviting the speaker:

- So, (name), the floor is yours.
- Now I'd like to hand over to you, Professor ...
- The stage is yours.

Reminding the speaker of the time limit:

- I'm sorry to interrupt you, but due to our tight schedule today speakers must stick to only ten minutes.
- I'm sorry to interrupt you, but you've just 3 minutes left to finish.

Thanking the speaker. Inviting questions:

- Thank you, (name), for this insightful presentation. Now, let's open the floor for questions.
- (Name), thank you for sharing your work with us. What questions do we have from the audience?
- Thank you, that's a fascinating topic. Now, who would like to ask a question?
- Thank you, that was a very thought-provoking presentation. Now, what questions or comments do we have for our speaker?
- Thank you for your engaging talk. Before we move on, let's take some questions from the audience. Who wants to start the discussion?

Encouraging participants to ask questions:

- As nobody seems to be ready to ask a question at the moment, I'll use this opportunity to ask the speaker if/why/how ...
- I think we need to break the ice. I have a question I'd like to ask.

Interruptions:

- I'd like to jump in and say that...
- May I just put a word in?

1.14 Look at these situations. Imagine that you are a chairperson. What phrases would you use to manage each of the following situations?

1. You're about to start the session. The room is full.
2. You've opened the discussion and nobody seems to be ready to ask a question.
3. A lively talk has just finished; you need to open the floor for another question and answer.
4. An audience member asks about a different research topic unrelated to the presentation.
5. Two participants raise their hands at the same time for a question.
6. The current speaker has already spoken 10 minutes over time.

Round-Up Discussion

1.15 Answer the following questions using the vocabulary you learnt in this unit.

1. What factors should you consider while choosing the conference?
2. What is the title of the conference you would like to attend?
3. What topics would you cover if you were to lead a panel discussion or workshop at the conference?
4. Is it easy for you to establish contacts at a conference? Are you usually the first to come and introduce yourself? What variants of 'conversation starters' can you offer to be used at the conference?

Final Homework Assignment

1.16 Research Real Conferences

Find information on three well-established conferences in your field. Complete the table below, then prepare a short in-class presentation about one of them using the vocabulary we've learned.

Conference Name	Host/ Organizer	Venue	Keynote Speaker(s) & Achievements	Themes & Fit with your Research	Affiliated Journals

1.17 Design Your Own Conference

Imagine you are organizing a new conference in your field. In a brief written outline or slide set, specify:

- Theme and title
- Dates and venue
- Keynote speakers
- Chairperson
- Session format (single-track or multiple-track)
- Social/entertainment events (e.g. welcome reception, cultural tour)
- Registration fees (standard and student rates)
- Abstract/paper submission deadline.

Be ready to resent your conference concept to the class, using key conference vocabulary (e.g. "plenary session," "peer review," "networking," etc.).

2. Giving a Paper at a Conference

Objectives:

By the end of this module, students will be able to:

1. Identify the key stages in preparing a conference paper.
2. Apply best practices for slide design and spoken delivery.
3. Engage the audience and manage attention effectively.
4. Apply techniques to understand and respond to questions clearly and confidently.

2.1 Warm-Up Discussion

1. Have you ever presented your research at an academic conference in English?
2. What might be difficult when presenting your paper in English, apart from the language difficulties you might have?

2.2 Key Tips for Conference Presentations

1. Preparation

- Choose a conference aligned with your research theme and audience profile.
- Transform your written paper into an engaging oral narrative—identify your core message.
- Research attendees' backgrounds to tailor examples and depth.
- Check all equipment in advance (laptop, projector, adapters) and save slides in multiple formats (PPTX, PDF).
- Rehearse thoroughly: time your talk, practice in front of a mirror or friend, record yourself to refine pace and clarity.

2. Structure & Content Design

- **Roadmap:** Open with a clear outline (*"My talk has three parts: background, methods, conclusions"*).
- **Beginning:** Hook listeners with a provocative question, anecdote, or striking fact.
- **Middle:** Limit yourself to two–three main points; use simple analogies to clarify complex ideas.
- **Conclusion:** Summarize key takeaways and end on a strong, positive note.
- Keep sentences short and signpost transitions (*"Moving on..."*, *"Next, we'll consider..."*).

3. Slide & Visual Guidelines

- Follow the "one visual per minute" rule — 8–10 slides for a 10-minute talk is ideal.
- Focus each slide on a single main idea; avoid dense text. Use bullets or prompts rather than full sentences.
- Ensure charts and graphics are legible from the back of the room. Highlight key data when referring (*"As you can see..."*).
- Include slide-based cues to remind you of your next point, not to replace eye contact with the audience.

4. Delivery & Audience Engagement

- Smile, speak clearly and confidently; don't rush.
- Maintain eye contact across the room to involve everyone, not just the front row.
- Use positive body language—upright posture, open gestures.
- Invite participation: pose a rhetorical question or ask for a quick show of hands.
- Anticipate questions; prepare concise answers and decide whether to take them during or after your talk.

5. Useful Phrases

Greeting and introducing yourself

- Good afternoon, everybody.
- My name is... and I'm a PhD candidate at the Department of of the Saint Petersburg State Institute of Technology.
- Let me introduce myself. My name is... and I'm from...
- It's a pleasure to welcome you today.

Introducing the topic

- I'm going to tell you about...
- I'm going to talk today about...
- My talk today is about my recent research investigating....

Signposting (giving a plan of your talk).

- I've divided my presentation into (three) parts. Firstly, I'll give you the background. Secondly, I'll discuss... Finally, I'll talk about...
- My talk is in three parts. I'll start with the background, move on to..., and finish with...
- I'll begin by explaining...
- To start with, I'll explain briefly...
- After that, I'll describe/go on to describe...
- Finally, I'll discuss/I'll conclude by discussing

Inviting questions

- If there's anything you're not clear about, feel free to ask me any questions.
- I'd be grateful if you could leave any questions to the end.
- If you have any questions, don't hesitate to ask.
- I'll be glad to answer any questions at the end of my talk.

Giving background information

- I'll give you some background
- Let's start with the background
- Let's begin by looking at...

Referring to audience's knowledge

- As you know,
- As you're aware

Referring to visuals

- If you look at the graph...
- Could I draw your attention to the chart?
- As you can see from this image,...

Switching between the parts of the presentation

- As I mentioned earlier,...
- As I have already said,...
- I'll deal with this point later.
- I will be returning to this point shortly.
- So now let's turn to the results.
- Next, we'll look at...

Concluding

- To sum up,...
- To summarize,...

- So, let me recap what I've said

Ending

- Thanks very much. Any questions?
- Well, that's all I have to say about.... Thank you for listening.
- That brings me to the end of my talk today.
- I would like to thank you for being such an attentive audience and I would be happy to answer any questions you may have.

2.3 Getting the Audience's Attention

Mini-lecture (1 min):

Winston Churchill was a great public speaker. A BBC presenter once sat next to him as the Prime Minister gave a speech in which he kept his audience hanging on every word. The presenter noticed that what appeared to be notes in his hand was in fact just a laundry slip. Later he mentioned this to Churchill. "Yes", said Churchill "It gave confidence to my audience."

Answer the question: Why might a speaker choose to use just a single prompt card instead of a full script, and how does this help the listeners and make the speaker seem more confident?

Practice Activity

2.4 Match each tip for Conference Presentation (1–9) with its illustrative example (A–I).

1. Choose an appropriate conference.
 2. Rewrite your essay for the oral medium.
 3. Begin your presentation with a roadmap.
 4. Keep your talk focused.
 5. Transitions should be clear.
 6. Follow a rule of thumb: no more than one visual for a minute of talk.
 7. Consider the audience.
 8. Find simple ways to discuss complex ideas.
 9. Keep to your time limit.
- A. Your subjects and verbs should be close to one another, your sentences brief.
 - B. Your work should fit the theme of the conference.
 - C. It is rude to other panelists to go over the allotted time.
 - D. Use simple analogies and easily grasped metaphors.
 - E. Visual aids should not overwhelm the audience and detract from the main point.
 - F. Find out about your audience's knowledge of the subject and their backgrounds.
 - G. Choose one idea and make a clear, focused argument backed up with valid evidence.
 - H. Preview the talk at the beginning and tell the audience exactly what you will be covering.
 - I. "Moving on to the next point ..." or "Now let's consider ..."

2.5 Experts say that the first few minutes of a presentation are the most important. If you are able to get the audience's attention quickly, they will be interested in what you have to say. Match the opening technique (A-D) to its example (1-4).

A. Ask a rhetorical question	B. Tell them a story
C. Start with an interesting fact	D. Give them a problem to think about

1. It might surprise you to learn that additive manufacturing has the potential to reduce material waste in aerospace by up to 80%, according to...

2. During my internship at a polymer plant, we faced a critical reactor failure. The solution we found using computational fluid dynamics was...
3. Imagine you are tasked with creating a secure, decentralized data storage system. What architectural decisions would you make?
4. Can we design a truly sustainable engine without considering the entire lifecycle impact?

2.6 Match the beginnings of the sentences (Column A) to their endings (Column B) to create English signposting sentences which correspond to the Russian sentences provided below.

Column A	Column B
1. Now, let's turn	a. let's <i>highlight</i> the main benefits again.
2. I wouldn't like	b. <i>go back to</i> what we've already covered.
3. This work is <i>significant</i>	c. to <i>focus on</i> the achieved results.
4. Let's	d. to <i>cover</i> the entire methodology in detail right now.
5. To summarize quickly,	e. given the widespread use of these materials in production.
6. Before I <i>move on to</i> the next key aspect,	f. let's <i>examine</i> potential problems and difficulties.
7. That concludes	g. to the next key aspect.
8. Now I'd like	h. I'd like to <i>summarize</i> what has been said.
9. To start,	i. the <i>review</i> of the equipment used.

1. Теперь давайте перейдем к следующему ключевому аспекту.
2. Мне бы не хотелось сейчас подробно освещать всю методологию.
3. Актуальность моей работы обусловлена широким использованием данных материалов в производстве.
4. Давайте вернемся к тому, что мы уже рассмотрели.
5. Чтобы быстро подытожить, еще раз выделим основные преимущества.
6. Прежде чем я перейду к следующему ключевому аспекту, я бы хотел подытожить сказанное.
7. Этим завершается обзор использованного оборудования.
8. Теперь я хотел бы сосредоточиться на достигнутых результатах.
9. Для начала рассмотрим потенциальные проблемы и трудности.

2.7 Fill in the gaps with the appropriate word from the list below and say what tips are to be remembered when presenting a research to the audience.

scope	tailor	stick	overwhelm	confidence	contact	clarity	supplemental
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Delivering a successful research presentation involves clear, concise, and logical communication. Here's how to make an impact:

Master your topic: Deep knowledge boosts (1) _____ and credibility.

Know your audience: (2) _____ the content to their knowledge level and expectations.

Structure logically: Follow the introduction-body-conclusion format, previewing and recapping key points for (3) _____.

Use visuals effectively: Employ visual aids sparingly (one per minute) to enhance, not (4) _____ and detract from the point.

Provide handouts: Offer concise summaries with contact information and (5) _____ details for lasting impact.

Begin with a roadmap: Outline the presentation's (6) _____ and objectives in advance.

Engage, don't read: Avoid reading slides; maintain eye (7) _____ and enthusiasm.

Stay focused: (8) _____ to the presentation you prepared and project excitement about your work.

Reading Comprehension Practice

2.8 Read the text about taking questions in English at international academic conferences and do the tasks after it. Pay special attention to the words in bold.

Taking questions

Taking questions in English at international academic conferences can be a **daunting** experience for non-native speakers due to **language barriers** and the pressure to respond clearly and confidently. Many speakers fear misunderstanding the questions because of **unfamiliar accents**, **idiomatic** expressions, or rapid speech, which can lead to **anxiety** about giving inadequate or incorrect answers. This fear is worsened by **concerns about** pronunciation and the possibility of **being judged on** language proficiency rather than content. To manage these challenges, presenters can use several techniques: asking for **clarification** or repetition of the question to ensure understanding, **paraphrasing** the question aloud before answering to **confirm accuracy**, and preparing key phrases or responses **in advance** to handle common queries. Additionally, acting calmly and confidently, using simple and clear language, and **taking brief moments** to formulate answers can help speakers communicate more effectively.

2.9 Answer the questions on the text “Taking Questions”.

1. Why can taking questions feel “daunting” for non-native speakers?
2. What three factors contribute to misunderstanding the questions?
3. Give at least three techniques the text recommends for managing Q&A (questions and answers).
4. How does paraphrasing the question help a speaker?
5. Why is it important to “take brief moments” before answering?

2.10 Decide if each statement is true or false. Correct the false ones.

1. Rapid speech from the audience can increase anxiety.
2. Preparing full scripts for answers is the best strategy.
3. Asking for repetition helps ensure understanding.
4. Speakers should avoid any pauses when answering.
5. Simple language improves audience comprehension.

2.11 Match each key term or phrase (1-6) to its definition (A-F).

1. Daunting
2. Idiomatic expression
3. Paraphrase
4. Anxiety
5. Clarification
6. Pace

- A. A short break to think before speaking
- B. A fixed phrase whose meaning isn't obvious from the words
- C. To express something in different words
- D. Feelings of worry or nervousness
- E. Something that makes you feel intimidated
- F. The speed of speech

2.12 Complete the sentences with the words from the box.

<i>idiomatic</i>	<i>anxiety</i>	<i>paraphrase</i>	<i>daunting</i>	<i>clarify</i>	<i>simple</i>
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1. Rapid questions can increase a speaker's _____.
2. Good presenters _____ questions aloud before answering.
3. Complex terms or _____ phrases may need explanation.
4. Asking "Could you _____ that point?" buys thinking time.
5. Using _____ vocabulary helps the audience follow.
6. For some, the Q&A feels _____.

2.13 Complete the following exercise using the words in the box. Then list the signposting phrases you could use to comment on your groupmates' questions during a discussion.

<i>asked</i>	<i>assure</i>	<i>get</i>	<i>glad</i>	<i>question</i>	<i>raises</i>	<i>recap</i>	<i>relevant</i>	<i>remember</i>	<i>words</i>
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Useful expressions: Commenting on questions	
Useful questions • That's a very good _____. • I'm _____ you raised that point.	Unnecessary questions • I think I covered that in Part Two, but let me just _____. • Well, you may _____ in the slide about growth forecast, I mentioned ...
Awkward questions • I'm afraid I don't have the figures with me. Can I _____ back to you on that? • So, in other _____ you're asking ... • I understand your position, but I can _____ you we've done our homework.	
Irrelevant questions • To be honest, I think that _____ a different issue. • I'm not sure that's really _____ to the discussion today.	No questions A question I'm often _____ is: 'Where do you see the firm in five years?'

Self-Reflection

2.14 Think of two questions you might be asked about your own research. Write full answers using:

- Simple language
- A paraphrase of the question at the start

Final Homework Assignment

2.15 Study the poster for the annual conference 'Science and Technology: 21st Century Advances'. Use the recommendations you have learnt in this module to make a presentation about your research for this conference.

SAINT-PETERSBURG STATE INSTITUTE OF TECHNOLOGY
(TECHNICAL UNIVERSITY)
FOREIGN LANGUAGES DEPARTMENT



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ALL POSTGRADUATES ARE WELCOME!

The time, venue and participation format will be announced later.

Part 3. Grammar

First-Semester Grammar: Tables and Diagrams with Practice Exercises

1. Verb Tenses (Active and Passive Voice). Types of Questions

Active Voice

	Время		
Характер действия	Past	Present	Future
Simple (Indefinite) факт	<i>yesterday, last week (year, summer), in 2002, two years ago, the other day, just now, used to + infinitive</i>	<i>usually, often, seldom, always, sometimes, every day (week, month), on Mondays, daily, never, regularly</i>	<i>tomorrow, next week, in a year, soon, in two days, the day after tomorrow, one of these days</i>
	He wrote a letter yesterday. Он писал (написал) письмо вчера.	He writes letters every day. Он пишет письма каждый день.	He will write a letter tomorrow. Он будет писать (напишет) письмо завтра.
Continuous (Progressive) процесс	<i>at 5 p.m., yesterday, when he came, still, while, just, as...as from...till</i>	<i>now, at the present moment, at this moment, right now, still, at 5 p.m.</i>	<i>at 5 p.m. tomorrow, when you come, from...till, still</i>
	She was writing a letter at five p.m. Она писала письмо в пять часов.	She is writing a letter (at the present moment). Я пишу письмо (в настоящий момент).	She will be writing a letter at five p.m. Она будет писать письмо в пять часов.
Perfect результат	<i>by 8 p.m. yesterday, by the time (when, before) we arrived, hardly... when, scarcely... then</i>	<i>already, yet, just, ever, never, recently, lately, several times, since, for, this is the first time, this week, today, so far, up to now</i>	<i>by 4 p.m. tomorrow, by the end of the year, by then, by the time (when, before) you come</i>
	I had written the letter by five p.m. Я (уже) написал письмо к пяти часам.	I have written the letter. Я (уже) написал письмо (к настоящему моменту).	She will have written the letter by five p.m. Она (уже) напишет письмо к пяти часам.
Perfect Continuous (Perfect Progressive) продолжительность	<i>for 2 hours when (before, until) he came, for 5 years before she resigned, by 5 p.m., by the time he met her</i>	<i>since 2 p.m., since I came here, for 2 hours already, for a long time, all day</i>	<i>for 2 hours when (before) he finds us, by 5 p.m., by the end of the year, by the time she arrives, when you come</i>
	I had been writing the letter for an hour when he came. Я писал письмо (уже) час, когда он пришел.	She has been writing the letter for an hour. Она пишет письмо (уже) час.	I will have been writing the letter for an hour when he comes. Я буду писать письмо (уже) час, когда он придет.

Passive Voice

	Время		
Характер действия	Past	Present	Future
Indefinite (Simple) факт	The article was published last year. Статья была опубликована в прошлом году.	Research results are usually published in academic journals. Результаты исследований обычно публикуются в академических журналах.	The paper will be published in a reputable journal. Статья будет опубликована в авторитетном журнале.
Continuous (Progressive) процесс	Was the file being downloaded while we were in the room? Файл скачивали, пока мы находились в номере?	Are the documents being printed now? Документы сейчас распечатывают?	-
Perfect результат	No arrangements had been made before we arrived. До нашего приезда не было сделано никаких приготовлений.	Our tests have already been checked. Наши тесты уже проверили.	The repairs will have been completed by the end of the day. К концу дня ремонтные работы будут завершены.

Types of questions

- General questions:** start with an auxiliary verb and require a response of "Yes" or "No."
Do you study at the Institute? – Yes, I do.
Have you ever been abroad? – No, I haven't.
Is there a continuous demand for the graduates of the Institute? – Yes, there is.
Does hydrogen occur in a free state in nature? – No, it doesn't.
- Alternative questions:** begin with an auxiliary verb and present a choice expressed with "or."
Will he come at 5 or at 6 o'clock?
Should we do this task orally or in written form?
Is hydrogen a heavy or a light substance?
- Special questions:** start with the specific interrogative word and require an auxiliary verb before the subject.
What were your reasons behind entering the Master's programme?
When did you take your last test?
Where do the graduates of the Institute work?
- Subject Special questions:** begin with Who, What, Which, How much/many and do not require an auxiliary verb.
What main factors affect the Institute's life?
Who takes a post-graduate course?
Which company won the contract?

How many people went to class yesterday?

5. **Tag/ Disjunctive questions:** start like a positive statement and contain a 'tail' formed by the corresponding auxiliary verb and a pronoun that replaces the subject. If the statement is affirmative, the 'tail' is negative, and vice versa.

The programmes of study are elaborated in accordance with the State Educational Standards, aren't they?

Undergraduate and postgraduate students spend a large amount of time in the library, don't they?

The applicants didn't take entrance examinations, did they?

Difficult Cases for Forming Tag Questions

1. Tag Questions with Negative Adverbs and Pronouns

When the first part of the sentence contains adverbs with negative meanings, it is considered negative. Therefore, there will be no negation in the tag. Negative expressions include: *no, neither of, none of, not many, not any, hardly (any), scarcely (any), little, few, never, barely, seldom, rarely, unlikely*; and the pronouns *nobody, anyone, none, nothing, nowhere*.

- *You can seldom see him reading, can you?*
- *Hardly any people can climb Mount Everest, can they?*
- *Few students passed the exam, did they?*
- *Nothing lasts forever, does it?*

2. Tag Questions with Indefinite Pronouns

If the first part of the sentence uses pronouns such as *everyone, everybody, someone, somebody, anyone, anybody, no one, or nobody*, then "they" is used in the tag.

- *Nobody lives here, do they?*
- *Anyone could do this, couldn't they?*

3. Tag Questions with the Pronoun "I"

When the first part contains "I", the tag will use "aren't I". The form "am I not" can also be used, but it is extremely rare. If the first part is negative (with "I"), use "am I" in the tag.

- *I am leaving tomorrow, aren't I?*
- *I am not late, am I?*

4. Tag Questions with "may"

If the first part contains "may," the tag will use "mightn't". The form "may it not" can appear, but it is very rare.

- *It may snow, mightn't it?*

5. Tag Questions with Imperative Sentences

If the first part contains an imperative (a request or command), use "will you" or "won't you" in the tag to make the request or command polite.

- *Open the door, will you (won't you)?*
- *Don't tell him about it, will you?*

6. Tag Questions with "let's" and "let"

If the first part includes "let's," the tag used will be "shall we?" Do not confuse "let's" with "let." If the question uses "let," then use "will you" or "won't you" in the tag.

- *Let's go for a walk, shall we?*
- *Let them go, won't you?*
- *Let me keep it, will you?*

7. Tag Questions with "this is" or "that is"

If the first part of the sentence starts with "this is" or "that is," then use "isn't it" in the tag.

- *That's better, isn't it?*

Exercises

1.1 Ask a question starting with a given question word or auxiliary verb. Define the type of the question.

1. Examples of software include web browsers, games, and word processors. (*What*)
2. Flash memory devices are popular ways to store data. (*Are*)
3. The history of Internet began in the US in 1969. (*When*)
4. They have already published the article. (*Have*)
5. There are a lot of things that we can do with the help of computers. (*What*)
6. She gives classes on chemistry every Tuesday at 4 p.m. (*Who*)
7. Hybrid cars are becoming popular. (*Why*)
8. Tractors have been used on farms since the start of the 20th century. (*How long*)
9. Around 1940, tractors began to use a hydraulic lift system. (*When*)
10. The results of the experiment met their expectations. (*Did*)

1.2 Ask a question to the underlined words.

1. Irrigation is necessary in parts of the world where rainfall is low.
2. They use a tunnel-boring machine to make deep tunnels in soft rock.
3. Plastics are used for fuel tanks because they are resistant to chemicals.
4. Polyester fibers are used a great deal in clothing.
5. Young people who have grown with PCs and mobile phones are often called the digital generation.
6. Teachers and students use projectors and interactive whiteboards to give presentations.
7. Silicon is the primary semiconductor in most electronics due to its unique electrical properties.
8. Engineers use finite element analysis (FEA) to predict a design's reaction to forces, heat, and vibration.
9. Photovoltaic cells convert sunlight to electricity via the photoelectric effect.
10. Modern aircraft wings use carbon fiber composites for their high strength-to-weight ratio.

***1.3 Ask disjunctive questions to the following statements.**

1. There are several types of chemical bonds.
2. The turbine generates electricity efficiently.
3. She can hardly calculate the tensile strength of the material.
4. The laboratory is equipped with advanced instruments.
5. Renewable energy sources are becoming more popular.
6. Let the simulation run for another hour.
7. That's an efficient algorithm.
8. The thermodynamic process can be complex.
9. Nobody knows how to minimize waste during production.
10. Research and development are vital for innovation.
11. The reaction rate may increase with temperature.
12. Few engineers use this software for design.
13. Everyone in this class has a strong background in fluid dynamics.
14. The viscosity of the liquid affects flow rate.
15. Let's add a catalyst to speed up the reaction.
16. The prototype went through extensive testing.
17. Everyone is happy about the decision.
18. Sorry. I hope I'm not too late.
19. I suppose I am more accurate in my measurements than George.
20. Nuclear fusion combines light nuclei to release immense energy.

1.4 Complete the sentences by putting the verb in brackets into the appropriate tense.

1. Civil engineers often (*to work*) on large construction projects that take several years to complete.
2. The design of the bridge (*to change*) after the initial reviews, so the team needs to submit an updated plan.
3. 'What she (*to do*)?' – 'She (*to prepare*) a presentation for the client meeting tomorrow.'
4. The number of renewable energy installations (*to increase*) steadily over the last decade, while fossil fuel usage (*to decline*) in many countries.
5. 'I can't talk right now. I (*to analyze*) the latest test results for my project.'
6. The research team (*to search*) for a solution to the problem of waste management in urban areas.
7. As the temperature (*to rise*), the pressure in the system increased as well, leading to potential safety risks.
8. The demand for electric vehicles (*to grow*) rapidly now, and manufacturers (*to invest*) heavily in new technologies.
9. Engineers (*to look for*) innovative ways to reduce energy consumption in buildings, which helps lower costs for homeowners.
10. Please be patient! I (*to finalize*) the report, and I will share it with you shortly.

1.5 Complete the sentences with either the Present Simple or the Future Simple to express a future action.

Rule: In English, you never use “**will**” in subordinate clauses introduced by **if, when, as soon as, while, till/until**, etc. To express future time in these clauses, use the Present Simple. Use “**will**” only in the main clause.

1. **If** you (*to spread out*) the x-axis, the data points (*to be clustered*) together (наложатся друг на друга).
2. A car (*not to have*) enough power to travel vertically **if** roads (*not to wind*) around mountains.
3. **If** you (*to use*) the pie chart, it (*to show*) how much money we're saving.
4. It (*to give*) you more power **if** the fulcrum (точка опоры рычага) (*to be closer*) to the load.
5. What you (*to do*) **when** summer (*to come*)?
6. He (*to visit*) the library more frequently **when** he (*to be*) a student.
7. **As soon as** he (*to purchase*) two ball bearings (шарикоподшипники), I (*to replace*) the broken ones.
8. Please don't forget to buy an inner race for a ball bearing **when** you (*to go*) to the hardware store.
9. You (*to convert*) 25 gallons into liters **if** you (*to multiply*) 25 by 3.78.
10. I (*to wait*) here **till** you (*to tell*) me the new measurements for the Grayline project.
11. **While** he (*to convert*) Fahrenheit into Celsius, I (*to record*) the liquid measurements in metric units.
12. You (*to have to*) use a flexible coupling (муфта) **if** the input and output shafts (*not to be*) in alignment (не сойдутся).

1.6 Complete the dialogue by putting the verb in brackets into the appropriate tense.

Engineer: Hi, I'm Ian Moore. I'm here for an interview.

Interviewer: It's nice to meet you, Ian. Tell me about your work experience.

Engineer: Well, I (a) (*to work*) for Osterbell Incorporated for twelve years.

Interviewer: What your job duties (b) (*to be*)?

Engineer: I (c) (*to assess*) old engine models. Then I (d) (*to make*) suggestions to improve them.

Interviewer: You (e) (*to design*) any engines?

Engineer: Not on my own. I (f) (*to design*) new parts for older models.

Interviewer: If we (g) (*to hire*) you, you'll design new engines as well.

Engineer: I always (h) (*to be excited*) about new projects, especially when they involve innovation.

Interviewer: That's good to hear! How many projects (i) (*to complete*) in your previous job?

Engineer: I (j) (*to complete*) over fifteen major projects, which included a lot of collaborative work.

Interviewer: And how long (k) (*to work*) in teams?

Engineer: I (l) (*to work*) in teams for most of my career, and I find it (*to help*) foster creativity and ideas.

Interviewer: Excellent! We value teamwork here. What skills (m) (*to develop*) during your time at Osterbell?

Engineer: I (n) (*to develop*) strong problem-solving skills and the ability to work under pressure, especially when deadlines are tight.

Interviewer: Sounds promising! If you (o) (*to join*) our team, I believe you (p) (*to bring*) valuable experience.

Engineer: I hope so! I always (q) (*to aim*) to contribute positively to my workplace.

1.7 In the following sentences underline the verbs and decide if they are active or passive.

1. A repeater boosts the electrical signals (ретранслятор усиливает электрические сигналы) so that longer cables can be used.
2. Open-air chemical storage has been constructed.
3. Nearly all paper can be recycled if it is sorted and contaminants are removed.
4. Geothermal energy is produced below the earth's surface.
5. The main sources of greenhouse gas emissions include fossil fuel generating plants and transportation vehicles.
6. Manufacturers choose plastic containers for many different reasons.
7. Oil was formed in underground rocks millions of years ago.

1.8 Describe the changes that have occurred in a manufacturing facility between 1995 and today. Use the Present Perfect Passive.

Example: Manual assembly lines have been automated with robotic assembly cells.

1995	today	verb
1. manual assembly lines	robotic assembly cells (цеха)	automate
2. hydraulic power tools	pneumatic power tools	replace
3. traditional welding techniques	laser welding systems	improve
4. a noisy fabrication area	sound-dampened zones	enclose
5. the batch production method	continuous flow system	convert
6. a limited spectroscopy lab	a comprehensive analytical suit	expand
7. an older filtration system	advanced membrane filters	upgrade

1.9 Make past tense questions or statements opening the brackets.

Example: 1. What was the primary material used for early aircraft frames?

1. What (be) the primary material (use) for early aircraft frames?
2. The prototype (not / test) extensively because the deadline (be) too tight.
3. The signal (disrupt) repeatedly while the technicians (calibrate) the antenna.
4. Why the assembly line (not / automate) before the new product (launch)?
5. How the first microchip (produce) in such a small size?
6. (be) the bridge (design) to withstand such strong winds?
7. The software (crash) as the engineer (install) the critical update.
8. How they (transmit) data before the internet (become) widespread?

9. (be) the faulty component (replace) before the system (restart)?
10. Who (discover) the element silicon, and when it (happen)?

1.10 Make sentences from the words in brackets. Sometimes the verb is active, sometimes passive.

1. There's somebody behind us. (I think/ we/ follow)
2. This lecture-room looks different. (you/ paint?)
3. My computer has been running very slowly recently. (I think /it / infect/ a virus)
4. My tractor (трактор) has disappeared. (I think /somebody/ take)
5. Tom gets a higher salary now. (he/ promote)
6. Ann can't use her office at the moment. (it/ redecorate)
7. The photocopier broke down yesterday, but now it's OK. (it/ work/ again; it/ repair)
8. The teacher has chosen the speakers. (two students/ select to present at the conference)
9. I registered as an author a week ago. (but no confirmation/ receive/ since then)
10. Abstracts should be submitted online. (the notification of acceptance or rejection/ send /via e-mail within a week)

1.11 A journalist is asking some questions. Complete the answers by putting the verb in brackets into the appropriate present tense in the active or passive.

1. A: How is the new software development progressing?
B: It ____ well. All the initial coding ____ now by the new junior developers. (progress) (do)
2. A: Are the user interface mockups (макеты) ready yet?
B: Not quite. They ____ currently ____ by the design team. (review)
3. A: Have you received feedback from the beta testers?
B: Yes, we have. Most of their suggestions ____ already ____ into the next build. (incorporate)
4. A: What kind of testing do you perform before launch?
B: We ____ extensive unit tests, and the system ____ also ____ through rigorous integration testing. (conduct) (put)
5. A: How long has the lead architect been working on this feature?
B: He ____ on it since the project started, but the core algorithm ____ only just _____. (work) (finalize)
6. A: Is the data migration process straightforward?
B: It ____ usually ____ smoothly, but large datasets ____ sometimes ____ with unexpected errors. (handle) (present)
7. A: When will the final project report be drafted?
B: It ____ after all the performance metrics _____. (draft) (collect)

1.12 Complete the following report of an accident which happened in a factory with the correct form of the verbs in brackets.

On Tuesday morning at approximately 10:15 AM, a critical error (a) ____ (detect) during a routine stress test on a new prototype. The engineer responsible, Mr. Davies, (b) ____ (monitor) the system's performance when he (c) ____ (observe) an unexpected surge in power consumption. He quickly (d) ____ (attempt) to override the test (прекратить испытание), but the system (e) ____ (become) unresponsive.

A loud warning alarm (f) ____ (sound), and the prototype (g) ____ automatically (shut down). The entire lab area (h) ____ (secure) (была оцеплена) within moments, and all personnel (i) ____ (instruct) to evacuate. By the time the safety team (j) ____ (arrive), the source of the anomaly (неисправности) (k) ____ already (identify) as a faulty power supply unit (блок питания).

Fortunately, no injuries (l) ____ (report). An internal investigation (m) ____ (begin) immediately to determine the root cause. It appeared that the faulty unit (n) ____ (install) only last week, and its specifications (технические характеристики) (o) ____ still (check) against the design requirements (на соответствие требованиям конструкции) when the incident occurred.

1.13 Some of the following sentences contain mistakes. Find the mistakes and correct them.

1. Have anyone had any experiences with nanoparticles sticking to glassware?
2. While they were carrying out tests in the laboratories, researchers were analyzing past results.
3. The results cannot applied to those who are older or have pre-existing health issues.
4. Did you ever ask a question on a science internet forum? If so, did your question answered?
5. The system didn't working because the loudspeaker had been wrongly connected.
6. Before factories were told to stop polluting the environment, waste was being dumped in rivers and in the sea.
7. Louis Pasteur was discovering the action of germs while he was studying fermentation in wines.
8. The production process had already been shut down when the leak in the fuel tank was found.
9. Nuclear energy began to be used from the mid-1950s.
10. In the second half of the 20th century, the electronics industry transforming the way we work in factories.

2. Relative Clauses

A **relative clause** is a part of a sentence that adds more information about a noun or pronoun. It starts with a "relative pronoun" (like *who*, *which*, *that*, *whose*, *whom*) or a "relative adverb" (like *where*, *when*, *why*).

The main difference between **defining** and **non-defining** relative clauses lies in the type of information they provide and how they are punctuated:



Defining Clauses

Provide essential information

The engineer that designed the circuit received an award.



Non-Defining Clauses

Add extra information

The main processor, which operates at 3.5 GHz, handles all core calculations.

Made with Napkin

Defining relative clauses	Non-defining relative clauses
Ограничительные	Распространительные
1) give essential information; no commas are used; introduced by 'who', 'which', 'that'; 'that' can replace 'who' or 'which'; 'who', 'which' or 'that' can be left out:	1) give extra, non-essential information; separated by commas; introduced by 'who', 'which'; 'that' cannot be used; relative pronouns can't be left out:

<i>The engineer who/that designed the circuit...; The job which/that she applied for...; ... conferences that are most appropriate...; ... fibers which have been carbonized... The assistant [that] we met was really efficient.</i>	<i>The new employee, who has a degree in ..., is...; An abstract, which will be peer-reviewed, should be submitted...; He had thousands of books, most of which he had read. The main processor, which operates at 3.5 GHz, handles all core calculations.</i>
2) can be introduced by when, where, why, whose, whom:	
<i>We provide a platform where researchers...; Do you know the reason why his proposal...? The student whose presentation we listen to ..</i>	<i>Conferences might be multiple-track, where there are several parallel sessions.</i>

Exercises

2.1 In the following text, underline the relative clauses and write defining (D) or non-defining (ND) beside each one.

The Future of Automation

The latest advancements in robotics, which are rapidly transforming industries worldwide, rely heavily on sophisticated AI algorithms. Engineers who design these complex systems often work in collaborative environments where interdisciplinary skills are highly valued. The new generation of robots that are equipped with advanced sensors can navigate challenging terrains, a capability which was once thought impossible. Many researchers, whose work focuses on human-robot interaction, believe that intuitive interfaces are crucial for widespread adoption. We recently visited a lab where a bipedal robot, which had just passed its final stability test, was learning to climb stairs. This particular robot, whose developers faced numerous challenges during its creation, represents a significant leap forward. Understanding the ethical implications, why these technologies are being developed, is as important as the technical challenges.

2.2 Choose the appropriate relative pronoun or adverb in each of the following sentences.

1. The CERN laboratory, *which/where* the World Wide Web was developed, is home to the Large Hadron Collider.
2. Tim Berners-Lee was the scientist *who/whom* developed the first web browser.
3. Consider the era *which/when* silicon-based microprocessors enabled the computing revolution.
4. Many startups prefer open-plan offices *where/which* collaborative problem-solving is encouraged.
5. *Blockchain* is a technology *where/that* records transactions across computers.
6. The Mars rover, 'Curiosity', *whose/which* primary mission is to assess habitability, sends back data.
7. The new composite material, *which/that* exhibits exceptional strength-to-weight ratio, is being developed for aerospace.
8. Catalysts are substances *who/which* increase the rate of a chemical reaction.
9. The structural analysis software, *whose/whom* algorithms simulate loads, helps engineers design buildings.
10. Green chemistry aims to design processes *that/where* reduce hazardous substances.
11. Engineers often face challenges in environments *when/where* space is limited.
12. The spectrometer is an instrument *which/who* measures the intensity of light.

2.3 Use the information in brackets to complete the following sentences.

Example: (The manufacturers provided some information.) We have used the information that ... We have used the information that the manufacturers provided.

1. (The new robotic arm can perform welding tasks.) The factory has invested heavily in the new robotic arm that _____.
2. (Data is stored on a cloud server.) The company relies on a cloud server where _____.
3. (Valves control the flow of liquids or gases.) Industrial systems often contain numerous valves which _____.
4. (Linus Torvalds created the Linux kernel.) Linus Torvalds, who _____, revolutionized the open-source software movement.
5. (The engineer's design blueprint requires approval.) The engineer whose _____, is awaiting feedback from the client.
6. (Microchips integrate millions of transistors.) Modern electronics depend on microchips that _____.
7. (The distillation column separates components based on boiling points.) The distillation column where _____, is critical for refining crude oil.
8. (The quality control system checks products.) Products that _____, rarely have defects.

2.4 Make sentences by matching the information in columns A-C and then linking it together using a relative clause with 'which' or 'who'.

Example: Penicillin, which was discovered by Alexander Fleming, revolutionized medicine by treating bacterial infections.

	A	B	C
	Subject	Additional information	Important information
	Penicillin	It was discovered by Alexander Fleming.	It revolutionized medicine by treating bacterial infections.
1	Marie Curie	It carries genetic instructions for all known organisms.	It has transformed global communication and commerce.
2	The Internet	He developed the theory of relativity.	She conducted pioneering research on radioactivity.
3	DNA	She was a Polish and French physicist and chemist.	He is considered one of the greatest physicists of all time.
4	Albert Einstein	It began as ARPANET for military and research purposes.	It led to the miniaturization of electronics.
5	The transistor	It converts light energy into chemical energy	It is vital for life on Earth.
6	Photosynthesis	It replaced vacuum tubes in electronic devices.	It forms a double helix structure.

2.5 Complete the sentences with 'who', 'which', 'that', 'whose', 'whom', 'when' or 'where'.

1. We recently hired a new research assistant, _____ possesses extensive experience in data analysis and visualization.
2. They invited me as a keynote speaker, _____ was a complete surprise.
3. The content of your presentation should have a logical flow, much like your research paper, _____ has an introduction, body and conclusion.
4. The library, _____ I usually study, is a quiet and relaxing place.
5. It is the year _____ the University was founded.

6. At the start of the conference it is a good idea to have an opening session ____ the chairperson can welcome everyone.
7. They sent me a rejection, ____ wasn't what I had expected.
8. The Universities ____ alumni include Nobel Prize laureates or Pulitzer Prize winners rank among the most prestigious academic institutions in the world.
9. There were a lot of scientists at the conference, many of ____ I had known for years.
10. Attend those conferences ____ have an established reputation.
11. It was the period ____ the first Universities appeared in Western Europe.
12. Biomimetic solar panels will be useful in developing areas ____ motor-based sun-tracking panels are not affordable.
13. Fog forms when the air cools to a point ____ it can no longer hold all of the water vapour it contains.
14. How can you find research papers ____ will be relevant to your area?

2.6 Correct the mistakes in these sentences.

1. The CEO of the company, that is a respected leader, delivered an inspiring speech.
2. You should explain why your thesis is original to scholars whom might be outside your specific area of research.
3. Following is a list of guidelines, that you should consider as you begin preparing your proposal.
4. We recognize that any successful program must be tailored for the institution when it is implemented.
5. Avoid overly specialized jargon what would only be familiar to participants in a subfield.
6. He is studying to become an engineer which is difficult.
7. It might be possible to develop a superhydrophobic material who has a surface covered in superhydrophilic bumps to trap water droplets in fog.
8. Prepare an effective presentation what will demonstrate your knowledge to the audience.
9. Narrowing the paper down to a manageable topic whose fits the time frame will make your proposal more likely to be selected.

3. Nominalization

Nominalization is turning verbs or adjectives into nouns or noun phrases.

	Verb/Adjective	Noun
Nominalization of Verbs	<i>The government <u>is introducing</u> new tax rules that should help those on low incomes.</i>	<i><u>The introduction</u> of new tax rules should help those on low incomes.</i>
Nominalization of Adjectives	<i>Liquid water must be <u>available</u> in substantial quantities. It is one of the major conditions for the development of life.</i>	<i><u>The availability</u> of liquid water in substantial quantities is one of the major conditions for the development of life.</i>

Nominalization is important for **academic writing**. It will:

- prevent you from repeating the same verb or other word again; your writing will thus be more creative, varied, and interesting.
- make your writing less personal as the focus will be on an action rather than who did it;
- shorten the sentences. It is important when you are trying to write a paper within a limited number of words.

Exercises

3.1 Read the extract from a student website to find the noun forms of the verbs in the box. Which words in the box use the same form for the verb and the noun?

analyse	collect	design	explain	form	observe	relate	run	vary
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The scientific method is a process in which experimental observations are used to answer questions. Scientists use the scientific method to search for relationships between items. That is, experiments are designed so that one variable is changed and the effects of the change observed. While the exact methodologies used vary from field to field, the overall process is the same. First, the scientist must define the question – what exactly they are trying to find out. Next comes the formation of a hypothesis, which is an idea or explanation for a situation based on what is currently known. The next stage of the method is the design of an experiment which will allow this hypothesis to be tested. Usually a primary run of the experiment is conducted, and any changes to the experimental setup made. In each experimental run, data collection takes place, followed by data analysis. Finally, the data is interpreted and from this, the scientist is able to draw conclusions.

3.2 Complete the table below using the extract from the following research paper to help you.

A promising candidate among the different adsorbent materials are activated carbons. Through activation, highly porous materials can be prepared. Due to their high porosity, activated carbon materials are able to adsorb large amounts of hydrogen. Following adsorption, hydrogen molecules can be found at two possible locations: (i) on the surface of the adsorbent, or (ii) as a compressed gas in the void space between adsorbent particles.

noun	verb	adjective
compression	compress	
	absorb	
	activate	
	-	porous

3.3 Complete the gaps in the table below.

Noun	Adjective	Noun	Adjective
approximation	approximate		particular
superiority		reason	
	strategic		synthetic
politics		economics/economy	
	industrial		cultural
exterior		average	
	high		reliable
heat		strength	
	confident		true
width		probability	
	necessary		long
danger		relevance	

3.4 Insert a suitable noun or adjective from the table in each sentence.

1. The bridge's _____ was thoroughly tested before its official opening.
2. _____ growth is vital for improving living standards.
3. The students were _____ that their project would be successful.
4. There is a strong _____ that fees will rise next year.
5. The students complained that the lecture was not _____ to their course.

6. The results are so surprising it will be _____ to repeat the experiment.
7. Regularly backing up computer files reduces the _____ of losing vital work.
8. Revising for exams is a tedious _____.
9. These data appear to be _____ and should not be trusted.
10. The _____ date of the founding of Rome is 750 BC.
11. The _____ consequences of the war were inflation and unemployment.
12. They attempted to make a _____ of all the different proposals.
13. The committee decided that the proposed timeline for the project was _____.
14. The _____ of the river makes it one of the longest in the country.

3.5 The gapped words below all describe physical or chemical properties of substances. The meaning of each word is given on the right. Complete the words with the correct vowels (a, e, i, o, u).

br_ttl_n_ss	how easily something can be broken
c_p_c_t_nc_	how well something holds an electrical charge
c_nc_ntr_t_n	how much of one substance is found in another
c_nd_ct_v_ty	how well something allows heat or electricity to go through it
d_ns_ty	how much mass a given volume of a substance has
fl_mm_b_l_ty	how easily something burns
l_m_n_nc_	how much light passes through or comes from a substance
m_ss	how much matter is in a solid object or in any volume of liquid or gas
p_rm_b_l_ty	how easily gases or liquids go through a substance
p_r_s_ty	how many small holes are in a substance
pr_ss_r_	how much force a liquid or gas produces when it presses against an area
r_ct_v_ty	how easily a chemical substance reacts
s_l_b_l_ty	how easily something can be dissolved to form a solution
v_l_c_ty	how quickly an object is travelling
v_sc_s_ty	how thick a liquid is
v_l_m_	how much space is contained within an object or solid shape

***3.6 Look at the following sentences and turn them into sentences with a nominalization.**

Example: The errors in the sentence were *removed*, which made it look much better. – *The removal* of the sentence errors made it look much better.

1. You should write how cellular components *are distributed* within a cell.
2. The number of applicants for admission *increased* significantly in 2025.
3. The University management *decided* to start the academic year in a hybrid format of training, which means the use of full-time technologies in combination with distance education and e-learning.
4. I *analysed* the data and it showed that uptake of the drug had increased.
5. Saint Petersburg State Institute of Technology *is introducing* a new course in autumn to foster inclusive design thinking among its engineering students.
6. She *was thinking* about all the work she had to do and this made her stressed.
7. The delegates at the conference questioned whether the clothes the speaker was wearing were *appropriate*.

3.7 Nominalize the italicized collective groups.

Example: The government needs to do more to help *homeless people*. – The government needs to do more to help *the homeless*.

1. *Poor people* often have a lower life expectancy than *rich people*.
2. The charity was set up to help *blind people*.
3. I've always found *French people* very friendly.

4. *Irish people* are proud of their sense of humour.

***3.8 Nominalization of phrasal verbs: change the phrasal verbs in italics into related nouns from the box and make other necessary changes.**

Nouns derived from phrasal verbs:

Verb+particle: build-up, roll-out, set-up, start-up, shutdown, take-off, touchdown, takeover

Particle+verb: input, output, inflow, outflow, overflow, upthrust, downthrust, inlet, outlet

In some cases, the verb takes the same form as the noun: *overhaul, upgrade, downgrade*

Example: Pressure *built up* (выросло) because the fluid *flowed in* too quickly and *flowed out* too slowly. – The *build-up of pressure* took place (occurred) because the *inflow of fluid* was too quick and the *outflow* too slow.

1. New software will be *rolled out* (внедрено) next week and the new system will *start up* soon.
2. Production slowed down soon after the new owner *took over* the factory.
3. Problems began soon after the aircraft *took off* and continued until it *touched down*.
4. Your equipment is very well *set up* (установлено, настроено), but the wiring needs to be *upgraded*.
5. With a computer system, if you *put* garbage *in*, the system will *put* garbage *out*.

4. Countable & Uncountable Nouns

The choice of singular and plural can be confusing in various situations, such as in the use of countable and uncountable nouns.

Uncountable nouns and irregular plurals usually have no final 's':

*Most students receive free **tuition**.*

*The main export is tropical **fruit**.*

I. Uncountable Nouns

Uncountable nouns are words that:

- do not have a plural form;
- cannot be used with 'a' or 'an' before them;
- demand a question 'how much', not 'how many';
- are used with expressions like, *some, little, much (of), (a) little of, a lot of, a piece of, most of, amount of, etc.*

Common examples of uncountable nouns

Type of Noun	Examples
Abstract concepts and physical phenomena	work, research, academia, advice, information, evidence, equipment, news, software, knowledge, advice, travel, electricity, money, logic, gravity, acceleration, pollution, feedback, traffic, radiation, biomass, lightning, progress, staff
Substances, materials and foods	air, water, wood, rubber, paper, stone, oil, blood, algae, mud, grass, seaweed, graphite, clay, quartz, rice, flour, meat
Elements, chemicals and gases	helium, iron, copper, hydrochloric acid, calcium carbonate, carbon monoxide, methane
Disciplines and fields	biology, chemistry, physics, mathematics, geography, psychology, economics, aquaculture, trigonometry

Examples:

- Academia often sets very high standards for publication.
- **X** There are ~~many academies~~ around the world.
- **✓** Instead say “many academic institutions” or “many universities”.

Many of uncountable nouns can be used with the following expressions:

A piece of advice/ evidence

An item of equipment/ work

A lot of research/ evidence/ equipment/ work

The amount of (not ~~the number of~~) research/ evidence/ equipment/ work

Not much research/ evidence/ equipment/ work

Six members of staff, many **staff members** => many **staff***

*‘Staff’ can be used as a collective noun that takes a plural verb (meaning “many employees”).

While “much stuff” (много персонала) isn't strictly incorrect in all contexts, “many staff” (много сотрудников) is more common when referring to the number of people in the group.

II. Nouns with Irregular Plurals

Some nouns in the academic English language which come from foreign languages (mainly Latin or Greek) have special plurals. **Examples:**

analysis – analyses

appendix – appendices [əˈpendɪsiːz] – приложение (к книге, документу и т.п.)

axis – axes: *time is represented by X axis; the point where the two axes meet is ...*

bacterium – bacteria

basis – bases

criterion – criteria

crisis – crises

formula – formulae /ˈfɔː.mjə.liː/ or formulas

fungus – fungi /ˈfʌŋ.ɡaɪ/

hypothesis – hypotheses

index – indices (or indexes)

matrix – matrices or matrixes

nucleus – nuclei /ˈnjuː.kli.ə/

phenomenon – phenomena

radius – radii: *the radius was measured as ...; the radii of the circular are ...*

series – series: *the objects were lined up in a series...; there are two series of lectures ...*

species – species

thesis – theses

stimulus – stimuli

stratum – strata

vertex – vertices: *each vertex represents 108 degrees in the...; The polygon had 5 vertices...*

virus – viruses

datum – data*

medium – media*: *Twitter is a good **medium** for promoting...; there are several **media** ...*

* **Data** is originally the plural of *datum* which is rarely used now. In modern English *data* can be used either as an uncountable noun (*this data is...*) or as a plural (*these data are...*).

* **Media** is originally the plural form of *medium*. The plural expression *the media* (meaning ‘radio, TV, the internet...’) is now quite often used as an uncountable noun with a singular verb.

III. Some Nouns can be both Countable and Uncountable

It depends on the meaning of the noun

	Countable Nouns	Uncountable Nouns
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Data	<i>These data illustrate why... The data from the samples are shown ...</i>	Our data is represented by...
Light	<i>Both of the lights need to be replaced. (лампа, источник света)</i>	Houseplants need the optimum amount of light to thrive. (свет)
Room	<i>We have two spare rooms. (комната)</i>	<i>There isn't much room for the equipment. (место, пространство)</i>
Paper	<i>I've read three papers on polymer synthesis. (статья; газета)</i>	Paper is made from wood pulp and degrades over time. (бумага)
Industry	<i>Many industries rely on fossil fuels. (отрасль промышленности)</i>	Government often regulates industry . (промышленность, производство)
Steel	<i>Some steels are used in the manufacture of medical instruments. (сплав стали)</i>	Is that really tempered steel ? (сталь)
Depression	<i>There are stories of great new eras and of looming depressions. (период депрессии)</i>	This depression is probably inevitable. (депрессия; понижение)
Time	<i>Loading and discharge times (сроки погрузки) for parts of the cargo...</i>	It is as old as time . (категория времени)
Interest	<i>She developed an interest in genetics. (интерес)</i>	The bank is paying 4 per cent interest . (процентная ставка)

IV. Quantifiers A few, a little / few, little

	Countable Nouns	Uncountable Nouns
мало, недостаточно	<i>Few rooms have such good modern equipment.</i>	<i>Little research has been done in this area.</i>
	<i>Isolated parts of the ocean contain fewer pollutants.</i>	<i>Isolated parts of the ocean contain less pollution.</i>
некоторое количество	<i>A few students have already submitted their works.</i>	<i>A little research has already been carried out in this area.</i>
	<i>We discovered a significant number of microorganisms in the lake.</i>	<i>We discovered a significant amount of green algae in the lake.</i>

Much, many, a lot of, lots of

	Negative sentences and questions	Positive sentences
Much Many	<i>How much calcium carbonate has been dissolved? How many students graduate each year?</i>	<i>after words like so, too, as: Too much money has been spent on this project. Take as much time as you need. I feel so much gratitude to you. They had so much so a lot fun in Spain. in a formal style 'much' and 'many' are used in positive sentences too: Much has been said about ... Many scientists believe ...</i>
A lot of/ lots of	<i>We don't have a lot of/ much time so we'll have to be quick. We don't have lots of time (less common in negative statements)</i>	<i>They have met lots of people. They have a lot of friends. I found a lot of information on the Internet. Much information ('much' is less common in positive statements in informal/general</i>

	English)
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Important Points to Understand:

- "A lot of / lots of" are the generally safe choices for both countable and uncountable nouns in positive statements. "Lots of" is slightly more informal.
- "Much / many" are most common in questions and negative statements.
- "Much / many" can also be used in positive statements, especially in formal contexts, or when preceded by "so," "too," or "as."
- In informal negative statements, all four (*much, many, a lot of, lots of*) can technically be used, though "much/many" and "a lot of" are more frequent than "lots of."

Exercises

4.1 In the following sentences, choose the correct alternative.

1. Little/ few news about the accident was released.
2. He established three successful *businesses/ business* in 2025.
3. Substantial *experiences/ experience* of report writing *are/ is* required.
4. It is often claimed that *travel broadens/ travels broaden* the mind.
5. *Paper was/ papers were* very expensive in the twelfth century.
6. How *much advice/ many advices* was he given before applying to this company?
7. She had *little interest/ few interests* outside her work.
8. *Irons were/ iron was* first powered by electricity in the twentieth century.

4.2 Choose the appropriate word to complete each sentence.

1. There aren't _____ errors in the measurement data.
A. much
B. many
C. a lot of
2. How _____ voltage do we need to power this circuit?
A. many
B. much
C. a lot of
3. The laboratory doesn't have _____ microscopes available today.
A. many
B. much
C. a lot of
4. _____ technical knowledge is required for this experiment.
A. A lot of
B. Much
C. Many
5. Are there _____ chemicals left in the storage room?
A. a lot of
B. much
C. many
6. The report contains _____ useful information about the new algorithm.
A. many
B. much
C. a lot of
7. During the test, we observed _____ anomalies in the process.
A. much
B. many
C. a lot of

8. Unfortunately, _____ equipment was damaged in the fire.
 A. a lot of
 B. many
 C. much

4.3 In each sentence, choose the correct alternative.

- The lab purchased *an item of equipment* / *one equipment* for the new experiment.
- Scientists have conducted *many researches* / *a lot of research* into renewable energy sources this decade.
- The project's success depends on *the amount of work* / *the number of work* put in by the entire team.
- We employ *many staff* / *much staff* in our engineering department.
- Before starting the new design, the junior engineers received *few advices* / *little advice* from the senior team.
- The system analyzes *the amount of traffic* / *number of traffic* on the network to prevent bottlenecks.
- There was *so a lot of* / *so much* data generated that it crashed the analysis software.
- The team spent *a lot of* / *much* time optimizing the algorithm.
- The new process produced *less defects* / *fewer defects* than the old one.
- They encountered *a little problems* / *a few problems* during the testing phase.

4.4 For each sentence, choose the correct alternative (A, B, C, or D).

- After the system crash, there was so _____ data recovered that we had to restart the simulation.
 A. few B. fewer C. little D. less
- The new algorithm requires _____ processing power than the old one to achieve the same results.
 A. fewer B. less C. little D. least
- The experiment generated less feedback from users, indicating very _____ issues.
 A. few B. fewer C. a little D. little
- Despite extensive efforts, the team made _____ progress on the complex design problem last week.
 A. few B. fewer C. little D. less
- Due to automation, the factory now requires _____ staff to operate, leading to higher efficiency.
 A. few B. fewer C. little D. less
- There are _____ known cases of this specific software bug, making it difficult to diagnose.
 A. little B. less C. few D. least
- To reduce environmental impact, the new process generates _____ pollution compared to traditional methods.
 A. fewer B. fewest C. less D. least
- The engineer found _____ information about the specific material's properties in the old database.
 A. few B. fewer C. little D. less
- After optimizing the network, there is _____ traffic congestion during peak hours.
 A. fewer B. fewest C. less D. least
- In some remote areas, _____ electricity is available, limiting technological development.
 A. few B. fewer C. little D. less
- The physicist discussed how _____ gravity would affect the behavior of liquids in space.
 A. fewer B. fewest C. little D. less
- Among all the prototypes, this one showed the _____ defects during stress testing.
 A. fewer B. fewest C. less D. least

13. Even with advanced filtration, there is still _____ radiation leakage from the containment unit.
A. few B. fewer C. little D. less
14. For the chemical reaction, you need _____ hydrochloric acid than you used in the previous trial.
A. fewer B. less C. little D. least
15. This new material uses _____ wood and more recycled composites in its construction.
A. fewer B. less C. little D. least
16. Compared to the old model, the new engine produces _____ acceleration with the same fuel input.
A. fewer B. less C. little D. least

4.5 Correct the mistakes in the following sentences if there are any.

1. I asked my friend for some advices about the best way to write a critical review.
2. Much progress has been made in a research into cancer.
3. There is much traffic in the city.
4. A research was carried out on the problem.
5. I have much time to study because I am not working right now.
6. My teacher gave me several good advices that I'll never forget.
7. Do you need information on this? I have few books on the topic if you would like to borrow them.
8. There was some important news about the conference.
9. Few is known about the real nature of the Multiverse.
10. The data suggests that the number of waste water produced in the UK in 2030 will be more than in 2029.
11. We spotted few typos in the Introduction and Discussion.
12. We also suggest that the authors dedicate little space to explaining where they think their model could be applied.
13. In fact, there is very little information on this topic.
14. There were less problems in the second test than in the first.
15. I have only few attempts left to finish this exam.
16. Few students than ever are studying languages at universities.

***4.6 Find the mistake in each of the following sentences (1-5) taking into account the following five areas of difficulty they refer to.**

Sentences (1-5)	Five areas of difficulty:
1. The proposal has both advantages and disadvantage.	Two linked nouns should agree.
2. A majority of children in Thailand is vaccinated against measles.	"A majority of" + a plural noun takes a plural verb.
3. There are few young people in rural area.	General statements normally use the plural.
4. Many places are experiencing an increase in crimes.	Some nouns in a general sense are uncountable and do not take a final 's.'
5. Each companies have their own policies.	'Each' or 'every' is followed by a singular noun and verb form.

5. Prepositions

The choice of preposition in a clause often depends on the adjective, verb or noun which comes before it.

I. Verb + preposition

Verb	Preposition
<i>care, complain, hear, know, learn, say something, talk, think, warn, wonder, worry, write</i>	about
<i>aim, fire, laugh, look, point, shout, yell</i>	at
<i>choose, decide, differentiate, distinguish</i>	between
<i>aim, apologise, apply, forgive, hope, long, prepare, search, wait, watch, wish, ask, apply, prepare, thank, pay</i>	for
<i>benefit, gather, isolate, learn, prevent, protect, separate, stop, recover, save, suffer</i>	from
<i>accuse, assure, convince, smell, approve, die,</i>	of
<i>concentrate, count, depend, insist, rely, agree</i>	on
<i>apologise, appeal, explain, present, speak, talk, write, listen, introduce, refer, respond,</i>	to
<i>agree, be, deal, go out, play, provide, stain, stay</i>	with
<i>embed, invest, involve, specialize, succeed, believe, arrive</i>	in
<i>place, load, divide</i>	into
<i>pump, transform, respond</i>	to
<i>surround, form</i>	by

II. Adjective + preposition

Adjective	Preposition
<i>anxious, annoyed, concerned, depressed, excited, upset, worried</i>	about
<i>amazed, annoyed, astonished, awful, bad, clever, efficient, excited, good, skilled, surprised, terrible, useless</i>	at
<i>amazed, annoyed, astonished, concerned, disturbed, exited, impressed, inspired, shocked, surprised</i>	by
<i>bad, concerned, good, responsible</i>	for
<i>useful, disappointed, interested</i>	in
<i>frightened, scared, terrified</i>	of
<i>aware, clever, cruel, generous, good, kind, mean, nasty, nice, polite, rude, selfish, true, typical, unkind</i>	of
<i>applicable, compared, cruel, good, kind, mean, nasty, nice, polite, rude, unkind</i>	to
<i>annoyed, angry, bored, concerned, disappointed, fed up, impressed, obsessed, pleased, satisfied, wrong</i>	with
<i>based</i>	on

III. Noun + preposition

Noun	Preposition
<i>anything, information, nothing, something</i>	about
<i>excuse, explanation, ideas, in return, need, reasons, reputation, responsibility</i>	for
<i>advance, change, decrease, development, drop, experience, fall, increase, rise, position</i>	in

<i>aim, experience, knowledge, result, understanding</i>	of
<i>research, effect, emphasis, impact, influence,</i>	on
<i>in association, experience</i>	with

Exercises

5.1 Underline the prepositions in the following text (ignoring to + infinitives).

The aim of this article is to explore developments in the chemical industry of Russia during the late twentieth and early twenty-first centuries. Growth within the sector was influenced by advances in technology as well as by increasing demand from domestic and international markets. After the dissolution of the Soviet Union, Russian chemical production underwent a process of restructuring, which led to greater integration with global supply chains. Throughout this period, companies invested in modern facilities and placed emphasis on sustainability in response to new environmental regulations. In summary, the article seeks to highlight the industry's evolution in relation to broader economic changes and Russia's position in the world market.

5.2 Complete the following text by choosing the most suitable preposition for each blank (a-q) from the box below.

among	across	between(x2)	during	for	from (x3)	in (x2)	of (x2)	on	to
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This research aims to investigate the impact (a) ___ digital literacy (b) ___ academic performance (c) ___ university students. It uses a mixed-methods approach, combining quantitative analysis (d) ___ survey data collected (e) ___ 500 undergraduates (f) ___ various disciplines, and qualitative insights derived (g) ___ focus group discussions. The study was conducted (h) ___ 2022 and 2023, capturing trends (i) ___ a critical period (j) ___ remote learning integration. Key findings indicate a strong correlation (k) ___ access (l) ___ digital resources and higher GPA*, particularly (m) ___ students (n) ___ STEM** fields. However, a notable disparity emerged (o) ___ students (p) ___ different socioeconomic backgrounds. Further research is needed to explore the mediating factors and develop targeted interventions (целенаправленные меры) to enhance digital equity (q) ___ educational settings.

* GPA: Grade Point Average, a numerical average of a student's academic grades.

**STEM: Science, Technology, Engineering, and Mathematics, a broad category encompassing these academic disciplines.

5.3 Complete the following phrases with the correct preposition.

(a) _____ the whole	(e) in support _____
(b) point _____ view	(f) _____ the other hand
(c) in respect _____	(g) _____ order to
(d) _____ spite of	(h) in accordance _____

5.4 When you are reading scientific literature, it is useful to record any verb + preposition combinations you find. Read the summary of an experiment and find at least five verb + preposition combinations.

(1) Multi-anvil high pressure apparatus was used to generate the high pressure and temperature for the specimen. (2) The powdered mineral sample was placed into a tube of rolled rhenium. (3) The rhenium was loaded into a ceramic octahedron. (4) Two tungsten-rhenium thermocouple leads were attached to the octahedron, which was surrounded by a set

of eight tungsten carbide cubes. (5) The cubes were placed into the space formed by six secondary anvils in the press, which was pumped up to the correct pressure. (6) The anvils transform the directed force of the hydraulic press to hydrostatic pressure on the sample. (7) Heating of the tube was carried out using an electrical current conducted through the anvils. (8) Temperature was controlled with a programmable temperature controller. (9) Energy diffraction patterns were collected using a germanium solid state detector. (10) The patterns were analysed to work out the material's strength.

5.5. Complete the sentences by choosing the correct preposition (a, b or c).

1. First, leucine (лейцин) *was separated* _____ other amino acids in the protein hydrolysate.
a for b from c with
2. Many bacteria, such as *Bacillus* spp., *can be isolated* _____ insects.
a from b to c with
3. Subsequently, the specimens *were embedded* _____ methylmethacrylate.
a at b in c to
4. The sections *were stained* _____ uranylacetate and lead citrate solutions.
a with b to c under
5. MIP-1 β was added to wells which had been *coated* _____ BSA.
a by b from c with
6. pH is *maintained* _____ 6.5, using Waterlife Buffer and Sera pH Minus.
a at b in c to

5.6. Complete the sentences with the correct preposition.

1. Perovskite shows no change _____ yield strength in response to temperature up to 873 K.
2. The supercritical fluid was pumped _____ the vessel which contained the solid solute.
3. We promise to respond _____ all e-mails within three days.
4. I'm doing an engineering degree, specializing _____ information technologies.
5. Our staff will provide you _____ audio guides on request.
6. I'm applying _____ internships this summer to get some work experience.
7. Solar cells that track the sun are probably more efficient _____ generating power than those in a fixed position.
8. Surface engineering of biomaterials is aimed _____ modifying the biological responses while still maintaining the mechanical properties of the implant.

Second-Semester Grammar: Tables and Diagrams with Practice Exercises

1. Modals of Deduction (Supposition)

Modal verbs	Meanings	Translation	Lexical equivalents (обязательны для заучивания!!!)	Examples
<i>may, might, could</i>	light supposition (uncertainty)	возможно; может быть	<i>perhaps, maybe, possibly, I'm not sure, I doubt; it seems; it looks like; it's not completely definite</i>	<i>She may (might) come any minute.</i> <i>It could be hard to do post-doctoral research.</i>
<i>must</i>	strong supposition (not used in negative sentences and with reference to	должно быть; вероятно; скорее всего; наверняка	<i>probably, evidently, surely, obviously, apparently, certainly to be (not) likely, to be sure, no doubt,</i>	<i>He must be delivering a lecture now.</i> BUT: <i>Evidently, she <u>didn't</u> know what to do next in her</i>

	<i>the future)</i>			work.
<i>can't/ couldn't</i>	strong negative supposition	Не может быть, чтобы...	<i>I don't believe it, I greatly doubt, impossible, unlikely, improbable, incredible, doubtful, is it possible?</i>	<i>It can't be true. Can it be true? Could such materials be informative?</i>
<i>Can...?/ Could...?</i>	disbelief, surprise, doubt; (not used in affirmative sentences)	Неужели ...?		

Note 1:

To express **negative strong supposition**, use:

1) lexical equivalents of *must*:

- Она, наверняка, **не** знала, какое направление подготовки выбрать. – **Evidently**, she *didn't know which qualification would be best for her.*

2) the verb '**to fail**':

- Он, вероятно, ничего **не** выяснил. – He **must have failed** to find it out.
- Она, должно быть, **не** совладала со своими чувствами. – She **must have failed** to control her feelings.

3) negative prefixes or suffixes or words with the opposite meaning:

- Он, должно быть, меня **неверно** понял. – He must have **misinterpreted** my words.
- Наверно, он **не** чувствовал уверенности. – He must have felt **insecure**.
- Должно быть, он **не** знал, какую возможность упускает. – He must have been **unaware** of what opportunity he was losing.

4) pronoun '**no**':

- Они наверняка **не** верят в то, что он добьется успеха. – They must have **no** confidence in his ability to succeed.

Note 2:

To express **strong supposition with reference to the future**, use: **to be likely, to be sure, probably**, etc.

- Он, должно быть, выберет карьеру ученого. – He **is likely/sure** to choose a career in science.
- Она, вероятно, продолжит свое образование за границей. – She will **probably** continue her education abroad.

Infinitive forms after modal verbs of deduction

Modal verbs	Forms of Infinitive	Examples
<i>may, might, could</i>		<i>He may/might not have noticed you.</i> <i>He may/might have overslept the lecture.</i> <i>He may/might still be waiting for us.</i> <i>You could be right.</i> <i>I could have attended the seminar if I had hurried.</i>
<i>must</i>	<i>do</i> <i>be doing</i> <i>have done</i> <i>have been doing</i>	<i>She doesn't answer the phone, she must be having lunch.</i> <i>She hasn't arrived yet. She must have missed the train.</i> <i>You look tired. You must have been working for a long time.</i> <i>This contribution is likely/sure to give us initial support.</i> <i>Evidently, she didn't know about my request.</i>
		<i>It's only 10 o'clock. He can't/couldn't be at home.</i>

Can? Could? can't, couldn't		<i>It can't/couldn't have been John you saw this morning. He had been busy all this time writing an executive summary.</i>
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Exercises

1.1 Choose the correct option and explain the meaning of the modal verb.

- 1) It may be him. He's about that height. (a)
It may be him. He's about that height. (b)
It may to be him. He's about that height. (c)
- 2) You could to be right. Let's check on the internet. (a)
You can be right. Let's check on the internet. (b)
You could be right. Let's check on the internet. (c)
- 3) It's six o'clock. She must have left work by now. (a)
It's six o'clock. She must have leave work by now. (b)
It's six o'clock. She must leave work by now. (c)
- 4) The reaction can't being complete. The yield is unexpectedly low. (a)
The reaction can't be complete. The yield is unexpectedly low. (b)
The reaction don't can be complete. The yield is unexpectedly low. (c)
- 5) Joe might be to conduct a stress test at the moment. (a)
Joe might be conducting a stress test at the moment. (b)
Joe might be conduct a stress test at the moment. (c)
- 6) That virus may of come from an app. (a)
That virus may have come from an app. (b)
That virus may to have come from an app. (c)
- 7) This writing is terrible. That can being an 'e', but I'm not sure. (a)
This writing is terrible. That can be an 'e', but I'm not sure. (b)
This writing is terrible. That could be an 'e', but I'm not sure. (c)
- 8) I don't believe it! You can't had failed your exam! (a)
I don't believe it! You can't have failed your exam! (b)
I don't believe it! You can't failed your exam! (c)

1.2. Replace the underlined words indicating light supposition (uncertainty) with appropriate modal verbs (may, might, could). Use perfect infinitive after the modal verb if the action refers to the past.

1. Perhaps the contamination in the sample came from the faulty equipment.
2. It seems that the vibrations in the machine are caused by an unbalanced rotor.
3. I'm not sure, but possibly the server is overloaded due to the increased traffic.
4. It looks like the reaction yield will be lower than expected.
5. It's not completely definite, but the project timeline will be delayed because of material shortages.
6. Maybe the stress fracture (разрушение под напряжением) was caused by metal fatigue (усталость металла).

7. I doubt the software is compatible with the new operating system.
8. Perhaps the temperature sensor is giving incorrect readings.
9. It seems that the bearing (подшипник) is worn out because of improper lubrication.
10. Possibly, the database server went down as a result of a power outage.

1.3. Paraphrase the following sentences using modal verb 'must' and appropriate infinitive forms (be, be doing, have done, have been doing) to express the meaning of strong supposition.

1. Evidently, the chemical reaction is exothermic, given the rapid temperature increase.
2. Surely, the machine is undergoing maintenance, as it is not operating.
3. It's certain that the network administrator installed the security patch (исправление для системы безопасности) last night.
4. Certainly, the engineers were testing the new catalyst all day yesterday, as the equipment was on all the time.
5. There's no doubt that the structural design is sound, given the rigorous testing.
6. Apparently, a technician recalibrated the robot arm (перекалибровал манипулятор робота) yesterday before the error occurred.
7. Obviously, the server is experiencing heavy traffic, judging by the slow response times.
8. Probably, the leak was detected a long time before the alarm went off.
9. It's likely that the bearing is defective, because of the noise it is making.
10. No doubt, the system crashed due to memory overflow (переполнение памяти).

1.4. Replace the underlined lexical equivalents with appropriate modal verbs expressing different degrees of certainty.

1. It's possible that the bridge design has a flaw, given the unexpected vibrations.
2. I'm sure that the reaction is complete, considering the color change and lack of further bubbling.
3. It's impossible that the system has been hacked; our firewall (брандмауэр) is impenetrable.
4. Perhaps the engine malfunction is due to low oil pressure.
5. Evidently, the new catalyst increased the reaction rate.
6. Is it possible that the server is down (отключен) for maintenance?
7. I greatly doubt that the material can withstand such stress.
8. Probably, there was an issue with the machine calibration that led to that error.
9. I am not sure, but there is a possibility that the server may be overloaded.

1.5. Paraphrase the following sentences to express the opposite meaning of strong negative supposition. Use can't/couldn't + appropriate infinitive form of the verb. Translate your sentences starting with the phrase «Не может быть, чтобы...»

1. The reaction must be proceeding at the optimal temperature.
2. The system must have detected the intrusion attempt.
3. The equipment must be operating according to the safety guidelines.
4. The firewall (брандмауэр) must be protecting the network from threats.
5. The machine must be properly lubricated.
6. The data backup must have been completed successfully.
7. The calculations must be accurate.

1.6. Write a modal verb of deduction to fill in the gaps.

1. He's assembled a complex machine. He _____ be really intelligent.
2. I'd provide more data here if I were you. It *looks like* it _____ sound unconvincing.
3. It _____ be him. He's away doing practical fieldwork until next week.

4. Hey guess what?! We _____ get funding for an academic project! It's *not completely definite* yet though.
5. I can't believe you've been analyzing the data all this time! You _____ be really bored.
6. She has always enjoyed using advanced simulation software. She _____ have given that up.
7. The funders have refused our grant proposal. – They _____ have found the problem raised in your proposal not very vital. I'm *nearly certain* of it.
8. They were listening to me very attentively. – *I'm not sure*, but you _____ have included the hook of the project into your report.
9. This project _____ be attractive for partners. It is not backed up by statistics.
10. _____ such format enable greater conversation between session participants? I *don't believe* it.
11. She has always liked teaching. Why hasn't she chosen academia? – *Perhaps* in industry she _____ also be supervising junior researchers.
12. I am working as a research assistant in the lab. – You _____ be doing your Master's here, part-time.

1.7. Match the sentences that have the same meaning.

1. <i>I'm sure</i> this investigation is credible.	a) This investigation <i>might not</i> be credible.
2. <i>I'm sure</i> this investigation <i>isn't</i> credible.	b) This investigation <i>must</i> be credible.
3. <i>It's possible</i> that this investigation is credible.	c) That <i>must</i> have been a credible investigation.
4. <i>Maybe</i> this investigation isn't credible.	d) This investigation <i>can't</i> be credible.
5. <i>I'm certain</i> this investigation <i>wasn't</i> credible.	e) The investigation <i>could</i> have been credible.
6. <i>Perhaps</i> that was a credible investigation.	f) This investigation <i>may</i> be credible.
7. It was <i>definitely</i> a credible investigation.	g) This investigation <i>can't</i> have been credible.

1.8. Revision. What are the lexical equivalents of modal verbs:

- a) may, might, could;
- b) can't;
- c) must?

1.9. *Translate the following Russian sentences into English.

Part I. Negative Suppositions

Each Russian sentence below expresses a negative strong supposition. Translate the sentences into English using one of the following methods:

- a) Lexical equivalent of 'must', e.g., *probably, evidently, surely, obviously, apparently, certainly (Evidently, she didn't know).*
- b) *Must + have failed to do* (with reference to the past)
- c) *Must + the word with negative prefix/suffix/opposite meaning (must be unaware)*
- d) *Must + verb + pronoun 'no' + noun (must have **no** confidence)*

1. Наверняка, этот датчик неисправен.
2. Вероятно, система не обнаружила вирус.

3. Должно быть, он не понял инструкцию.
4. Наверное, она не знает о проблеме с кодом.
5. Скорее всего, реакция не завершилась.
6. Наверняка, он не проверил расчеты.
7. Вероятно, в сплаве нет нужного элемента.

Part 2: Future Suppositions

Translate the following Russian sentences expressing a strong supposition about the future into English. Use the phrases: to be likely, to be sure, will probably, etc. in your translation.

1. Наверняка, он выберет эту специальность.
2. Вероятно, она продолжит обучение в магистратуре.
3. Должно быть, эта технология станет популярной.
4. Наверное, он получит высокую оценку за проект.
5. Скорее всего, компания разработает новое программное обеспечение.
6. Наверняка, этот материал найдет широкое применение в строительстве.
7. Вероятно, новый метод очистки воды будет эффективным.

2. Conditional Sentences

Conditional sentences are sentences that express a condition (“if” clause) and its result (main clause). They link a hypothetical or real situation with its outcome (result).

Types of Conditionals

Types	Use/ Structure	If-clause	Main Clause
Zero Conditional	Facts and general truths If + Present Simple, Present Simple	If you heat metal, If the power supply fails ,	it expands . the server doesn't work .
First Conditional	Real or likely future situations If + Present Simple, will + Base Verb	If the pressure is too high, If you don't add a catalyst,	the valve will open . the reaction won't speed up .
Second Conditional	Hypothetical present/ future situations If + Past Simple, would (could, might) + Base Verb	If we had a better reactor, If the software <u>was/were</u> bug-free, If the program <i>didn't have</i> errors,	the process would (could, might) be safer. it would (could, might) run perfectly. it would (could, might) be much faster.
Third Conditional	Hypothetical past situations If + Past Perfect, would (could, might) have + Past	If they <i>had controlled</i> the temperature, If the bolt <i>had been tightened</i> ,	the experiment would (could, might) have succeeded . it would (might) not have come loose.

	<i>Participle</i>	
Mixed Conditional	Hypothetical past + present consequence If + Past Perfect, would (could, might) + Base Verb	If they had used stronger material, the bridge would (could, might) be safe now. If the file hadn't been deleted , the program would (could, might) open now.
	Hypothetical present + past consequence If + Past Simple, would (could, might) have + Base Verb	If he wasn't/weren't so lazy, he wouldn't (might not) have failed the exam. If she spoke more confidently, her presentation would (could, might) have made a stronger impact.

Equivalents of “If” in Conditional Clauses

Equivalent	Meaning	Usage/Rules	Example
unless	‘if not’	Already contains negation; avoids negatives.	<i>Unless</i> he calls, we’ll leave. <i>Unless</i> you hurry, we’ll be late.
in case	precaution for problems	Used only for precautions against potential problems.	Take an umbrella <i>in case</i> it rains.
as long as	‘on condition that’/‘only if’	Emphasizes that something is true only if a specific condition is met. Formal. It shouldn’t be used to express time duration.	I’ll stay <i>as long as</i> you need me. I’ve been here <i>as long as</i> three hours.
provided that/ providing that	‘on condition that’/‘only if’	Used in formal contexts to introduce stipulations (requirements).	You can join the team <i>provided that</i> you pass the test. We’ll fund the project <i>providing that</i> you submit a detailed plan.
Suppose/ supposing	‘What if?’	Introduce hypothetical or imagined conditions. Commonly used in questions or informal contexts.	<i>Suppose</i> we leave early, would we make it? <i>Supposing</i> your paper is accepted, will you participate in the conference with it?

Comma Usage in Conditional Sentences

Structure pattern	Comma Rule	Example

1. If-clause first If + condition, + result	Use a comma	If the data are inconclusive, we will repeat the experiment.
2. Main clause first Result + if + condition	No comma	We will repeat the experiment if the data are inconclusive.

Notes:

- With other conditional equivalents (e.g., *provided that*, *supposing*, *in case*), follow the same pattern: *Supposing we obtain ethical approval, we can begin interviews next month.*
- For long or complex conditional clauses, you may optionally use a comma for clarity, even when the ‘if’ clause follows: *The committee will review the proposal, if it includes all required appendices.*

Exercises

2.1 Zero Conditional. Complete the sentences using the correct form of the verb.

1. If sunlight (*to hit*) a solar panel, it (*to generate*) electricity.
2. Provided carbon dioxide (*to be mixed*) with water, it (*to create*) carbonic acid.
3. If a part (*not to fit*), the machine (*to stop*).
4. The equipment (*to function*) better if it (*to be maintained*) regularly.
5. If light (*to pass*) through a prism, it (*to be separated*) into different colors.
6. If a website (*to have*) many users, it (*to require*) more server capacity.
7. If the power source (*not to be connected*), the circuit (*not to work*).
8. If the pump (*not to create*) enough pressure, the fluid (*not to flow*).

2.2 First Conditional. Complete the sentences using the correct form of the verb.

1. If we (*to use*) self-healing concrete in bridges, they (*to last*) longer.
2. Production lines (*to be*) faster and more efficient if we (*to use*) advanced robotics in manufacturing.
3. If phone screens (*to be made*) with self-healing polymers, they (*not to be scratched*) easily.
4. If you (*to use*) a weak password, your account (*to be*) easier to hack.
5. If artificial intelligence (*to be integrated*) into healthcare, diagnoses (*to be*) more accurate.
6. If we widely (*to adopt*) renewable energy technologies, pollution levels (*to decrease*) significantly.
7. Traffic flow (*to improve*) and accidents (*to be reduced*) if autonomous vehicles (*to become*) common.

2.3 Combine these sentences into one using if, unless, in case, provided, as long as to form a First Conditional sentence.

MODEL: I may finish my work early. I will go to the seminar. – *If I finish my work early, I will go to the seminar.*

1. The machine may break down. We should have a backup plan ready.
2. The team will probably meet the deadline. They will receive a bonus.
3. We must follow the instructions. We should avoid any errors.
4. The parts may not arrive on time. The project will be delayed.
5. The temperatures may rise significantly. The chemical reactions will speed up.

2.4 Second Conditional. Complete the sentences using the correct form of the verb.

1. If smart grids (*to be fully implemented*), energy distribution (*to be*) more reliable and efficient.

2. I (*to consider*) alternative solutions if I (*to have*) more budget flexibility.
3. If buildings (*can repair*) themselves completely, we (*not to need*) as many maintenance workers.
4. If you (*to have*) NASA's internet speed, you (*can download*) movies almost instantly.
5. If the designers (*to be allowed*) to work with a larger budget, they (*to create*) more innovative solutions for the project.
6. If the material (*to be*) less brittle, it (*not to crack*) under stress.

2.5 **Third Conditional**. Complete the sentences using the correct form of the verb.

1. The system (*to work*) properly, if the technician (*to install*) the correct software.
2. If the team (*to work*) together better, they (*can complete*) the project ahead of schedule.
3. If the engineers (*to have*) more funding, they (*may develop*) a more advanced prototype.
4. If the prototype (*to be built*) according to specifications, it (*to pass*) the tests.
5. The sample (*not to degrade*) if it (*to be stored*) at a lower temperature.
6. If the engineer (*to know*) about the failure earlier, the accident (*can be prevented*).

2.6 Complete the sentences using the third conditional.

1. The team didn't complete the project on time. They missed the deadline.
= If they had completed the project on time, _____ the deadline.
2. You didn't study for the exam. You failed.
= You wouldn't have failed if _____ for the exam.
3. The software didn't run correctly. The developers didn't test it properly.
= The software could have run correctly if _____ it properly.
4. The engineer didn't submit the report. The project got delayed.
= If the engineer had submitted the report, _____ delayed.
5. The laboratory didn't have enough equipment. The experiment was inconclusive.
= The experiment would have been conclusive if _____ enough equipment.
6. You didn't back up your data. You lost important files.
= You wouldn't have lost important files if _____ your data.
7. The mechanic didn't fix the issue on time. The machine broke down.
= The machine wouldn't have broken down if _____ the issue on time.
8. They didn't consider the material properties. The structure failed.
= The structure would not have failed if _____ the material properties.

2.7 **Mixed Conditional**. Complete the sentences using the correct form of the verb.

1. If the project manager (*to approve*) the changes last week, we (*to use*) the new design now.
2. If scientists (*not to develop*) self-healing materials, we (*not to have*) the potential for self-repairing structures now.
3. If engineers (*to perfect*) self-healing polymers years ago, our phone screens (*to be*) more durable now.
4. The internet (*not to be*) as developed today if the first website (CERN) (*not to be created*) in 1992.
5. If the specifications (*to be*) clearer, the team (*not to make*) so many mistakes during the assembly process.

2.8 Divide the following sentences into those belonging to real condition and unreal condition:

Real Condition	Unreal condition

1. The project will be delayed *unless* I finish this report on time.
2. The system is not working properly *if* the indicators are red.

3. *If* I were in charge of the design, I would use a more sustainable material.
4. *If* we had upgraded the software last year, we wouldn't have faced these issues.
5. *Provided* a substance is cooled to absolute zero, its molecular motion stops completely.

2.9 Discuss the difference between real and unreal conditions:

1. Can we use “would” in real conditions?
2. In which type of condition do we typically use “will”?
3. In which type of condition are present tenses typically used?
4. Can present tenses be used in unreal conditions?
5. What conjunctions can introduce conditional clauses?
6. When do we use ‘were’ in unreal condition?
7. What type of modal verbs do we typically use with unreal conditions?
8. What form will be used in the third conditional: *would conduct* or *would have conducted*?
9. What form will be used in the second conditional: *knew* or *had known*?
10. Give full forms instead of contracted in the sentence: *If he'd stayed here last week, we'd have met him.*

2.10 Choose the correct answer:

1. What would you do if you _____ a malfunction in the system?
 - a) discover
 - b) discovered
 - c) will discover
2. If I _____ the advanced software, I will improve the accuracy of the project.
 - a) use
 - b) used
 - c) will use
3. The project _____ delayed if the engineers don't submit their designs on time.
 - a) was
 - b) will be
 - c) would be
4. If I _____ you, I would check the pressure levels before proceeding.
 - a) am
 - b) were
 - c) will be
5. You don't need to install the updates as long as you _____ the latest version of the software.
 - a) have
 - b) had
 - c) will have
6. If I had known about the new regulations, I _____ made different design choices.
 - a) will have
 - b) would have
 - c) had
7. I'll submit the report provided I _____ the final results.
 - a) have
 - b) will have
 - c) had
8. If we _____ the safety protocols in advance, we wouldn't have had the accident.
 - a) had followed
 - b) followed
 - c) will follow

9. The engineers would have completed the project earlier if they _____ the necessary resources available on time.
 - a) had had
 - b) have
 - c) will have
10. If the temperature _____ too high, the reaction will not occur as expected.
 - a) is
 - b) were
 - c) would be
11. If you _____ the code before submission, you would have avoided many bugs.
 - a) test
 - b) would test
 - c) had tested
12. They would have successfully developed the product if they _____ the initial feedback sooner.
 - a) received
 - b) receive
 - c) had received
13. If I didn't have multiple deadlines this week, I _____ the opportunity to collaborate on the new research project.
 - a) will take
 - b) would take
 - c) had taken
14. The team _____ more efficient designs if they had access to better simulation tools.
 - a) would create
 - b) will create
 - c) created
15. If the client had provided the necessary data, we _____ the analysis much earlier.
 - a) could complete
 - b) can complete
 - c) would have completed

***2.11 Paraphrase the if-clause, by omitting "if" and using subject-auxiliary inversion.**

MODEL:

If I were in your position, I would pay close attention. – Were I in your position, I would pay close attention.

If it had been raining, we would have used our umbrellas. – Had it been raining, we would have used our umbrellas.

If the pressure had increased, the system would have failed.

1. If the machine were calibrated properly, it would operate more efficiently.
2. If we had conducted more tests, we would have ensured better accuracy.
3. If the components were compatible, we could assemble the device without issues.
4. If the experiment had considered all variables, the results would have been more reliable.
5. If the project had been completed on time, we would have met the client's expectations.

2.12 Match the beginnings (1-6) of the sentences with their endings (a-f).

1. If iron is exposed to oxygen and moisture,	a. unless you had been expelled for academic failure.
2. As long as the components fit together correctly,	b. we will be able to reduce workplace accidents.
3. Provided a circuit is closed,	c. you wouldn't have been so stressed at the last minute.

4. If we implement the new safety measures,	d. it rusts.
5. You could be an engineer,	e. electric current flows through it.
6. If you were more organized,	f. the assembly will proceed smoothly.

2.13 Choose the correct form of the verb to complete the following conditional sentences.

- Provided they (*to use*) these materials, the cost of production would not be expensive.
a) use b) used c) would use
- I wouldn't have lost my job if I (*not to be*) late for work.
a) were not b) had not been c) did not be
- If you (*to heat*) water it boils and turns to steam.
a) heat b) will heat c) heated
- The building will be durable if you (*to use*) these materials.
a) used b) would use c) use
- If robots became common in the home, people (*to have*) more free time.
a) would have b) will have c) had
- If you (*to prepare*) everything yesterday, you wouldn't be in a hurry right now.
a) had prepared b) prepared c) would prepare
- If they had known about this new discovery earlier, they (*to apply*) the method in their investigation.
a) had applied b) would apply c) would have applied
- If you (*to mix*) equal parts of gasoline and frozen orange juice concentrate, you can make napalm.
a) mix b) will mix c) would mix
- If analytical balance were used, the results (*to be*) much more precise.
a) were b) would be c) will be
- If the water (*to be purified*) carefully, they would not have repeated the experiment.
a) had been purified b) was purified c) is purified
- If we looked at water in a glass, it (*to appear*) optically uniform.
a) will appear b) appeared c) would appear
- Provided they (*to take into account*) the weight before heating, they would have made fewer mistakes.
a) take into account b) had taken into account c) will take into account
- If water were allowed to run back into the generator, an explosion (*may result*).
a) might result b) may result c) might have resulted
- Our grandparents (*to face*) the same problems if they had had the same technology as us.
a) would have faced b) will not raise c) would not raise
- Many teenagers would never become interested in computers if they (*not to play*) video games.
a) don't play b) did not play c) had not played
- If I (*to have*) a thousand years to live, I would become an expert in many fields.
a) had b) had had c) will have
- If the wind hadn't been so strong, the bridge (*not to collapse*).
a) would not have collapsed b) will not collapse c) had not collapsed
- Provided zinc were heated with sulphuric acid, the metal (*to replace*) hydrogen.
a) had replaced b) would have replaced c) would replace
- If I slept more, I (*not to be*) so exhausted.
a) would not have been b) would not be c) were not
- If Sia (*to learn*) everything by heart, she would have passed the exam.
a) had learnt b) learned c) would learn

3. Participle vs. Gerund

I. Participles

Participles in English may be of two kinds:

Participle I – причастие I (причастие настоящего времени) – The Present Participle

Participle II – причастие II (причастие прошедшего времени) – The Past Participle

Participle I (V+ing) – действительное причастие – действие выполняет сам предмет	Participle II (V+ed/ V3-form) – страдательное причастие – выражает пассивное и завершенное действие
<i>discussing</i> – обсуждающий <i>writing</i> – пишущий <i>summarizing</i> – суммирующий <i>referring</i> – сверяющий(ся), относящий(ся) <i>reporting</i> – передающий (сообщение) <i>organizing</i> – систематизирующий <i>socializing</i> – налаживающий контакты	<i>discussed</i> – обсужденный <i>written</i> – написанный <i>summarized</i> – суммарный, подытоженный <i>referred</i> – сверяемый, отнесенный <i>reported</i> – переданный (о сообщении) <i>organized</i> – систематизированный <i>socialized</i> – подготовленный к общению

Forms of Participle I

Participle I unlike Participle II has several forms:

Simple Active	<i>writing</i>	Recognizing a problem, the scientist makes the first step to its solution.
Simple Passive	<i>being written</i>	Being written in technical language, the manual was challenging for beginners to understand.
Perfect Active	<i>having written</i>	Having reported of his discovery of rays of unknown nature, Bequerel excited the curiosity of Marie Curie.
Perfect Passive	<i>having been written</i>	Having been written by industry experts, the report provided valuable insights into best practices.

Notes:

- Simple forms denote a *simultaneous* action (одновременное с глаголом-сказуемым действие);
- Perfect forms denote a *prior* action (действие, предшествующее тому, что выражено сказуемым) when there is a lapse of time between the two actions (когда между двумя действиями достаточно долгий промежуток времени) or when the action denoted by the participle is durative.

Having seen the student only once, the teacher didn't recognize him.

Not having seen her for a long time, I didn't recognize her.

Simple forms are mostly used to express immediate priority (немедленное предшествование) with the following verbs:

- verbs of sense perception (глаголы восприятия): *Hearing my voice, he looked back.* (Not ~~Having heard~~ my voice...)
- verbs of motion (глаголы движения): *Entering the room, she turned on the computer.* (Not ~~Having entered~~ the room...)
- some terminative verbs (некоторые предельные глаголы): *Taking the pencil, he began writing formulae.* (Not ~~Having taken~~ ...)

Participial Constructions

The Objective Participial Construction (Complex Object)	The Subjective Participial Construction (Complex Subject)	The Nominative Absolute Participial Construction	The Prepositional Absolute Participial Construction
used after verbs of: 1) sense perception; 2) mental activity; 3) wish; 4) <i>to have</i> and <i>to get</i> .	used with verbs of sense perception in the passive voice	used as an adverbial modifier of: 1) time; 2) cause; 3) condition; 4) attendant circumstances.	used as an adverbial modifier of: 1) time; 2) cause; 3) attendant circumstances.
<i>Harry heard <u>her walking</u> towards the classroom. I want the work done immediately.</i>	<i><u>He</u> was seen <u>entering</u> the laboratory.</i>	<i><u>These formalities completed</u>, they decided to apply for research funding.</i>	<i><u>With this done</u>, she corrected the order of information in a citation.</i>

The Nominative Absolute Participial Construction consists of two interdependent elements, nominal and verbal, which are in a predicative relation. The *nominal* element is a noun in the common case or a pronoun in the nominative case. The *verbal* element is participle. The nominal and the verbal elements make a syntactical complex functioning as a detached **adverbial modifier**.

The Nominative Absolute Participial Construction is used as an adverbial modifier of:

- 1) time: *These formalities completed, they decided to apply for research funding.*
- 2) cause: *The equipment being too old, we had to postpone the tests.*
- 3) condition: *Circumstances permitting, we'll start tests tomorrow.*
- 4) attendant circumstances: *Any moving object is able to do the work, the quantity of kinetic energy depending on its mass and velocity.*

The construction discussed is always separated from the rest of the sentence by commas.

In **the Prepositional Absolute Participial Construction** a noun (or a pronoun) is introduced by the preposition (mostly 'with'). The construction usually performs the function of an adverbial modifier of:

- 1) time: *With this done, she corrected the order of information in the citation.*
- 2) attendant circumstances: *She went on reading with her eyes fixed on the pages of the book.*
- 3) cause: *I can't work in such conditions with all this noise going on.*

II. Gerund

Gerund is a verb form that functions as a noun in a sentence. It is formed by adding the suffix *-ing* to the stem of the verb, and coincides in form with Participle I (the present participle). Gerund and the present participle have the same forms but are used differently.

Gerund:

- functions as a **noun** – is doing a noun job (naming an activity or thing)
***Quoting** too much was something he wanted to avoid.*
- can perform the functions of **subject and object** in the sentence (unlike Participle I)
***Speaking** is a whole lot easier than **doing**.
He liked neither **reading** aloud nor **being read** aloud to.*
- in other syntactical functions is **preceded by a preposition**.
Sharon studied those field-specific terms before using them in her essay.

Present Participle (Participle I):

- Modifies a noun describing something (is used as an attribute)
*The **shining** stars lit up the sky.*
- Forms part of a continuous verb
*She is **presenting** her paper now.*

Gerund of transitive verbs **has special forms** for the active and the passive voice.

Forms of Gerund

Simple Active	<i>speaking</i>	<i>Chemical methods of purifying water are extensively made use of.</i>
Simple Passive	<i>being spoken</i>	<i>We know of copper being used as a conductor.</i>
Perfect Active	<i>having spoken</i>	<i>He gave no sign of having seen or heard her.</i>
Perfect Passive	<i>having been spoken</i>	<i>The machine broke on account of its having been made of brittle material.</i>

Notes:

- Simple forms denote a *simultaneous* action (одновременное со сказуемым действие);
 - Perfect forms denote a *prior* action (действие, предшествующее тому, что выражено сказуемым). But there are exceptions:
- 1) **Simple forms** sometimes express a **prior action**:
 - a) after prepositions *on, upon, after, without*:
On writing the exercise, I began to doubt if it was correct. (Not On ~~having written~~...)
 - b) after the verbs *to remember, to excuse, to forgive, to thank*.
Please forgive me for not writing to you. (Not ...for not ~~having written~~...)
 - 2) **Active forms** sometimes express a **passive action**:
 - a) after verbs *to want, to need, to deserve, to require*:
This equipment required upgrading. (Not ...required ~~being upgraded~~.)
 - b) after the adjective *worth*:
These findings are worth publishing. (Not ...worth ~~being published~~.)

The Use of Gerund

Gerund is always used after:

1. Such verbs as: <i>admit, advise, anticipate, appreciate, avoid, consider, delay, deny, discuss, dislike, enjoy, excuse, fancy, finish, forgive, imagine, include, involve, justify, keep, mention, mind, miss, postpone, practice, recall, recollect, recommend, regret, resent, resist, risk, suggest, tolerate, understand, stop</i>	<i>You can hardly avoid taking part in the event. Her work involves filing and typing letters. I'll risk trying my hand at it. Would you mind shortening this part?</i>
2. Verbs followed by prepositions: <i>accuse of, apologize for, approve of, blame for, forgive for, congratulate on, depend on, hear of, inform of, insist on, object to, prevent from, result in, succeed in, thank for, think of</i>	<i>They succeeded in discovering the structure of this protein. Thank you for coming. I apologize for being so awkward. Are you thinking of carrying out the experiment again?</i>
3. Nouns used with prepositions: <i>chance of, difficulty in, experience in, idea of, importance of, interest in, means of, opportunity of, point in/of, preparation for, process of, reason for, sense of, skill in, way of</i>	<i>He has no intention of staying. The importance of being reliable. There are different ways of solving this problem. Do you have any reason for saying such a</i>

	<i>thing?</i>
4. Word combinations: <i>be afraid of, be angry for, be busy, be capable of, be disappointed with, be engaged in, be fond of, be good/clever at, be grateful for, can't stand/bear, can't help, feel like, be keen on, be pleased at, be proud of, be sorry for, be sure of, be surprised at, be worried about, be worth, be responsible for, be used to, be no use</i>	<i>It's no use worrying about it. He is interested in developing the project. He is proud of having won the grant. He is keen on making graphs. He didn't feel like presenting his findings in front of that audience.</i>
5. Prepositions: <i>after, before, besides, instead of, in spite of, on, without, by</i>	<i>They have been working for eight hours without having a break. He went to his office in spite of being ill.</i>

Exercises

3.1 Choose the correct alternative to complete the sentences using Participle I or Participle II:

1. Carried out/ Carrying out many experiments, you learn a lot in your field of science.
2. Look at the sections *highlighted/ highlighting* the words that you could use in your own project summary.
3. The equipment *installed/ installing* modernized our laboratory.
4. At the Institute there are several subjects *studied/ studying* optionally.
5. The experiment *referred /referring* to above can be easily made.
6. When *cooled/ cooling* water expands slightly.
7. Students *taken/ taking* exams next week should come to the dean's office.
8. While *made/ making* this experiment, we noticed that the substance changed its colour.
9. The experiment *spoken/ speaking* of is to be carried out again.
10. Substances *thought/ thinking* of as radioactive should be treated carefully.

3.2 Open the brackets, using the correct form of Participle I (Indefinite or Perfect, Active or Passive).

1. (*To state*) the aims of the research, the applicant explained why the topic was worth investigating.
2. (*To correct*) by the teacher, the papers were returned to the students.
3. (*To say*) what the expected outcomes of the research were, he outlined the procedures to follow.
4. (*To sing*) everywhere, the school anthem is very popular.
5. (*To learn*) the new words, he didn't find it difficult to translate the text.
6. (*To submit*) to the journal several weeks before, the paper has not yet been published.
7. (*To look*) at the list of possible headings for a CV, she was puzzled, never (*to apply*) for a job in science before.
8. (*To wait*) for half an hour in the hall, he was invited into the office.
9. (*To study*) what kind of information is to be included under each heading, he was able to say how a resume should be organized.
10. (*To number*) below from 1 to 6, the items presented the correct order of information in a citation.

3.3 Complete the sentences containing the Nominative Absolute Participial Construction by choosing the correct variant.

1. All the members ... their places, the conference began.
 - a. having taken
 - b. having taking
 - c. have taken

2. The relationship between the two concepts ..., she produced detailed graphical profiles.
 - a. having examined
 - b. having been examined
 - c. has been examined
3. The paper ... to the journal, he started preparing his presentation for the conference.
 - a. having been submitted
 - b. having been submitting
 - c. having submitted
4. It ... too difficult to verify, they stopped searching for proofs.
 - a. had been
 - b. been
 - c. being
5. There ... too many weak points in the project, they delayed delivering a presentation of their research proposal.
 - a. having being
 - b. being
 - c. have been
6. I didn't ask questions during the panel session, talking in front of the audience ... the hardest for me.
 - a. having been
 - b. being
 - c. have been
7. Negotiations ... , we'll have to make a new plan.
 - a. failing
 - b. having failed
 - c. being failing
8. Time ... , I will write a full-length paper.
 - a. permitting
 - b. being permitting
 - c. being permitted
9. The keynote speakers ... very late, we had to postpone the plenary session.
 - a. having coming
 - b. being coming
 - c. coming
10. My publication plan ... , I started looking for the journals affiliated with the conference.
 - a. been made
 - b. having been made
 - c. having been making

***3.4 Replace the underlined clauses by absolute participial constructions.**

MODEL: The science internet forum was very interesting and we decided to ask several questions. – The science internet forum being very interesting, we decided to ask several questions.

1. When the post for an online forum was written, she read it carefully to be sure that it was free of spelling errors.
2. As the research was complete, she started selecting a conference to present her findings.
3. He avoided overly specialized jargon; his audience followed his arguments with great interest.
4. Her analogies were precise. She found simple ways to discuss complex ideas.
5. Annie was presenting her point clearly and consistently. Duck was critical.
6. Practice improves writing. Write often.
7. As the delicate point had been settled, she could relax.

3.5 Complete the sentences with the correct gerunds from the box.

<i>making including anticipating</i> <i>contacting recovering delivering writing describing</i>
--

1. This scientist is fond of ____ lengthy quotes into his essays.
2. On ____ all his visual data, he made corrections to some figures.
3. Students liked their teacher's manner of ____ lectures.
4. Harvard neuroanatomist gave a talk on her experience of ____ from a stroke.
5. Her task for today was ____ several journals.
6. Prepare for the conference by ____ questions and criticism.
7. Many people think that they can't improve their ____.
8. ____ clear transitions is very important when writing a conference paper.

3.6 Choose the correct preposition and complete the sentence. Find sentences with gerundial constructions.

1. Her supervisor insisted ____ presenting the findings at the conference.
a) on b) of c) for
2. She apologized ____ disturbing us.
a) to b) about c) for
3. I object ____ using overly specialized jargon here.
a) to b) against c) from
4. I feel ____ using a metaphor here to make the audience remember the point.
a) like b) as if c) about
5. I am used ____ proofreading my papers carefully.
a) of b) to c) at
6. Do you think she will agree ____ his joining our project?
a) by b) with c) to
7. He was suspected ____ cheating in the exam.
a) in b) of c) for
8. The students complained ____ having boring lessons with that teacher.
a) at b) on c) about
9. The Dean's office should do something to prevent these students ____ skipping classes.
a) from b) against c) of
10. The teacher was not surprised ____ his doing badly in the exam.
a) at b) about c) of
11. Everybody was sure ____ her fulfilling all the necessary criteria for this job.
a) to b) in c) of
12. Using modern installations and techniques the scientists succeeded ____ working out a complicated engineering problem.
a) at b) to c) in
13. His research resulted ____ establishing a new principle.
a) on b) at c) in
14. Science aims ____ understanding the laws of the material world.
a) to b) at c) on
15. Traditionally chemists were concerned ____ measuring the properties of matter.
a) about b) with c) of

3.7 Use the proper preposition and put the verb in the correct form.

MODEL: A quantum chemist is interested ... (to construct) adequate mathematical models of atomic and molecular structure.

A quantum chemist is interested in constructing adequate mathematical models of atomic and molecular structure.

1. She apologized ... not *(to do)* a literature review.
2. I congratulated Ann ... *(to publish)* her first article.
3. Professor E. was the first to see the advantages of the new approach and insisted ... *(to make use of)* it to interpret the results.
4. Do you think she would object ... *(to describe)* her experimental set-up to colleagues?
5. I don't feel ... *(to plan)* a career in science.
6. I'm looking forward ... *(to answer)* all your questions about my poster presentation.
7. I wonder what prevented him ... *(to summarize)* all this information efficiently.
8. The supervisor didn't approve ... my *(to change)* experimental data.
9. After a long time I succeeded ... *(to make)* corrections to the graphs in my research.
10. Forgive me ... *(to interrupt)* you, but I must ask you a question.
11. Adequate theories often prevented scientists ... *(to make)* many useless experiments.
12. This group of engineers is responsible ... *(to modernize)* the laboratory equipment.
13. The advent of electronic computers aided ... *(to free)* man's brain from the labour of measurement and computation.
14. The success of any research depends largely ... *(to define)* its objective.

3.8 Identify what verbal the sentence contains – gerund or participle?

1. *Giving* an effective elevator speech is a crucial skill.
2. What is the problem that you are *addressing* in your research?
3. He ruined his eyesight by *playing* computer games.
4. In your poster refer to *supporting* charts and images.
5. *Speaking* is a whole lot easier than doing.
6. *Tailoring* the speech to the audience can turn a dissertation-length discussion into a two minute pitch.
7. Researchers are notorious for *using* jargon.
8. Spend time *planning* and *laying out* the format of your poster.
9. Nobody was surprised at John's *being* absent.
10. The handouts should have a professional, *appealing* look.
11. Analogies and strong images are effective ways of *capturing* attention.
12. *Odour-sensing* robots offer many benefits over the current use of animals in similar roles.
13. Aren't you afraid of *telling* the truth?
14. Do you have experience of *delivering* a poster presentation?
15. In the course of the *coming* week, I have three interviews to go to.

4. Infinitive

Infinitive is the non-finite form of the verb (неличная форма глагола) which names a process in a most general way. It is the initial form of the verb and its special marker is the particle "to": *to come, to inform, to know*.

I. Forms of Infinitive

Infinitive has 6 forms:

Simple Active	to do	<i>What methods will be used to prove research hypotheses?</i>
Simple Passive	to be done	<i>The experiment to be carried out is very important.</i>
Perfect Active	to have done	<i>I am glad to have taken your advice.</i>
Perfect Passive	to have been done	<i>The letter must have been delivered yesterday.</i>

Continuous	to be doing	<i>The majority of students seem to be taking at least one online course.</i>
Perfect Continuous	to have been doing	<i>For 10 days we seemed to have been working in vain.</i>

Notes:

- Simple forms (active and passive) denote a *simultaneous* action (одновременное со сказуемым действие);
- Perfect forms (active and passive) denote a *prior* action (действие, предшествующее тому, что выражено сказуемым).

Negative Infinitive Form

The negative infinitive is formed by placing '**not**' before the full infinitive: *not to do, not to come, not to follow*

Simple Active	not to do	<i>She decided not to attend the conference.</i>
Simple Passive	not to be done	<i>These documents are not to be distributed without permission.</i>
Perfect Active	not to have done	<i>I would have preferred not to have known the results yet.</i>
Perfect Passive	not to have been done	<i>The experiment was not to have been conducted without approval.</i>
Continuous	not to be doing	<i>He seems not to be following the speaker's argument.</i>
Perfect Continuous	not to have been doing	<i>They appear not to have been working on the draft for long.</i>

Notes:

- Avoid splitting **to** and **not** (i.e., **not to...** is more formal than **to not...**)
- Negative infinitives allow you to express prohibition, refusal, preference, or regret:
Employees are not to use personal devices during the meeting.
I'd prefer not to answer speculative questions.

II. The Use of Infinitive

Infinitive is used after the following verbs:

<i>agree, aim, appear, arrange, ask, choose, claim, decide, demand, deserve, expect, fail, hope, intend, learn, manage, need, offer, plan, prepare, pretend, promise, refuse, seem, tend, threaten, volunteer, want, wish, would like, would love.</i>	<i>The engineers managed to fix the complex issue within the deadline.</i> <i>The company plans to launch the updated software.</i> <i>They hope to finish the prototype by the end of the month.</i> <i>The system appeared to be working correctly after the maintenance.</i> <i>The technician needs to calibrate the equipment before starting the test.</i>
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Exercises

4.1 Use the appropriate form of the infinitive:

1. He wants his thesis (*to check*) by the professor.
2. I am glad (*to organize*) the results and discussion sections yesterday.
3. This plant is known (*to produce*) tractors.
4. He wants his project (*to implement*).
5. He seems (*to know*) French very well: he is said (*to spend*) his youth in Paris.
6. We are happy (*to invite*) to the conference.
7. That firm is reported (*to conduct*) negotiations for the purchase of chemicals.

8. The field that was abandoned is *(to take over)* by IBM.
9. Although it is a fine organization it cannot *(to expect)* to provide the atmosphere of intellectual fertility.
10. I want *(to inform)* of her arrival.
11. At that time in 1938 after the death of Rutherford brilliant physicists of Cambridge were known *(to establish)* scientific schools of their own.
12. He is known *(to work)* on the problem for many years.
13. The representative of the firm asked for the documents *(to send)* by e-mail.
14. He seems *(not to follow)* the speaker's argument.

4.2 Insert an active or a passive infinitive form of the following verbs: to invite, to use, to praise, to bother

1. She was delighted ... to the conference as a keynote speaker.
2. She was delighted ... her colleagues to the conference.
3. I hate ... you but I can't build a scatter plot alone.
4. She hated ... with questions concerning the cost of her project.
5. The room ... as a laboratory should be spacious enough.
6. His intention is ... this room as a laboratory.
7. She was anxious for us ... her contributions to research findings.
8. Her contributions to research findings should ...

***4.3 Replace the subordinate clauses by infinitives.**

1. I am proud *that I have met the deadline by completing my research on time.*
2. I am glad *that I have earned a degree from an accredited college.*
3. There is a lift in the dormitory *which takes us to the upper stories.*
4. I'm sorry *that I didn't come on Friday.*
5. We have an equipped laboratory *where you can work.*
6. We received a contribution from our partners *which will give us initial support and cover salary expenses.*
7. He bought some new equipment *which will be installed in our laboratory.*

4.4 Choose the correct form of the verb in brackets (Gerund or Infinitive) to complete each sentence.

1. The engineers decided *(use)* _____ a new material for the prototype.
2. The team is considering *(implement)* _____ a new software update.
3. We need *(calibrate)* _____ the sensor before the experiment.
4. The technician managed *(repair)* _____ the faulty equipment.
5. The company plans *(develop)* _____ a more efficient production process.
6. Avoid *(touch)* _____ the wires when the power is on.
7. They postponed *(test)* _____ the system until next week.
8. The researcher hopes *(publish)* _____ the results of the study.
9. It is important to learn *(use)* _____ the new software correctly.
10. We don't mind *(work)* _____ overtime to finish the project.
11. The committee recommended *(approve)* _____ the proposal for the new project.
12. The manufacturer offers *(provide)* _____ technical support for the equipment.
13. The software seems *(work)* _____ efficiently on the new hardware.
14. They postponed *(release)* _____ the new version of the software due to some bugs.
15. We appreciate *(receive)* _____ your feedback on the design.

5. Complex Object

Complex Object (The Objective-with-the-Infinitive Construction)

Examples: *The scientists knew these gases to be inert.*

I believed him (her, them, us, me) to know much in this field of science.

The Objective-with-the-Infinitive Construction is used after the verbs of:

1) sense perception (!!! after these verbs the infinitive is used without 'to')	hear, see, watch, feel, observe, notice	<i>They <u>heard</u> him walk up the stairs. I <u>watched</u> my colleague reinstall the application.</i>
2) mental activity	think, expect, understand, suppose, know, find, consider, believe, assume, mean	<i>I always thought the failure of our group to be a disaster for Princeton.</i>
3) declaring	say, ask, inform, confirm, declare, claim, report	<i>The scientist reported his thesis to be published soon.</i>
4) wish and intention	want, wish, desire, would like,	<i>My parents wanted me to enter the technical university.</i>
5) feeling and emotion	like, dislike, love, hate	<i>Technicians hate the equipment to malfunction during critical tests.</i>
6) order and permission	order, command, require, allow, !!! let (+ infinitive without 'to')	<i>The system commands the robot arm to move. The software update <u>let</u> users access new features.</i>
7) compulsion	have, urge, cause !!! make (+infinitive without 'to')	<i>The intense heat <u>made</u> the metal expand.</i>
8) some verbs with prepositions	rely on, count upon	<i>I count upon him to help. I rely upon you to do it in time.</i>

The for-to-Infinitive Construction

Examples: *It is easy for you to say.*

The temperature was too low for the reaction to proceed.

For the theory to be proved correct it was necessary to implement it.

He asked for the letter to be sent.

Exercises

5.1 Put "to" before the infinitive where it is necessary. Point out the Objective-with-the-Infinitive Construction. State after what verbs it is used.

1. The manager allowed the engineers (use) _____ the new software.
2. We heard the alarm (sound) _____ when the temperature exceeded the limit.
3. The power surge caused the system (shut down) _____.
4. The engineers saw the pressure (increase) _____ rapidly during the test.
5. We expect the new material (perform) well under these conditions.
6. You can feel the vibration (decrease) _____ as the machine warms up.
7. The supervisor made the technician (check) _____ the wiring.
8. The security system let the authorized personnel (access) _____ the data.
9. The data confirmed the hypothesis (be) _____ correct.
10. We noticed the system (respond) _____ faster after the upgrade.
11. They would like the machine (operate) _____ continuously.

12. They watched the robot arm (*perform*) _____ the assembly task.
13. The intense pressure made the material (*deform*) _____.
14. The updated protocol requires the users (*authenticate*) _____ their access.
15. Researchers observed the chemical reaction (*change*) _____ colour over time.

5.2 Paraphrase using the Objective-with-the-Infinitive Construction.

MODEL: *They supposed that the machine was working correctly. – They supposed the machine to be working correctly.*

1. They believed that the new software would improve efficiency.
2. The engineer confirmed that the design was feasible.
3. She reported that the calculations were accurate.
4. The team assumed that the process was too slow.
5. We found that the results were consistent with our hypothesis.
6. The project manager wanted that we start the testing phase.
7. They expect that the project will be completed on time.
8. He confirmed that the equipment was functioning properly.
9. We waited that they would begin the conversation.
10. The technician thought that the system was running smoothly.

5.3 Complete the sentences using the for-to-Infinitive Construction.

1. There was no reason for (she)
2. It was necessary for (they)
3. There were no exceptions for (he)
4. It is advisable for (our company)
5. It is obligatory for (the students)
6. It seemed natural for (they)

6. Complex Subject

I. Complex Subject (The Subjective-with-the-Infinitive Construction)

Examples: *The changes are reported to have already taken place.*

He is expected to present his project in an orderly and consistent way.

The Subjective-with-the-Infinitive Construction is used with:

• some verbs in the Passive Voice:		
1) sense perception	<i>hear, see, watch, feel, observe</i>	<i>Vapor was seen to disappear.</i>
2) mental activity	<i>think, expect, understand, suppose, know, find, consider, believe, assume, mean</i>	<i>Laser is known to be used in CD/DVD drives.</i>
3) declaring	<i>say, ask, inform, declare, claim, report</i>	<i>His article is reported to be published soon.</i>
4) compulsion	<i>make</i>	<i>He was made to admit his fault.</i>
• some verbs in the Active Voice:	<i>to seem / to appear; to happen / to chance; to prove / to turn out;</i>	<i>He doesn't seem to know this subject.</i>
• some modal phrases:	<i>to be sure, to be likely / unlikely; to be certain.</i>	<i>He is unlikely to answer the questions about OS functioning.</i>

II. Bare Infinitive

In modern English Infinitive is chiefly used with the particle **to** – the formal sign of Infinitive. Still there are cases when bare infinitive is used:

1. After auxiliary verbs: *He didn't **mind** working overtime to finish the project.*
2. After modal verbs: *I could **use** Python when I was 15.*
3. After the verbs of sense perception (**see, hear, feel, watch, notice**, etc.) and the verbs **make** (=force) and **let** in the Objective-with-the-Infinitive Construction:

*I've never seen him **use** this new application.*

*If he calls, let me **know**.*

*Please, don't make me **update** the video card drives now!*

NB! After these verbs in the Passive the to-infinitive is used:

*The component was made **to withstand** extreme temperatures.*

*The data was seen **to fluctuate** unexpectedly during the test.*

4. After the verbs **need** and **dare** (in the modal meaning):

*You needn't calibrate the sensor if it was done yesterday. (But this is less common than
You don't need to calibrate...)*

*He daren't try to fix the circuit without proper insulation. (But this is less common than
He doesn't dare to try...)*

5. After the expressions **had better** (... лучше бы), **would rather/sooner** (... бы, пожалуй), **cannot but** (не могу не...), **why (not)** (почему бы не...), etc.:

*I would sooner/rather **delay** the test than risk inaccurate results.*

*You had better **check** the power supply before turning on the equipment.*

*Given the evidence, we cannot but **conclude** that the sensor is faulty.*

*The experiment didn't work. Why not **try** a different catalyst?*

6. After formal words **than, rather than, but, except**:

*I'll do anything but **work** with children.*

*It's better to measure the voltage directly than **rely** on the theoretical value.*

*They chose to repair the component, rather than **replace** it.*

*The instructions told us to check everything except **calibrate** the pressure gauge.*

Exercises

6.1 Find the Subjective-with-the-Infinitive Construction. State with what verbs or phrases it is used. Translate the sentences into Russian.

1. The findings of this experiment are expected to contribute significantly to future research.
2. The data appears to indicate that renewable energy sources are becoming more viable.
3. The project guidelines seem to be a clear outline of the requirements.
4. The specifications happen to include a new safety feature.
5. His calculations are believed to be the most accurate in the industry.
6. The estimate might turn out to be higher than originally anticipated.
7. The new software doesn't seem to have made the process more efficient.
8. The prototype is supposed to function without any errors.
9. Who is known to have developed the most effective design for this type of system?
10. The machinery was reported to operate efficiently under various conditions.
11. He has been known to propose innovative solutions to complex problems.

***6.2 Paraphrase using the Subjective-with-the-Infinitive Construction.**

MODEL: *They supposed that the machine was working correctly. – The machine was supposed to be working correctly.*

1. They saw that the engineer was designing a new machine.
2. We know that the power plant operates efficiently.
3. They expected that the team would complete the project on time.
4. It is believed that the new material is more durable.

5. It is said that the last project was a great success.
6. It is reported that the researchers discovered a new energy source.
7. I am sure that the technician will fix the issue quickly.
8. We believe that the new design improves safety.
9. The manager will probably approve the budget today.
10. We think that the students are learning the new software successfully.

6.3 Find the Objective-with-the-Infinitive and the Subjective-with-the-Infinitive Constructions. Translate the sentences into Russian.

1. The results of this work are likely to be of considerable public interest.
2. We expect them to focus on all aspects of Engineering and Technology.
3. Greece appeared to be the only country with a steady downward trend.
4. Examples from the past prove a forecast for 30 years to be sometimes possible and fruitful.
5. The conference title seems to be an announcement of a regular event.
6. I believe the conference to be a great opportunity to share knowledge and expertise in the field of Engineering.
7. They let her talk before pressing her with questions.
8. I didn't really expect him to write such a definitive paper on nanotechnology.
9. We are looking for new solutions and tools that will allow us to cope with methodological challenges.
10. The scientist declared the task to be a serious one.
11. I just happened to see her name in the paper.
12. She thought him to be well-read in this field.
13. I expect their team to win the grant.
14. His work appeared to be a definitive one.
15. Do you know him to be a good specialist in nanotechnology?
16. I said it might turn out to be a terrible mistake.

6.4 Put "to" before the infinitive where it is necessary.

1. He asked the professor ... let him ... present his research at the conference.
2. You must make him ... correct the error in the measurement calculation.
3. She was made ... convert the measurements from metric units to imperial units.
4. He is not sure that it can ... be done, but he is willing ... try.
5. Let me ... help you with your work.
6. She asked me ... read the proposal carefully and ... point out the unique features of the project.
7. You ought ... specify who might be interested in the research results.
8. I looked for the article everywhere but could not ... find it.
9. He was seen ... leave the Institute.
10. We had ... refer to visuals.
11. The professor told me not ... refer to figures in my abstract.
12. Have you heard him ... recommend changes to proposal submission?
13. You had better ... pay special attention to the methods used.
14. I would rather ... check it twice.
15. We shall take care so as not ... use informal vocabulary.

Appendices

Appendix A. Numbers and Basic Math

Symbol	Interpretation	Example
=	equals, is, comes to	$1/5 = 0.2$ One fifth equals zero point two.
+	plus, add	$A + B = C$ A plus B equals C. Add A and B to get C.
-	minus, subtract	$A - B = C$ A minus B is C. Subtract B from A to get C.
x	multiplied by, times	$A \times B = C$ A multiplied by B comes to C.
$\div, /$	divided by, over	$A \div B = C$ A divided by B equals C. $A / B = C$ A over B is C.
10^n	to the power of n to the n^{th} power	$10^5 = 100,000$ Ten to the power of five is one hundred thousand. Ten to the fifth power is one hundred thousand.
2^2	squared	$2^2 = 4$ Two squared makes four.
2^3	cubed	$2^3 = 8$ Two cubed is 8.
0.1	decimal number (tenth)	2.1 Two point one
0.01	decimal number (hundredth)	2.19 Two point one nine
0.001	decimal number (thousandth)	2.193 Two point one nine three
$1/4$	fraction	$\frac{1}{4} + \frac{2}{5} = \frac{13}{20}$ One fourth plus two fifths equals thirteen twentieths.
Σ	sum	$\sum_{i=1}^n Fi$ the sum of forces Fi from $i = 1$ to $i = n$.
$\int$	integral	$\int f(x)dx$ the integral of the function $f(x)$ with respect to x

Appendix B. SI Units

Symbol	Name	Measurement
s	second	time
m	meter	length
kg	kilogram	mass
A	ampere	electric current
K	kelvin	thermodynamic temperature
mol	mole	amount of substance
cd	candela	luminous intensity
Hz	hertz /hɜ:ts/	frequency
N	newton	force
Pa	pascal	pressure, stress
J	joule /dʒu:l/	energy, work, or amount of heat
W	watt /wɒt/	power
C	coulomb /'ku:.lɒm/	electric charge
V	volt	voltage, electric potential difference
F	farad /'fær.æd/	capacitance
Ω	ohm /əʊm/	electrical resistance
S	siemens	electrical conductance
Wb	weber	magnetic flux
T	tesla	magnetic flux density
H	henry	inductance
°C	degree Celsius	Celsius temperature
Bq	becquerel	activity referred to a radionuclide
lm	lumen	luminous flux
lx	lux	illuminance
Gy	gray	absorbed dose, kerma
Sv	sievert /'si:.vɜ:t/	dose equivalent
kat	katal	catalytic activity

Appendix C. How to Read Formulas in Mechanics and Engineering

Formula	Reading	Explanation
$F = ma$	F equals m a	Force equals mass times acceleration (Newton's second law).
$\Sigma = \frac{F}{A}$	Sigma equals F over A	Stress (sigma) equals force (F) divided by area (A).
$E = \frac{\sigma}{\epsilon}$	E equals sigma over epsilon	Young's modulus of elasticity (E) equals stress (sigma) divided by strain (epsilon).
$KE = \frac{1}{2}mv^2$	KE equals one-half m v squared	Kinetic energy (KE) equals one-half times mass (m) times velocity (v) squared.
$PE = mgh$	PE equals m g h	Potential energy (PE) equals mass (m) times the acceleration due to gravity (g) times height (h).
$\tau = r \times F$	Torque equals r cross F	Torque (tau) equals the cross product of the position vector (r) and the force vector (F). It's important to state "cross" here, as opposed to other multiplication operations, since the order matters, and cross product indicates the vector product.
$I = \int r^2 dm$	I equals the integral of r squared dm	Moment of inertia (I) equals the integral of the square of the distance (r) from the axis of rotation times the differential mass (dm).
$\dot{x} = \frac{dx}{dt}$	x dot equals dx over dt	The time derivative of x (x dot) equals the derivative of x with respect to time (t).
$\sum_{i=1}^n F_i = F_1 + F_2 + \dots + F_n$	Summation from i equals one to n of F sub i equals F one plus F two plus up to F_n	The sum of forces F sub i , from i equals one to n , equals the sum of all individual forces.
$\frac{d^2y}{dx^2} + y = 0$	d squared y over dx squared plus y equals zero	The second derivative of y with respect to x , plus y , equals zero (a second-order differential equation).

Appendix D. How to Read Chemical Formulas and Reactions

Element Symbols: When reading chemical formulas, the letters representing elements are pronounced according to the English alphabet.

The symbol of an element represents one atom of that element: Ba = one Barium atom.

Uppercase and lowercase letters are very important: O = one Oxygen atom; S = one Sulfur atom; OS = one Sulfur with one Oxygen; Os = one Osmium atom.

If possible, try always to name the compound rather than read the symbols. For example: NaCl – sodium chloride, KOH – potassium hydroxide, H₂SO₄ – sulphuric acid.

Other Symbols Used in Chemical Reactions:

Symbol	Meaning	Example
Subscript (нижний индекс)	The number of atoms of the element in the molecule	H ₂ O = two hydrogen and one oxygen atoms; Cl ₂ = two chlorine atoms
Superscript (верхний индекс)	The number of charges present	Ca ²⁺ = calcium ions have two positive charges
Single arrow (→)	"give," "pass over," "lead to", "produce", "form"	CaS + H₂O → CaO + H₂S When calcium sulphide <u>combines with</u> water, calcium oxide <u>and</u> hydrogen sulphide <u>are formed</u> .
Plus sign (+) on the left side of the equation	"reacts with", "combines with"	
Plus sign (+) on the right side of the equation	"and"	
(s)	solid phase	CaCO₃ (s) + H₂CO₃ (aq) → Ca(HCO₃)₂ (aq) <u>Solid</u> calcium carbonate reacts with carbonic acid <u>in water</u> to form calcium bicarbonate, which is much more soluble so becomes <u>dissolved in water</u> . Or: <u>Solid</u> calcium carbonate combines with <u>dissolved</u> carbonic acid forming <u>dissolved</u> calcium bicarbonate.
(l)	liquid phase	
(g)	gas phase	
(aq)	aqueous phase (dissolved in water)	
Double arrow (↔)	"forms and is formed from", meaning that forward and reverse reactions are happening at the same rate.	N₂ (g) + 3H₂ (g) ↔ 2NH₃ (g) Nitrogen and hydrogen <u>form and are formed from</u> ammonia.
Coefficient	A number placed before the chemical formula indicates the number of molecules.	CH₄ + 2O₂ → CO₂ + 2H₂O <u>One</u> methane molecule reacts with <u>two</u> oxygen molecules producing (→) <u>one</u> carbon dioxide molecule and <u>two</u> water molecules.
Parentheses (круглые скобки)	The molecule contains many similar groups of atoms.	B(OH)₃ The molecule contains one atom of boron and three atoms each of oxygen and hydrogen. The oxygen and hydrogen atoms are bonded together, they occur in a repeating group and the group repeats three times.

Examples

Chemical Formula/Equation	Pronunciation	Explanation
H ₂ O	[ˈeɪtʃ ˈtuː ˈou]	H two O (water)
HNO ₃	[ˈeɪtʃ ˈen ˈou ˈθriː]	H N O three (Nitric acid)
AgNO ₃	[ˈeɪdʒiː ˈen ˈou ˈθriː]	A G N O three (silver nitrate)
C + O ₂ → CO ₂	C plus O two give CO two	One atom of carbon reacts with one two-atom molecule of oxygen and produces (yields) one molecule of carbon dioxide.
2H ₂ + O ₂ → 2H ₂ O	[ˈtuː ˈmɒlɪkjuːlz əv ˈeɪtʃ ˈtuː plʌs ˈou ˈtuː gɪv ˈtuː ˈmɒlɪkjuːlz əv ˈeɪtʃ ˈtuː ˈou]	Two two-atom molecules of hydrogen react with one two-atom molecule of oxygen and produce two molecules of water.
N ₂ + 3H ₂ ↔ 2NH ₃	[ˈen ˈtuː plʌs ˈθriː ˈmɒlɪkjuːlz əv ˈeɪtʃ ˈtuː ˈfɔːm ənd aː ˈfɔːmd frəm ˈtuː ˈmɒlɪkjuːlz əv ˈen ˈeɪtʃ ˈθriː]	N two plus three molecules of H two form and are formed from two molecules of NH three. (Or: Nitrogen plus three molecules of hydrogen form and are formed from two molecules of ammonia.)
Zn + H ₂ SO ₄ → ZnSO ₄ + H ₂	[ˈzɛn plʌs ˈeɪtʃ ˈtuː ˈɛs ˈou ˈfɔː ˈgɪvz ˈzɛn ˈɛs ˈou ˈfɔː plʌs ˈeɪtʃ ˈtuː]	One atom of zinc reacts with one molecule of sulphuric acid producing one molecule of zinc sulphate and one molecule of hydrogen.
CH ₄ + 2O ₂ → CO ₂ + 2H ₂ O	[ˈsiː ˈeɪtʃ ˈfɔː ˈplʌs ˈtuː ˈmɒlɪkjuːlz əv ˈou ˈtuː ˈgɪvz ˈsiː ˈou ˈtuː ˈplʌs ˈtuː ˈmɒlɪkjuːlz əv ˈeɪtʃ ˈtuː ˈou].	C H four plus two molecules of O two gives C O two plus two molecules of H two O. (Or: Methane reacts with two molecules of oxygen to produce carbon dioxide and two molecules of water.)
H ⁺ + NaHCO ₃ → Na ⁺ + H ₂ CO ₃ → Na ⁺ + H ₂ O + CO ₂	[ˈhaɪdrədʒən ˈaɪən ˈplʌs ˈen ˈeɪ ˈeɪtʃ ˈsiː ˈou ˈθriː ˈgɪvz ˈneɪtriəm ˈaɪən ˈplʌs ˈeɪtʃ ˈtuː ˈsiː ˈou ˈθriː ˈgɪvz ˈneɪtriəm ˈaɪən ˈplʌs ˈeɪtʃ ˈtuː ˈou ˈplʌs ˈsiː ˈou ˈtuː].	Hydrogen ion plus N A H C O three gives sodium ion plus H two C O three gives sodium ion plus H two O plus C O two. (Or: A hydrogen ion reacts with sodium bicarbonate to produce a sodium ion and carbonic acid, which then decomposes into a sodium ion, water, and carbon dioxide.)
4HCl + O ₂ → 2Cl ₂ + 2H ₂ O	[ˈfɔː ˈmɒlɪkjuːlz əv ˈeɪtʃ ˈsiː ˈel ˈplʌs ˈou ˈtuː ˈgɪvz ˈtuː ˈmɒlɪkjuːlz ˈsiː ˈel ənd ˈtuː ˈmɒlɪkjuːlz əv ˈeɪtʃ ˈtuː ˈou].	Four molecules of H C L plus O two gives two molecules C L and two molecules of H two O. (Or: Four molecules of hydrochloric acid plus oxygen give two molecules of chlorine and two molecules of water.)
AcOH ↔ AcO ⁻ + H ⁺	[ˈeɪ ˈsiː ˈou ˈeɪtʃ ˈfɔːmz ənd ɪz ˈfɔːmd frəm ˈeɪ ˈsiː ˈɔ ˈksɪdʒən ˈaɪən ˈplʌs ˈhaɪdrədʒən ˈaɪən].	A C O H forms and is formed from A C O oxygen ion plus hydrogen ion. (Or: Acetic acid dissociates into acetate ion and hydrogen ion, and acetate ion and hydrogen ion react to form acetic acid.)

Appendix E. Abbreviations and Acronyms

Types of Abbreviations and Acronyms:

- **Acronyms:** Abbreviations formed from the first letters of words, pronounced as a word (e.g., NASA, NATO, SCUBA, laser, radar, RAM, CAD, etc.). Acronyms for international standards bodies and technical organizations are frequent, e.g., ISO [ˈaɪsoʊ] – International Organization for Standardization.
- **Initialisms:** Abbreviations formed from the first letters of words, pronounced letter by letter (e.g., USB, CPU, IT, CD). Your understanding of your technical field will help you recognize initialisms (e.g., atomic force microscope (AFM), transmission electron microscope (TEM), field emission scanning electron microscope (FE-SEM), water

contact angle (WCA), voice over Internet protocol (VoIP), metal-organic frameworks (MOF), finite element analysis (FEA), computational fluid dynamics (CFD), etc.

- **Latin Abbreviations:** Abbreviations derived from Latin phrases (*e.g.*, *etc.*, *i.e.*, *e.g.*, *et al.*).
- **Graphical Abbreviations:** Abbreviations which present shortened forms of written words or phrases for the economy of space and effort. They are only written in abbreviated way, but pronounced fully (*pp* (pages), *ll* (lines), *ltd* (limited), *Vol.* (volume), *Fig./Figs.* (figure/s), *Rev.* (revised), *Suppl.* (supplement). Here belong forms of address (*Mr.*, *Mrs.*, *Dr.*); scientific degrees (*BA*, *PhD*); units of measurement (*e.g.*, *m*, *kg*, *s*, *N*, *Pa*, *V*, *A*, *W*, Ω). Many of the Latin abbreviations are also graphical: *cf.* (*confer*) – compare, *e.g.* (*exempli gratia*) – for example, *i.e.* (*id est*) – that is, *viz.* (*videlicet*) – namely.

Latin Abbreviations and Terms

Latin abbreviation	Full Latin word	English equivalent	Use
cf.	confer	compare	Jones 1992, Smith 2003; cf. on methodology Harris 2005.
c./ca	circa	around, approximately	The fossil remains date back to ca. 150 million years ago
e.g.	exempli gratia	for example	Several factors influence plant growth (<i>e.g.</i> , soil quality, water availability, and sunlight exposure).
et al.	et alii	and other people	Johnson et al. (2021) demonstrated a novel approach to gene editing in mammalian cells.
etc.	et cetera	and others, and the rest	All of the objects in our solar system (planets, comets, etc.) orbit the sun.
i.e.	id est	that is, in other words, namely, which is to say	The treatment was administered orally, <i>i.e.</i> , the patients ingested the medication in pill form.
ibid.	ibidem	in the same place	The earlier study showed similar trends (<i>ibid.</i> , p. 45), confirming the initial hypothesis.
id.	idem	the same person	The methodology described by Thompson (<i>id.</i>) was adapted for this

			experiment.
in vitro	in vitro	outside the body, in artificial conditions, in a test tube	The drug's efficacy was first tested in vitro using cultured human cells.
in vivo	in vivo	inside a living body	Subsequent in vivo studies in mice confirmed the compound's therapeutic potential.
N.B.	nota bene	note well, pay attention	N.B.: All measurements were taken at room temperature to ensure consistency.
sic	sic	thus, so	The original text states, "The results was [sic.] significant," indicating a grammatical error in the source.
vs. / v.	versus	against, opposed to	The study compared the effects of drug A vs. drug B on tumor growth rates.

A List of Popular Abbreviations Used in Chemistry

Abbreviation	Full name	Russian equivalent
b.p.	boiling point	температура кипения
DCM	methylene chloride/ dichloromethane	дихлорметан
DMSO	dimethyl sulfoxide	ДМСО, диметилсульфоксид
EPA	Environmental Protection Agency (USA)	Агентство по охране окружающей среды (США)
fp	freezing point	температура застывания/замерзания
FT	Fourier Transform	преобразование Фурье
IR	infrared	инфракрасное излучение
IUPAC	International Union of Pure and Applied Chemistry	Международный союз теоретической и прикладной химии (ИЮПАК)
LC	Liquid Chromatography	жидкостная хроматография
M	molar	молярный
mp	melting point	температура плавления
MS	Mass Spectrum	масс-спектр
NMR	nuclear magnetic resonance	ядерный магнитный резонанс
OSHA	Occupational Safety and Health Administration (USA)	Управление по охране труда (США)

PCR	polymerase chain reaction	ПЦР
ppb	parts per billion	частей на миллиард
ppm	parts per million	частей на миллион
ppt	parts per trillion	частей на триллион
PVC	polyvinyl chloride	ПВХ
SEM	scanning electron microscopy	сканирующая электронная микроскопия
TLC	Thin layer chromatography	тонкослойная хроматография
UV	ultraviolet	ультрафиолетовое излучение
vol	volume	объем
WHO	World Health Organization	Всемирная Организация Здравоохранения (ВОЗ)
w/v	weight per volume	вес на единицу объема
XPS	X-ray photoelectron spectroscopy	рентгеновская электронная спектроскопия

Appendix F. Transcription of Personal Names

There are different ways to write Russian personal names in English. Usually, how a Russian name is spelt in English depends on tradition and personal preferences. When you publish research articles, try to spell your name the same way every time so there is no confusion. We suggest you use the following table to help you spell your name in English:

Russian names - Advisable English variant of spelling

Russian letters	English equivalents	Russian names - Advisable English variant of spelling
А а	A a	Анна Александрова - Anna Aleksandrova
Б б	B b	Борис Бабушкин - Boris Babushkin
В в	V v	Виктор Васильев - Viktor Vasilyev
Г г	G g	Галина Георгиева - Galina Georgiyeva
Д д	D d	Денис Дудырев - Denis Dudyrev
Е е	E e (after consonants), Ye ye (after vowels, in the initial position)	Елена Еремеева - Yelena Yeremeyeva, Сергей Лебедев - Sergey Lebedev
Ё ё	Yo yo / E e	Пётр Фёдоров - Pyotr Fyodorov / Petr Fedorov
Ж ж	Zh zh	Жанна Стриж - Zhanna Strizh
З з	Z z	Зоя Язова - Zoya Yazova
И и	I i	Ирина Ильина - Irina Ilyina
Й й	Y y	Андрей Зайцев - Andrey Zaytsev
К к	K k	Карина Акопян - Karina Akopyan
Кс/кс	Ks / X, ks / x	Алексей – Aleksey/Alexey, Ксения – Ksenia/Xenia
Л л	L l	Леонид Ковалёв - Leonid Kovalyov
М м	M m	Мария Туманова - Maria Tumanova
Н н	N n	Наталья Аничкина - Natalia Anichkina
О о	O o	Олег Строев - Oleg Stroyev
П п	P p	Павел Лапшин - Pavel Lapshin
Р р	R r	Руслан Нуриев - Ruslan Nuriyev
С с	S s	Светлана Стасова - Svetlana Stasova
Т т	T t	Тамара Терентьева - Tamara Terentyeva
У у	U u	Ульяна Бушуева - Ulyana Bushuyeva
Ф ф	F f	Руфина Фролова - Rufina Frolova
Х х	Kh kh	Михаил Тихонов - Mikhail Tikhonov
Ц ц	Ts ts	Цаплин, Кузнецов - Tsaplin, Kuznetsov
Ч ч	Ch ch	Вячеслав Чистович - Vyacheslav

		Chistovich
Ш ш	Sh sh	Шамиль Шарипов – Shamil Sharipov
Щ щ	Shch shch	Щукин, Татищев - Shchukin, Tatishchev
Ъ	-	Объедалов - Obyedalov
Ы	y	Матыцин, Малышев - Matytsin, Malyshev
Ь	-	Ольга Касьянова - Olga Kasyanova
Э э	E e	Элла Этуш - Ella Etush
Ю ю	Yu yu	Юлия Тюрина – Yulia Tyurina
Я я	Ya ya / Ia ia	Яна Смелянская - Yana Smelyanskaya, Татьяна Яковлева - Tatiana Yakovleva
Endings		
-ий	-y, -iy, -i	Дмитрий Петровский – Dmitry Petrovsky / Dmitriy Petrovskiy / Dmitri Petrovski
-ый	-y, -yy	Белый – Bely, Belyy
-ия	-ia, -iya	Мария – Maria / Mariya, Анастасия – Anastasia / Anastasiya
-аев, -иев, -оев	-ayev, -iyev, -oyev / -aev, -iev, -oev	Кабаев – Kabayev / Kabaev, Алиев – Aliyev / Aliev, Строев – Stroyev / Stroev

Appendix G. English Names of Faculties and Departments of the Saint Petersburg State Institute of Technology

Russian Name	English Translation
1 факультет - Химии веществ и материалов	Faculty 1 - Chemistry of Substances and Materials
Кафедра технологии электрохимических производств	Electrochemical Production Technology Department
Кафедра общей химической технологии и катализа	General Chemical Technology and Catalysis Department
Кафедра химической нанотехнологии и материалов электронной техники	Chemical Nanotechnology and Materials for Electronics Department
Кафедра физической химии	Physical Chemistry Department
Кафедра неорганической химии	Inorganic Chemistry Department
Кафедра аналитической химии	Analytical Chemistry Department
Кафедра химической технологии тугоплавких неметаллических и силикатных материалов	Chemical Technology of Refractory Non-Metallic and Silicate Materials Department
Кафедра химии и технологии электровакуумных материалов на базе ОАО «Светлана-Рентген»	Chemistry and Technology of Electro-Vacuum Materials Department (in co-op with OAO Svetlana-Roentgen)
Кафедра технологии неорганических веществ	Inorganic Substance Technology Department
Кафедра материаловедения и технологии высокотемпературных материалов и изделий на базе ООО «Вириал»	Materials Science and Technology of High-Temperature Materials and Products Department (in co-op with OOO Virial)
Кафедра физико-химического конструирования функциональных материалов на базе Физико-технического института им. А. Ф. Иоффе Российской академии наук	Physico-Chemical Design of Functional Materials Department (in co-op with the Ioffe Institute)
2 факультет – Химической и биотехнологии	Faculty 2 – Chemical and Biotechnology
Кафедра молекулярной биотехнологии	Molecular Biotechnology Department
Кафедра технологии микробиологического синтеза	Microbiological Synthesis Technology Department
Кафедра химии и технологии синтетических биологически активных веществ	Chemistry and Technology of Synthetic Biologically Active Substances Department
Кафедра химии и технологии материалов и изделий сорбционной техники	Chemistry and Technology of Sorption Materials and Products Department
Кафедра химической технологии органических красителей и фототропных соединений	Chemical Technology of Organic Dyes and Phototropic Compounds Department
Кафедра технологии нефтехимических и углехимических производств	Technology of Petrochemical and Coal Chemical Industry Department
Кафедра химической технологии полимеров	Department of Chemical Technology of Polymers
Кафедра ресурсосберегающих технологий	Resource-Saving Technologies Department
Кафедра органической химии	Organic Chemistry Department
3 факультет – Механический	Faculty 3 – Mechanical Engineering
Кафедра оборудования и робототехники	Equipment and Robotics of Plastics

переработки пластмасс	Processing Department
Кафедра оптимизации химической и биотехнологической аппаратуры	Optimization of Chemical and Biotechnological Equipment Department
Кафедра процессов и аппаратов	Processes and Apparatuses Department
Кафедра механики	Mechanics Department
Кафедра инженерного проектирования	Engineering Design Department
Кафедра теоретических основ материаловедения	Theoretical Foundations of Materials Science Department
Кафедра химии, физики и биологии наноразмерного состояния на базе Института химии силикатов Российской академии наук	Nanochemistry, Nanophysics, and Nanobiology Department (in co-op with Institute of Silicate Chemistry of the Russian Academy of Sciences)
4 факультет – Информационных технологий и управления	Faculty 4 –Information Technology and Control
Кафедра автоматизации процессов химической промышленности	Automation of Chemical Industry Processes Department
Кафедра автоматизированного проектирования и управления	Computer-Aided Design and Control Systems Department
Кафедра системного анализа и информационных технологий	System Analysis and Information Technology Department
Кафедра математики	Mathematics Department
5 факультет – Инженерно-технологический	Faculty 5 – Engineering Technology
Кафедра химии и технологии органических соединений азота	Chemistry and Technology of Organic Nitrogen Compounds Department
Кафедра химии и технологии высокомолекулярных соединений	Chemistry and Technology of High-Molecular Compounds Department
Кафедра химической энергетики	Chemical Energetics Department
Кафедра высокоэнергетических процессов	High-Energy Processes Department
Кафедра инженерной радиоэкологии и радиохимической технологии	Engineering Radioecology and Radiochemical Technology Department
Кафедра радиационной технологии	Radiation Technology Department
Кафедра технологии редких элементов и наноматериалов на их основе	Rare Elements Technology Department
Кафедра общей физики	Department of General Physics
Кафедра мехатронных технологических комплексов	Mechatronic Technological Complexes Department
Кафедра инженерной защиты окружающей среды	Environmental Engineering Department
Кафедра основ военной подготовки	Basics of Military Training Department

Appendix H. Translation of Names of Companies, Research and Educational Institutions

You should always check the company's or institution's official website for the official English name they use in international contexts. If you have to translate the name yourself, remember the following recommendations:

- The type of business organization (e.g., ООО, ЗАО, ОАО) is not translated but transliterated together with the company's name. For example, ООО «Самсон-Мед» is written as ООО Samson-Med, not as LLC or Ltd.
- Do not use inverted commas (quotation marks) around company names in translation.
- Capitalize major words (e.g., Research Institute of Food Additives)

Here is a list of some Saint Petersburg scientific and educational institutions and their English translations that you may find useful:

Russian Name	English Translation
Санкт-Петербургский государственный технологический институт (технический университет)	The St. Petersburg State Institute of Technology (Technical University)
Санкт-Петербургский государственный университет	Saint Petersburg State University (SPbU)
Институт эволюционной физиологии и биохимии им. И. М. Сеченова Российской академии наук	I.M. Sechenov Institute of Evolutionary Physiology and Biochemistry
Федеральное государственное бюджетное учреждение научно-исследовательский институт синтетического каучука имени академика С.В. Лебедева	Federal State Budgetary Institution 'S.V. Lebedev Research Institute of Synthetic Rubber
НИИ гриппа имени А.А. Смородинцева	A.A. Smorodintsev Research Institute of Influenza
Институт химии силикатов имени И.В. Гребенщикова РАН	I.V. Grebenshchikov Institute of Silicate Chemistry
Институт электрофизики и электроэнергетики РАН	Institute of Electro-physics and Electric Power RAS
Санкт-Петербургский государственный химико-фармацевтический университет	Saint Petersburg State Chemical and Pharmaceutical University
Балтийский федеральный университет имени Иммануила Канта	Immanuel Kant Baltic Federal University
Государственный оптический институт имени С.И. Вавилова	S.I. Vavilov State Optical Institute
Научно-исследовательский институт гигиены, токсикологии и профпатологии (НИИГТП)	Research Institute of Hygiene, Occupational Pathology and Human Ecology (RIHTOP)
«Национальный медицинский исследовательский центр имени В.А. Алмазова» Минздрава России	Federal State Budgetary Institution 'Almazov National Medical Research Centre' of the Ministry of Health
Институт высокомолекулярных соединений Российской академии наук	Institute of Macromolecular Compounds of Russian Academy of Sciences
Санкт-Петербургский горный университет императрицы Екатерины II	Empress Catherine II Saint Petersburg Mining University
Радиевый институт имени В. Г. Хлопина	V.G. Khlopin Radium Institute
Физико-технический институт им. А.Ф. Иоффе Российской Академии Наук	The Ioffe Physical-Technical Institute of the Russian Academy of Sciences

Appendix I. Capitalization in Academic Writing

Always capitalize:

- People's names and titles (*Dr. Jane Smith, Albert Einstein*)
- Job titles and positions when they come before a name but not alone or after it (*Dr. Kellog, President Washington, Professor Collins. But: Peter Collins, professor*)
- Major words in names of schools, universities, departments, government agencies and official bodies (*University of Florida, Department of Mechanical Engineering, Environmental Protection Agency*)
- First and major words in book, article, document, and paper titles and headings (*Fabrication of Highly Porous Platinum by Cyclic Electrodeposition of PtCu Alloys*). But: do not capitalize them in reference lists!
- First and major words in titles of periodicals (*Biochemical Journal, Australian Journal of Chemistry, Annual Reports Section A*)
- Sections and subsections of a scientific paper and all references to figures and tables (*Introduction, see Figure 4, Appendix A, Table 1, Graph*)

Do not capitalize:

- Names of chemical elements: *tungsten, oxygen, nitrogen, californium, curium, francium*
- Names of academic subjects (*algebra, art, history*) unless they are languages (*English, Spanish, French*)

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AI tools were used in the editing process and also to compose the text titled ‘Annual Academic Conferences’